

R codes for Yang et al. The global loss of floristic uniqueness

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Introduction

This document includes the R codes used for the main analysis in *Yang et al. The global loss of floristic uniqueness*. The document is structured following the order: main figures → supplementary tables → supplementary figures. To save space, some figure-styling codes that are not important in presenting the results are not shown. To reduce the size of the document, the large figures are not shown. These include Fig.3a,b,d,e, Fig.4, Supplementary Fig.2, Supplementary Fig.5, Supplementary Fig.7a,b,e,f, Supplementary Fig.12. All the other figures are produced during the knitting of the original Rmarkdown document. All figures except the Fig.1 and Supplementary Fig.1 could be reproduced with the relevant R codes and data. For more questions about the code and data, please contact **Qiang Yang** (qiang.yang@uni-konstanz.de).

Required R packages

```
> Packages_ <- c(
+   "tidyverse", "magrittr", "grid", "gridExtra", "wesanderson", "ape",
+   "scales", "rgdal", "sf", "viridis", "ggrepel", "gtools", "betapart",
+   "ggExtra", "ggeffects", "ggpubr", "broom", "sjlabelled",
+   "mgcv", "nlme", "ciTools", "ggtree", "ggtreeExtra", "castor",
+   "ggnewscale", "FactoMineR", "factoextra", "formula.tools", "spdep",
+   "scico"
+ )
> invisible(lapply(Packages_, library, character.only = TRUE))
```

Fig. 2

define a function for making the hex plot

```
> library(RColorBrewer)
>
> f.hex <- function(dat) {
+   dat %>%
+     ggplot(aes(x = native,
+               y = native.and.naturalized)) +
+     geom_hex(bins = 50,
+              aes(fill = stat(count))) +
+     scale_fill_scico(palette = "bilbao",
+                     trans = "log10") +
+     geom_abline(
+       slope = 1,
+       intercept = 0,
+       color = "black",
+       size = 1,
+       linetype = "dashed"
+     ) +
+     coord_fixed(ratio = 1,
+                 xlim = c(0, 1),
+                 ylim = c(0, 1)) +
+     theme_bw()
+ }
```

define a function for making the histogram

```
> f.hist <- function(dat) {
+   dat %>%
+     ggplot(aes(x = homo.value)) +
+     geom_histogram(bins = 20,
+                    colour = "darkgray",
+                    fill = NA,
+                    size = 0.6) +
+     theme_classic() +
+     geom_vline(xintercept = 0,
+                color = "red",
+                size = .6)
+ }
```

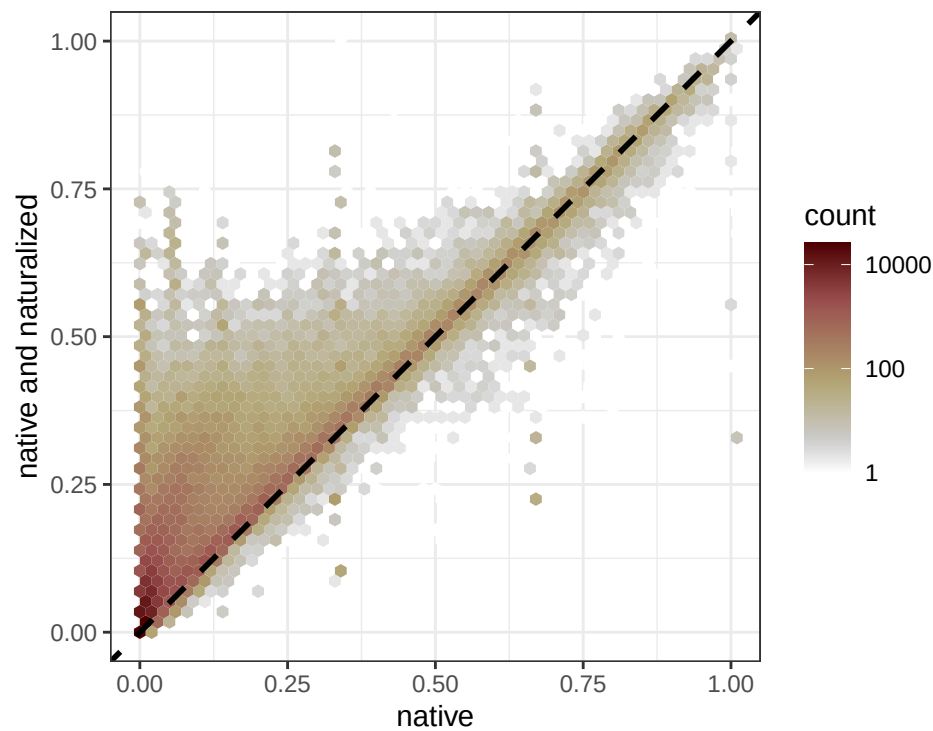
the hex plot in Fig. 2a

```
> load(file = "df.tidy.pairwise.dat.regional.level.Rdata")
> select <- dplyr::select
>
> df.tidy.pairwise.dat.regional.level %>%
+   select(beta.sim.native.identity,
+          beta.sim.native.and.naturalized.identity) %>%
+   rename(
```

```

+   native = beta.sim.native.identity,
+   native.and.naturalized = beta.sim.native.and.naturalized.identity
+ ) %>%
+ f.hex() +
+ scale_fill_scico(palette = "bilbao",
+                 trans = "log10",
+                 breaks = c(1, 100, 10000))
+ ) +
+ xlab("native") +
+ ylab("native and naturalized")

```



the hex plot in Fig. 2b

```

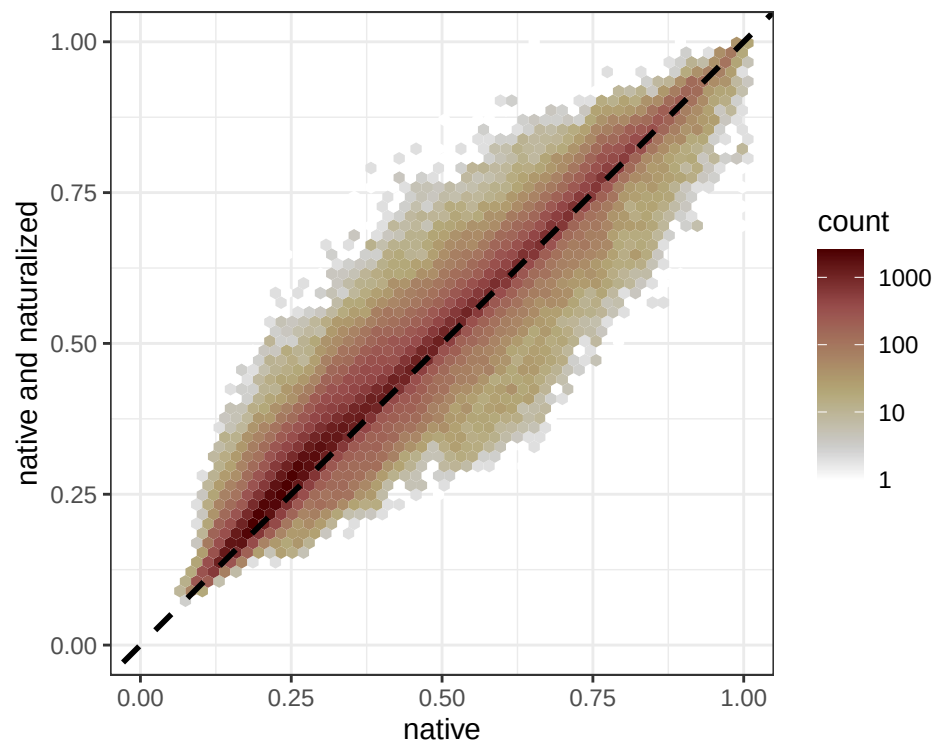
> df.tidy.pairwise.dat.regional.level %>%
+   select(
+     beta.sim.native.phylogeny,
+     beta.sim.native.and.naturalized.phylogeny
+   ) %>%
+   rename(
+     native = beta.sim.native.phylogeny,
+     native.and.naturalized = beta.sim.native.and.naturalized.phylogeny
+   )

```

```

+ ) %>%
+ f.hex() +
+ scale_fill_scico(palette = "bilbao",
+                 trans = "log10",
+                 breaks = c(1, 10, 100, 1000))
+ ) +
+ xlab("native") +
+ ylab("native and naturalized")

```



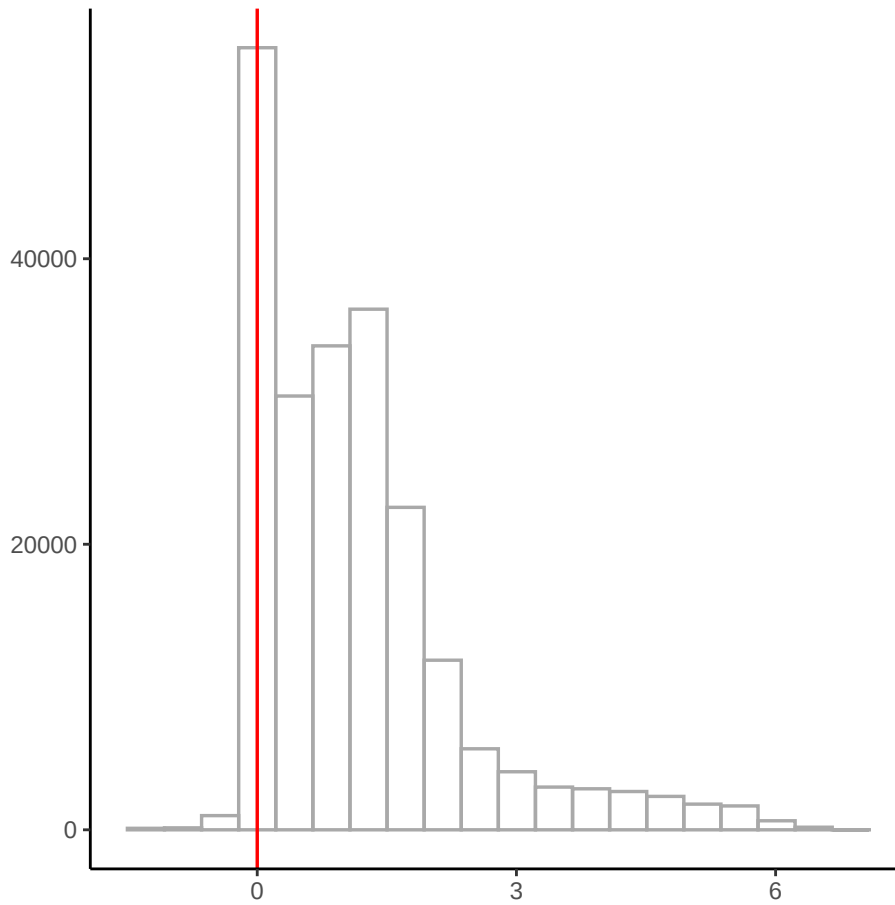
the histogram in Fig. 2a

```

> dat <- df.tidy.pairwise.dat.regional.level %>%
+   select(homo.sim_by.naturalized_delta.0.001_id) %>%
+   rename(homo.value = homo.sim_by.naturalized_delta.0.001_id)
> f.hist(dat) +
+   scale_x_continuous(breaks = c(0, 3, 6)) +
+   xlab("") +
+   ylab("")

```

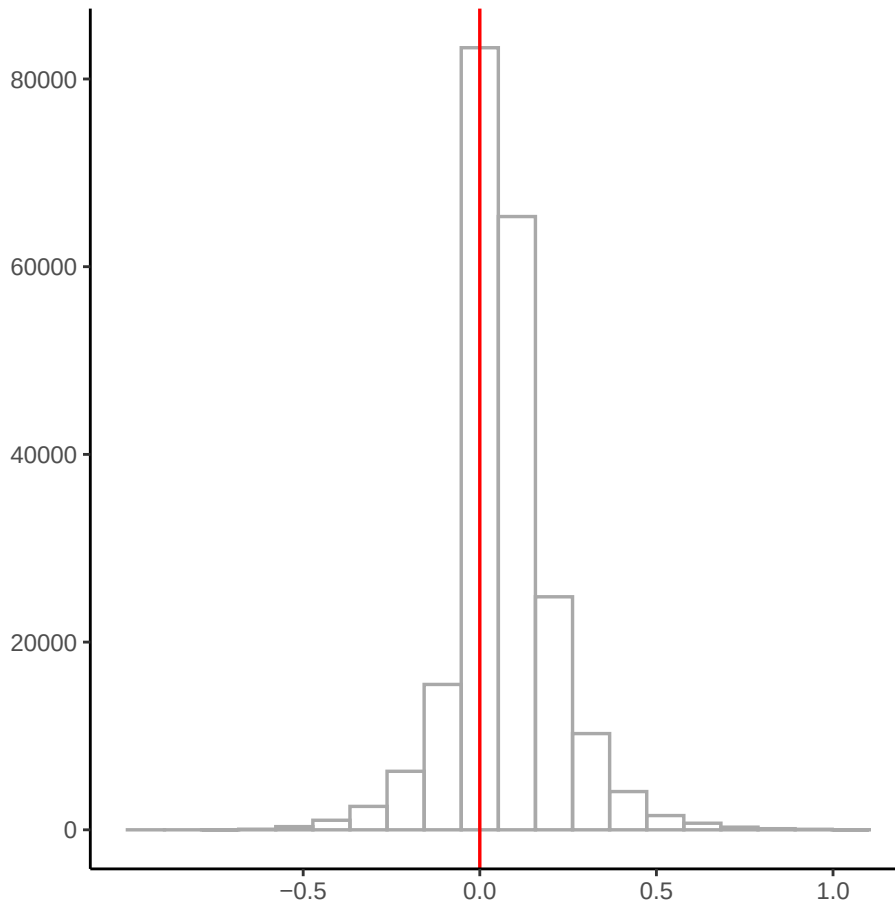
Fig. 2



the histogram in Fig. 2b

```
> dat <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(homo.sim_by.naturalized_delta.0.001_phylo) %>%
+   rename(homo.value = homo.sim_by.naturalized_delta.0.001_phylo)
> f.hist(dat) +
+   scale_x_continuous(breaks = c(-0.5, 0, 0.5, 1)) +
+   xlab("") +
+   ylab("")
```

Fig. 2



the hex plot in Fig. 2c

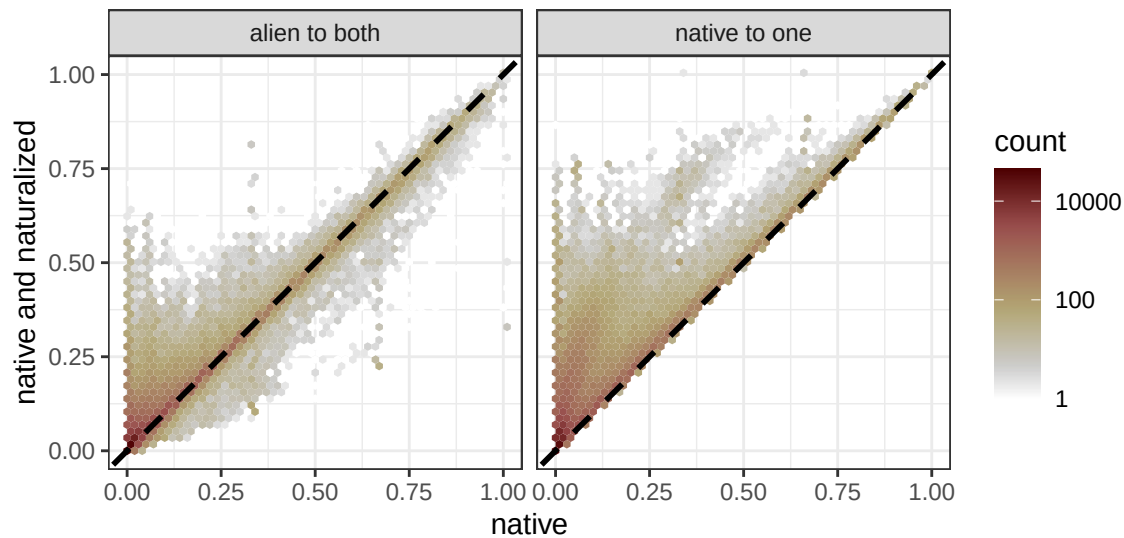
```
> df.plot.dat.identity <- df.tidy.pairwise.dat.regional.level %>%
+   select(beta.sim.native.identity:
+     beta.sim.native.and.outside.species.identity) %>%
+   mutate(row.id = 1:nrow(df.tidy.pairwise.dat.regional.level)) %>%
+   rename(
+     `native to one` = beta.sim.native.and.exchange.identity,
+     `alien to both` = beta.sim.native.and.outside.species.identity
+   ) %>%
+   gather(key = "group",
+     value = "Similarity",
+     -row.id,
+     -beta.sim.native.identity) %>%
+   filter(group %in% c("native to one", "alien to both"))
> df.plot.dat.identity %>%
+   ggplot(aes(x = beta.sim.native.identity,
+     y = Similarity)) +
+   facet_wrap(~group,
+     nrow = 1) +
+   geom_hex(bins = 50,
```



```

+       aes(fill = stat(count)) +
+   scale_fill_scico(palette = "bilbao",
+                   trans = "log10",
+                   breaks = c(1, 100, 10000)) +
+   geom_abline(slope = 1,
+               intercept = 0,
+               size = 1,
+               linetype = "dashed") +
+   coord_fixed(ratio = 1,
+               xlim = c(0, 1),
+               ylim = c(0, 1)) +
+   theme_bw() +
+   xlab("native") +
+   ylab("native and naturalized")

```



the hex plot in Fig. 2d

```

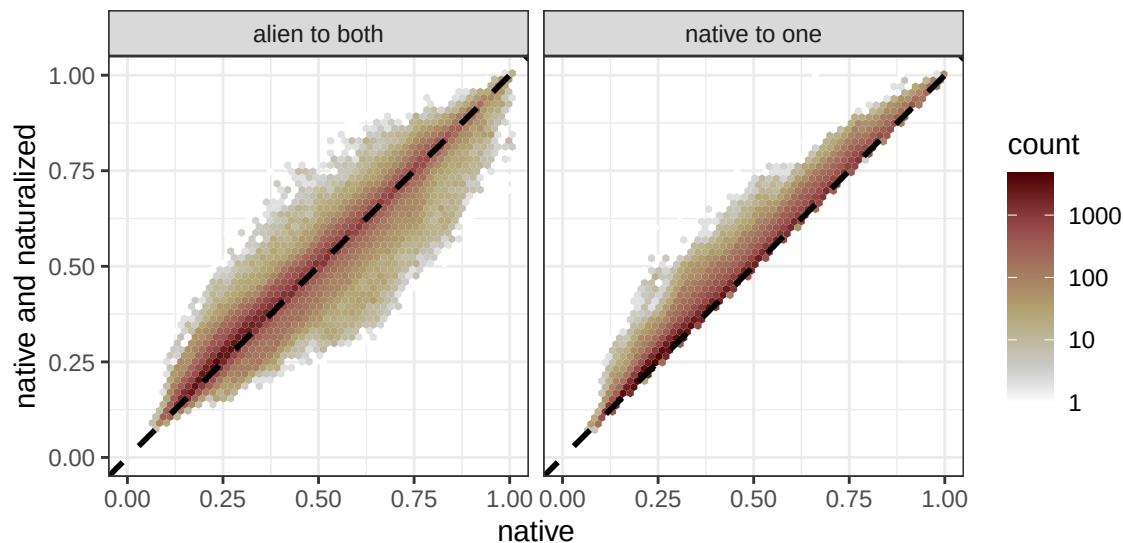
> df.plot.dat.phylogeny <- df.tidy.pairwise.dat.regional.level %>%
+   select(beta.sim.native.phylogeny:
+         beta.sim.native.and.outside.species.phylogeny) %>%
+   mutate(row.id = 1:nrow(df.tidy.pairwise.dat.regional.level)) %>%
+   rename(
+     `native to one` = beta.sim.native.and.exchange.phylogeny,
+     `alien to both` = beta.sim.native.and.outside.species.phylogeny
+   ) %>%
+   gather(key = "group",
+          value = "Similarity",
+          -row.id,
+          -beta.sim.native.phylogeny) %>%
+   filter(group %in% c("native to one", "alien to both"))
> df.plot.dat.phylogeny %>%
+   ggplot(aes(x = beta.sim.native.phylogeny,
+              y = Similarity)) +

```

```

+ facet_wrap(~group,
+           nrow = 1) +
+ geom_hex(bins = 50,
+          aes(fill = stat(count))) +
+ scale_fill_scico(palette = "bilbao",
+                  trans = "log10",
+                  breaks = c(1, 10, 100, 1000)) +
+ geom_abline(slope = 1,
+             intercept = 0,
+             size = 1,
+             linetype = "dashed") +
+ coord_fixed(ratio = 1,
+             xlim = c(0, 1),
+             ylim = c(0, 1)) +
+ theme_bw() +
+ xlab("native") +
+ ylab("native and naturalized")

```



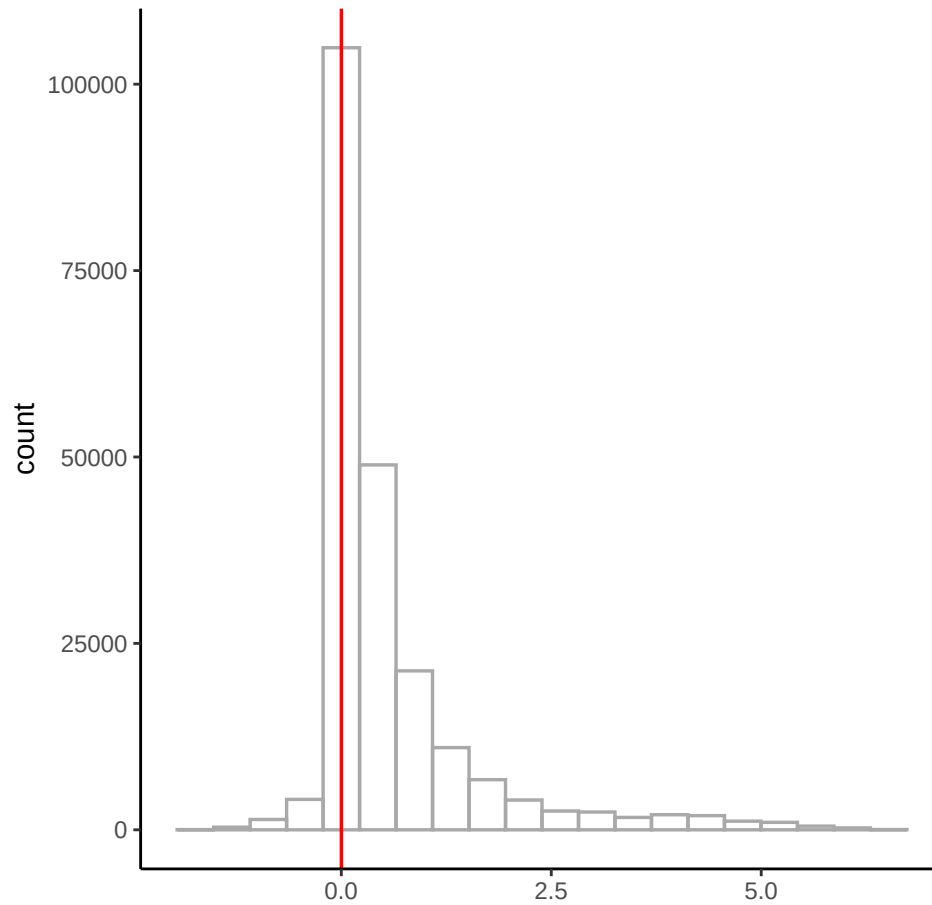
the histogram in Fig. 2c (left panel)

```

> dat.taxonomic.outside <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(homo.sim_by.outside.species_delta.0.001_id) %>%
+   rename(homo.value = homo.sim_by.outside.species_delta.0.001_id)
> f.hist(dat = dat.taxonomic.outside) +
+   scale_x_continuous(breaks = c(0, 2.5, 5)) +
+   xlab("")

```

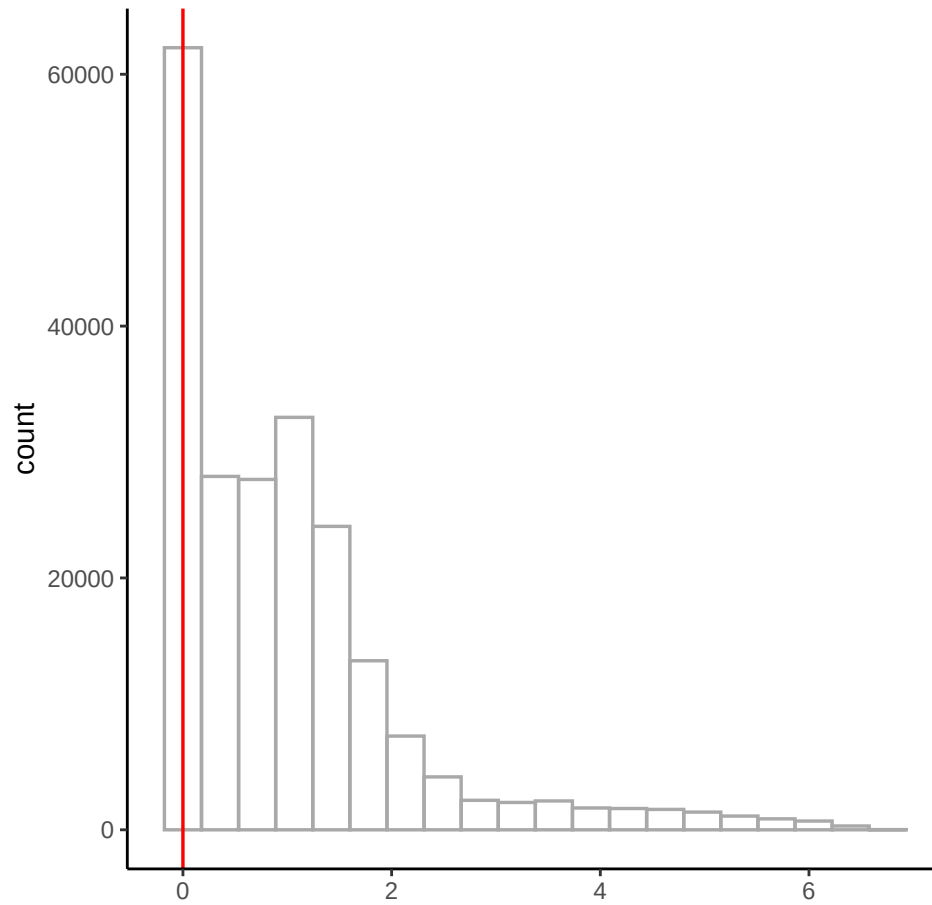
Fig. 2



the histogram in Fig. 2c (right panel)

```
> dat.taxonomic.within <- df.tidy.pairwise.dat.regional.level %>%  
+   dplyr::select(homo.sim_by.exchange_delta.0.001_id) %>%  
+   rename(homo.value = homo.sim_by.exchange_delta.0.001_id)  
> f.hist(dat = dat.taxonomic.within) +  
+   scale_x_continuous(breaks = c(0, 2, 4, 6)) +  
+   xlab("")
```

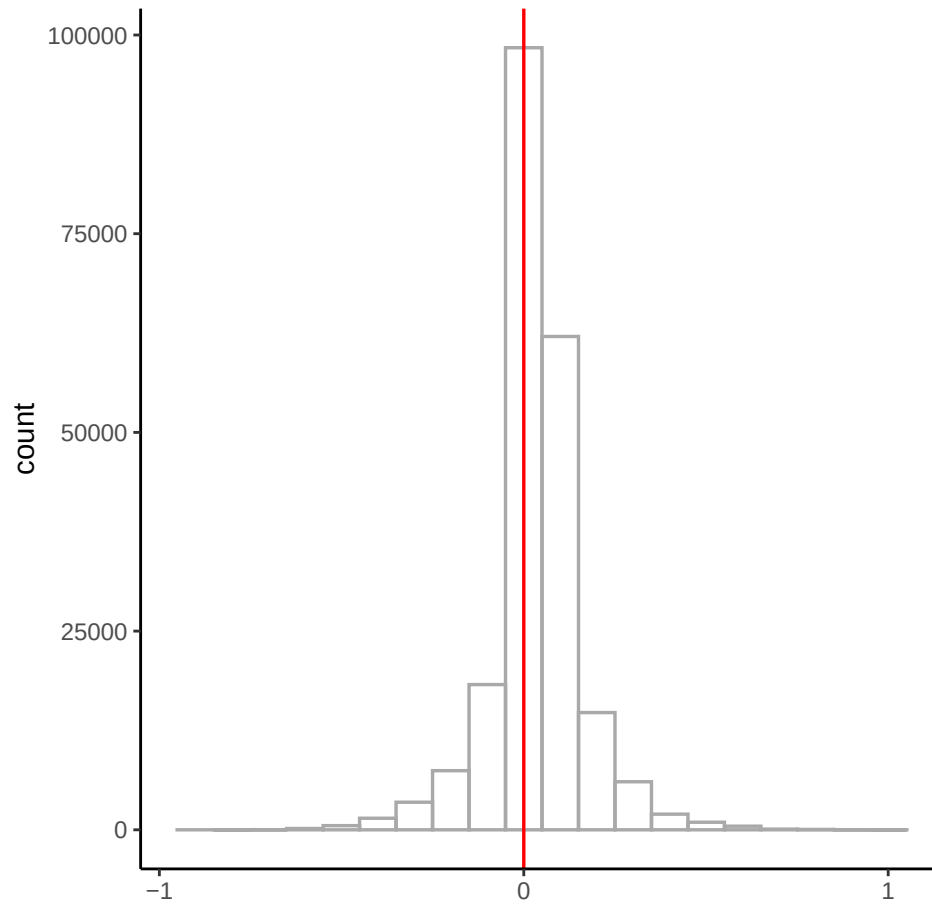
Fig. 2



the histogram in Fig. 2d (left panel)

```
> dat.phylogenetic.outside <- df.tidy.pairwise.dat.regional.level %>%  
+   dplyr::select(homo.sim_by.outside.species_delta.0.001_phylo) %>%  
+   rename(homo.value = homo.sim_by.outside.species_delta.0.001_phylo)  
> f.hist(dat = dat.phylogenetic.outside) +  
+   scale_x_continuous(breaks = c(-1, 0, 1)) +  
+   xlab("")
```

Fig. 2



the histogram in Fig. 2d (right panel)

```
> dat.phylogenetic.within <- df.tidy.pairwise.dat.regional.level %>%  
+   dplyr::select(homo.sim_by.exchange_delta.0.001_phylo) %>%  
+   rename(homo.value = homo.sim_by.exchange_delta.0.001_phylo)  
> f.hist(dat = dat.phylogenetic.within) +  
+   scale_x_continuous(breaks = c(0, 0.5, 1)) +  
+   xlab("")
```

Fig. 2

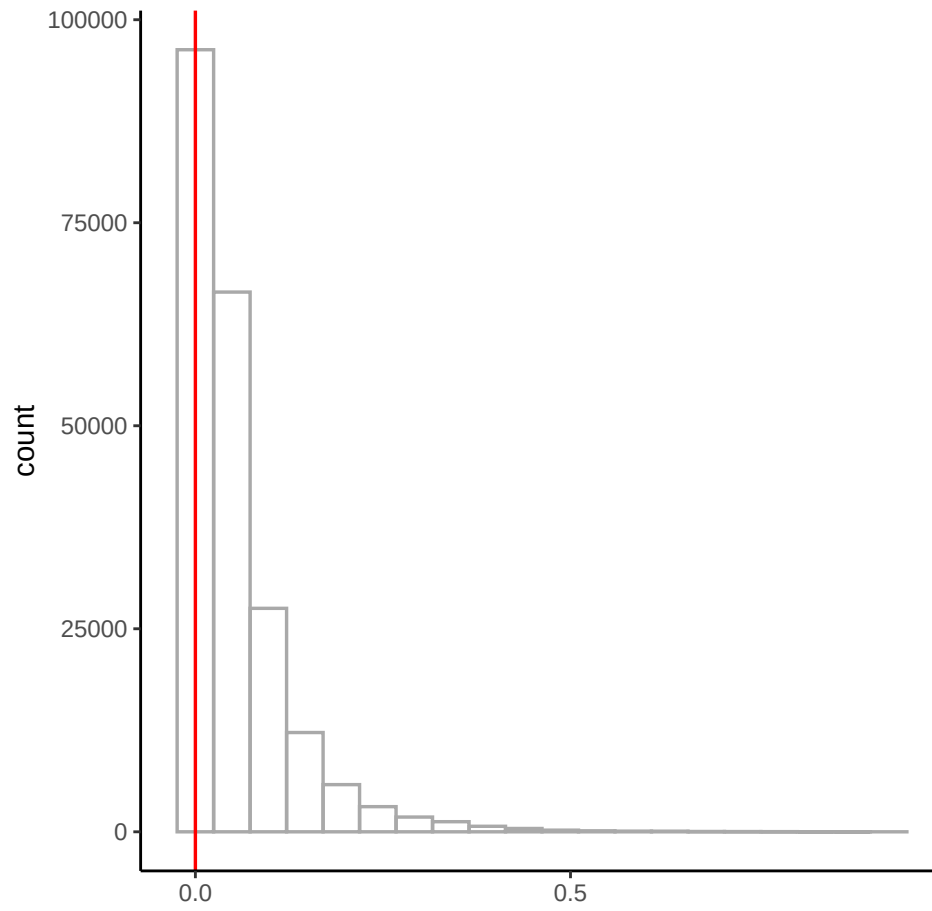


Fig. 3

Fig. 3a,b,d,e include many data points, making them very slow to open. I do not show them in the document, but the readers could reproduce them using the codes in this section if they are interested.

fitting the decay of similarity with geographic distance and climatic dissimilarity with GLMs

```
> load(file = "df.regional.coredata.Rdata")
> GLM.f <- function(response,
+                   predictor,
+                   formula_,
+                   newdata_) {
+   df <- df.regional %>%
+     dplyr::select(one_of(c(response,
+                             predictor))) %>%
+     na.omit()
+   colnames(df) <- c("y", "x")
+   fit.obs <- glm(formula = formula_,
+                  family = binomial(link = "log"),
+                  data = df)
+   predicted.obs <- predict(fit.obs,
+                            newdata = newdata_,
+                            type = "response")
+   out <- newdata_ %>% mutate(predicted.obs = predicted.obs)
+   return(out)
+ }
> mod.native.tax.sim.to.geo.d <-
+   GLM.f(
+     response = "beta.sim.native.identity",
+     predictor = "geographic_distance",
+     formula_ = y ~ x + 0,
+     newdata_ = data.frame(x = seq(1, 20001, length.out = 500))
+   )
> mod.natura.tax.sim.to.geo.d <-
+   GLM.f(
+     response = "beta.sim.native.and.naturalized.identity",
+     predictor = "geographic_distance",
+     formula_ = y ~ x + 0,
+     newdata_ = data.frame(x = seq(1, 20001, length.out = 500))
+   )
> mod.native.phy.sim.to.geo.d <-
+   GLM.f(
+     response = "beta.sim.native.phylogeny",
+     predictor = "geographic_distance",
+     formula_ = y ~ x + 0,
+     newdata_ = data.frame(x = seq(1, 20001, length.out = 500))
+   )
> mod.natura.phy.sim.to.geo.d <-
+   GLM.f(
+     response = "beta.sim.native.and.naturalized.phylogeny",
+     predictor = "geographic_distance",
+     formula_ = y ~ x + 0,
+     newdata_ = data.frame(x = seq(1, 20001, length.out = 500))
+   )
```

```

+ )
> mod.native.tax.sim.to.cli.d <-
+   GLM.f(
+     response = "beta.sim.native.identity",
+     predictor = "environmental_distance",
+     formula_ = y ~ x,
+     newdata_ = data.frame(x = seq(0, 15, length.out = 500))
+   )
> mod.natura.tax.sim.to.cli.d <-
+   GLM.f(
+     response = "beta.sim.native.and.naturalized.identity",
+     predictor = "environmental_distance",
+     formula_ = y ~ x,
+     newdata_ = data.frame(x = seq(0, 15, length.out = 500))
+   )
> mod.native.phy.sim.to.cli.d <-
+   GLM.f(
+     response = "beta.sim.native.phylogeny",
+     predictor = "environmental_distance",
+     formula_ = y ~ x,
+     newdata_ = data.frame(x = seq(0, 15, length.out = 500))
+   )
> mod.natura.phy.sim.to.cli.d <-
+   GLM.f(
+     response = "beta.sim.native.and.naturalized.phylogeny",
+     predictor = "environmental_distance",
+     formula_ = y ~ x,
+     newdata_ = data.frame(x = seq(0, 15, length.out = 500))
+   )

```

Fig. 3a

```

> df.plot <- df.regional %>%
+   select(
+     beta.sim.native.identity,
+     beta.sim.native.and.naturalized.identity,
+     geographic_distance
+   )
> colnames(df.plot) <- c("native", "native and naturalized", "x")
> df.plot %<>%
+   gather(key = "key",
+          value = "value", -x)
> df.plot %<>%
+   mutate(key = fct_relevel(key,
+                             c("native", "native and naturalized")))
> p.taxonomic.similarity.vs.geographic.distance <- df.plot %>%
+   ggplot(aes(x = x, y = value)) +
+   geom_point(aes(color = key),
+             alpha = 0.02,
+             pch = 1,
+             size = 0.2) +
+   theme_bw() +

```



```

+   theme(
+     legend.position = "top",
+     legend.title = element_blank()
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF")) +
+   ylab("Taxonomic similarity") +
+   xlab("Geographic distance (km)") +
+   ggtitle("a") +
+   scale_x_continuous(breaks = seq(0, 15000, 5000))
> # add fitting curves
> fit.dat <- mod.native.tax.sim.to.geo.d %>%
+   mutate(group = "native") %>%
+   bind_rows(mod.natura.tax.sim.to.geo.d %>%
+     mutate(group = "native and naturalized")) %>%
+   set_colnames(c("geographic_distance", ".fitted", "group"))
> p.taxonomic.similarity.vs.geographic.distance +
+   geom_line(
+     data = fit.dat,
+     aes(x = geographic_distance,
+         y = .fitted,
+         color = group),
+     size = 2
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF"))

```

Fig. 3b

```

> df.plot <- df.regional %>%
+   select(
+     beta.sim.native.identity,
+     beta.sim.native.and.naturalized.identity,
+     environmental_distance
+   )
> colnames(df.plot) <- c("native", "native and naturalized", "x")
> df.plot %<>%
+   gather(key = "key", value = "value", -x)
> df.plot %<>%
+   mutate(key = fct_relevel(key,
+     c("native", "native and naturalized")))
> p.taxonomic.similarity.vs.climatic.dissimilarity <- df.plot %>%
+   ggplot(aes(x = x, y = value)) +
+   geom_point(aes(color = key),
+     alpha = 0.02,
+     pch = 1,
+     size = 0.2) +
+   theme_bw() +
+   theme(
+     legend.position = "top",
+     legend.title = element_blank()
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF")) +
+   ylab("Taxonomic similarity") +

```

```

+   xlab("Climatic dissimilarity") +
+   ggtitle("b") +
+   scale_x_continuous(breaks = c(0, 4, 8, 12))
> # add fitting curves
> fit.dat <- mod.native.tax.sim.to.cli.d %>%
+   mutate(group = "native") %>%
+   bind_rows(mod.natura.tax.sim.to.cli.d %>%
+     mutate(group = "native and naturalized")) %>%
+   set_colnames(c("climatic_dissimilarity", ".fitted", "group"))
> p.taxonomic.similarity.vs.climatic.dissimilarity +
+   geom_line(
+     data = fit.dat,
+     aes(x = climatic_dissimilarity,
+         y = .fitted,
+         color = group),
+     size = 2
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF"))

```

Fig. 3d

```

> df.plot <- df.regional %>%
+   select(
+     beta.sim.native.phylogeny,
+     beta.sim.native.and.naturalized.phylogeny,
+     geographic_distance
+   )
> colnames(df.plot) <- c("native", "native and naturalized", "x")
> df.plot %<>%
+   gather(key = "key", value = "value", -x)
> df.plot %<>%
+   mutate(key = fct_relevel(key, c("native", "native and naturalized")))
> p.phylogenetic.similarity.vs.geographic.distance <- df.plot %>%
+   ggplot(aes(x = x, y = value)) +
+   geom_point(aes(color = key), alpha = 0.02, pch = 1, size = 0.2) +
+   theme_bw() +
+   theme(
+     legend.position = "top",
+     legend.title = element_blank()
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF")) +
+   ylab("phylogenetic similarity") +
+   xlab("Geographic distance (km)") +
+   ggtitle("d") +
+   scale_x_continuous(breaks = seq(0, 15000, 5000))
> # add fitting curves
> fit.dat <- mod.native.phy.sim.to.geo.d %>%
+   mutate(group = "native") %>%
+   bind_rows(mod.natura.phy.sim.to.geo.d %>%
+     mutate(group = "native and naturalized")) %>%
+   set_colnames(c("geographic_distance", ".fitted", "group"))
> p.phylogenetic.similarity.vs.geographic.distance +

```

```

+   geom_line(
+     data = fit.dat,
+     aes(x = geographic_distance, y = .fitted, color = group), size = 2
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF"))

```

Fig. 3e

```

> df.plot <- df.regional %>%
+   select(
+     beta.sim.native.phylogeny,
+     beta.sim.native.and.naturalized.phylogeny,
+     environmental_distance
+   )
> colnames(df.plot) <- c("native", "native and naturalized", "x")
> df.plot %<>%
+   gather(key = "key", value = "value", -x)
> df.plot %<>%
+   mutate(key = fct_relevel(key, c("native", "native and naturalized")))
> p.phylogenetic.similarity.vs.climatic.dissimilarity <- df.plot %>%
+   ggplot(aes(x = x, y = value)) +
+   geom_point(aes(color = key), alpha = 0.02, pch = 1, size = 0.2) +
+   theme_bw() +
+   theme(
+     legend.position = "top",
+     legend.title = element_blank()
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF")) +
+   ylab("phylogenetic similarity") +
+   xlab("Climatic dissimilarity") +
+   ggtitle("e") +
+   scale_x_continuous(breaks = c(0, 4, 8, 12))
> # add fitting curves
> fit.dat <- mod.native.phy.sim.to.cli.d %>%
+   mutate(group = "native") %>%
+   bind_rows(mod.natura.phy.sim.to.cli.d %>%
+     mutate(group = "native and naturalized")) %>%
+   set_colnames(c("climatic_dissimilarity", ".fitted", "group"))
> p.phylogenetic.similarity.vs.climatic.dissimilarity +
+   geom_line(
+     data = fit.dat,
+     aes(x = climatic_dissimilarity, y = .fitted, color = group), size = 2
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF"))

```

Analysis of the relationship between homogenization and geographic distance, climatic dissimilarity and administrative relationship

```

> load(file = "df.regional.coredata.Rdata")
> load(file = "new.data.predictors.Rdata")
> # MRM

```

```

> df.regional %<>%
+   select(homo.sim_by.naturalized_delta.0.001_id:administrative_relation)
> df.regional$administrative_relation[
+   df.regional$administrative_relation == "others"] <- "dependency"
> df.regional$administrative_relation[
+   df.regional$administrative_relation ==
+   "no administrative relation"] <- "no relation"
> df.regional %<>% na.omit()
>
> df.regional %<>%
+   mutate(administrative_relation = fct_relevel(
+     administrative_relation,
+     c("no relation", "dependency", "same country")
+   ))
> MRM.taxonomic <- lm(
+   homo.sim_by.naturalized_delta.0.001_id ~ geographic_distance *
+   environmental_distance * administrative_relation,
+   data = df.regional
+ )
> MRM.phylogenetic <- lm(
+   homo.sim_by.naturalized_delta.0.001_phylo ~ geographic_distance *
+   environmental_distance * administrative_relation,
+   data = df.regional
+ )
>
> # make predictions
> df.fitted.taxonomic <- new.data %>%
+   mutate(.fitted = predict(MRM.taxonomic, newdata = new.data))
> df.fitted.phylogenetic <- new.data %>%
+   mutate(.fitted = predict(MRM.phylogenetic, newdata = new.data))
>
> # make figures
> p.fitted.taxonomic <- df.fitted.taxonomic %>%
+   ggplot(aes(x = `geographic_distance`, y = `.fitted`)) +
+   geom_line(size = 1, aes(
+     linetype = `administrative_relation`,
+     color = as.factor(`environmental_distance`)
+   )) +
+   scale_color_manual(
+     values =
+     scico(n = 5, palette = "batlow")
+   ) +
+   scale_linetype_manual(values = c("solid", "dashed", "dotted")) +
+   theme_bw() +
+   scale_x_continuous(
+     breaks = seq(0, 20000, 5000)
+   )
> p.fitted.phylogenetic <- df.fitted.phylogenetic %>%
+   ggplot(aes(x = `geographic_distance`, y = `.fitted`)) +
+   geom_line(size = 1, aes(
+     linetype = `administrative_relation`,
+     color = as.factor(`environmental_distance`)
+   )) +

```

```
+ scale_color_manual(  
+   values =  
+     scico(n = 5, palette = "batlow")  
+ ) +  
+ scale_linetype_manual(values = c("solid", "dashed", "dotted")) +  
+ theme_bw() +  
+ scale_x_continuous(  
+   breaks = seq(0, 20000, 5000)  
+ )  
> grid.arrange(p.fitted.taxonomic, p.fitted.phylogenetic)
```

Fig. 3

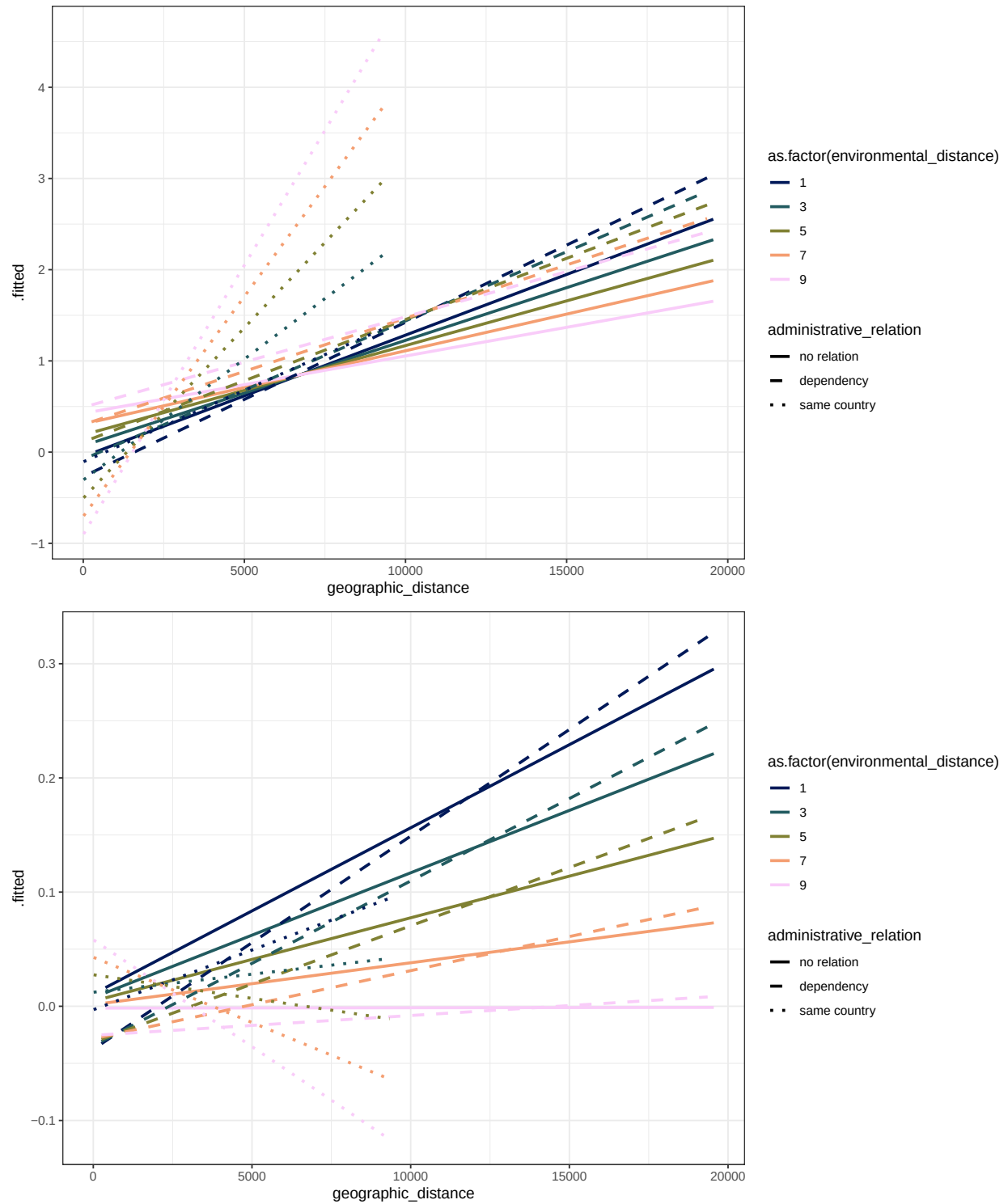


Fig. 4

the map

```
> load(file = "wrld.sf.Rdata")
> load(file = "wrld.regional.level.Rdata")
> load(file = "small.island.centroids.Rdata")
> # define color gradients
> colors1 <- rev(scico::scico(n=7, palette = "bamako")) # color gradients for taxonomic homogenization
> colors2 <- rev(c(scico::scico(n=6, palette = "bamako"), "#CC6699")) # color gradients for phylogenetic
>
> # make the map for mean taxonomic homogenization
> scale_manual_id <- list(
+   scale_fill_manual(values = colors1, drop = FALSE),
+   scale_color_manual(values = colors1, drop = FALSE)
+ )
>
> p.mean.id <-
+   ggplot() +
+   geom_sf(data = wrld.sf, fill = "gray", color = "gray") +
+   geom_sf(
+     data = wrld.regional.level %>% filter(island == 0),
+     aes(fill = mean.homo.id.group), color = NA
+   ) +
+   geom_sf(
+     data = wrld.regional.level %>% filter(island == 1),
+     aes(fill = mean.homo.id.group), color = NA
+   ) +
+   geom_sf(
+     data = small.island.centroids,
+     aes(color = mean.homo.id.group), shape = 1
+   ) +
+   coord_sf(crs = "+proj=wag4 +lon_0=0 +datum=WGS84 +units=m +no_defs") +
+   theme_minimal() +
+   scale_manual_id +
+   labs(title = "a")
>
> # make the figure for mean phylogenetic homogenization
> scale_manual_phylo <- list(
+   scale_fill_manual(values = colors2, drop = FALSE),
+   scale_color_manual(values = colors2, drop = FALSE)
+ )
>
>
> p.mean.phylo <-
+   ggplot() +
+   geom_sf(data = wrld.sf, fill = "gray", color = "gray") +
+   geom_sf(
+     data = wrld.regional.level %>% filter(island == 0),
+     aes(fill = mean.homo.phylo.group), color = NA
+   ) +
+   geom_sf(
+     data = wrld.regional.level %>% filter(island == 1),
```

```

+   aes(fill = mean.homo.phylo.group), color = NA
+ ) +
+   geom_sf(
+     data = small.island.centroids,
+     aes(color = mean.homo.phylo.group), shape = 1
+   ) +
+   coord_sf(crs = "+proj=wag4 +lon_0=0 +datum=WGS84 +units=m +no_defs") +
+   theme_minimal() +
+   scale_manual_phyl +
+   labs(title = "b")
+
>
> grid.arrange(p.mean.id, p.mean.phylo, nrow = 2)

```

regress the mean homogenization to variables characterizing the regions and the regional floras

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.mean.H.and.species.area.island.info.Rdata"))
> df <- df.mean.H.and.species.area.island.info %>% filter(new_region_id != 647)
>
> df.plot <- df %>%
+   na.omit() %>%
+   select(-contains("donor_n"))
> df.plot %<>% mutate(island = as.factor(island))
> df.plot %<>%
+   select(one_of(c(
+     "new_region_id", "mean H.tax", "mean H.phyl", "n_native",
+     "n_naturalized", "pd.native", "pd.naturalized",
+     "endemic.ratio", "donor per spp", "area", "island"
+   )))
>
> df.plot.log <- df.plot %>%
+   mutate(
+     `log(n_native)` = log(n_native),
+     `log(n_naturalized)` = log(n_naturalized),
+     `log(pd.native)` = log(pd.native),
+     `log(pd.naturalized)` = log(pd.naturalized),
+     `log(endemic.ratio+0.01)` = log(endemic.ratio + 0.01),
+     `log(donor_per_spp+1)` = log(`donor per spp` + 1),
+     `log(area)` = log(area)
+   )
>
> df2 <- df.plot.log
>
> # scale explanatory variables
> cols.to.scale <- setdiff(
+   colnames(df2),
+   c("new_region_id", "island", "mean H.tax", "mean H.phyl")
+ )
> df2 %<>% mutate_at(cols.to.scale, scale)
>
> # the regression models
>

```



```

> # add geographic coordinates
> library(spdep)
> load(file = paste0(data.path, "/sp.combined.regions.Rdata"))
> sp.combined.all.regions <- subset(
+   sp.combined.regions,
+   new_region_id %in% df2$new_region_id
+ )
> sp.combined.all.regions@data %<>% left_join(df2)
>
> # define the orig model
> mod.name_ <- c(
+   "H_tax; all regions; island-Y",
+   "H_phyl; all regions; island-Y"
+ )
> mod.formula_ <- list(
+   formula(`mean H_tax` ~ `log(n_native)` + `log(n_naturalized)` +
+     endemic.ratio + `log(donor_per_spp+1)` + `log(area)` + island),
+   formula(`mean H_phyl` ~ `log(pd.native)` + `log(pd.naturalized)` +
+     endemic.ratio + `log(donor_per_spp+1)` + `log(area)` + island)
+ )
> data_ <- list(
+   sp.combined.all.regions, sp.combined.all.regions
+ )
> df.mod <- data.frame(mod.name = mod.name_) %>%
+   as_tibble() %>%
+   mutate(
+     mod.formula = mod.formula_,
+     mod.data = data_
+   )
> # conduct the original regression
> df.mod %<>%
+   mutate(models = map2(
+     mod.formula, mod.data,
+     function(x1, x2) lm(formula = x1, data = x2@data)
+   ))
>
> # add an autocovariate to the model
> # I used the function autocov_dist to get the autocovariate
> # I tried different values for the argument in the function and choose nbs=100,
> # type="inverse.squared" and stype = "S"
> df.mod %<>%
+   mutate(autocovs = map2(
+     models, mod.data,
+     function(x1, x2) {
+       autocov_dist(x1$residuals,
+         xy = coordinates(x2),
+         nbs = 100,
+         type = "inverse.squared", zero.policy = TRUE,
+         style = "S", longlat = NULL
+       )
+     }
+   ))
>

```

```

> # add autocovariate to the data
> library(formula.tools)
> df.mod %<>%
+   mutate(mod.data = map2(
+     mod.data, autocovs,
+     function(x1, x2) {
+       x1@data %<>% mutate(autocov = x2)
+       return(x1)
+     }
+   ))
>
> # re-do the regression
> df.mod %<>%
+   mutate(mod.formula_new = map(mod.formula, function(x) {
+     x %>%
+       as.character() %>%
+       paste(., "+ autocov") %>%
+       as.formula()
+   })))
> df.mod %<>%
+   mutate(models_new = map2(
+     mod.formula_new, mod.data,
+     function(x1, x2) lm(formula = x1, data = x2)
+   ))
>
> library(broom)
> # the model coefficient and performance
> df.mod %<>%
+   mutate(
+     mod.coef.orig = map(models, tidy),
+     mod.coef.sac = map(models_new, tidy)
+   )
>
>
> df.coefficient <- df.mod %>%
+   select(mod.name, mod.coef.orig, mod.coef.sac) %>%
+   gather(key = "key", value = "value", -mod.name)
> df.coefficient %<>% unnest(cols = value)
>
> # rename the predictors
> df.rename <- data.frame(
+   term = c(
+     "(Intercept)",
+     "`log(n_native)`",
+     "`log(n_naturalized)`",
+     "endemic.ratio",
+     "`log(donor_per_spp+1)`",
+     "`log(area)`",
+     "island1",
+     "`log(pd.native)`",
+     "`log(pd.naturalized)`",
+     "autocov"
+   ),

```

```

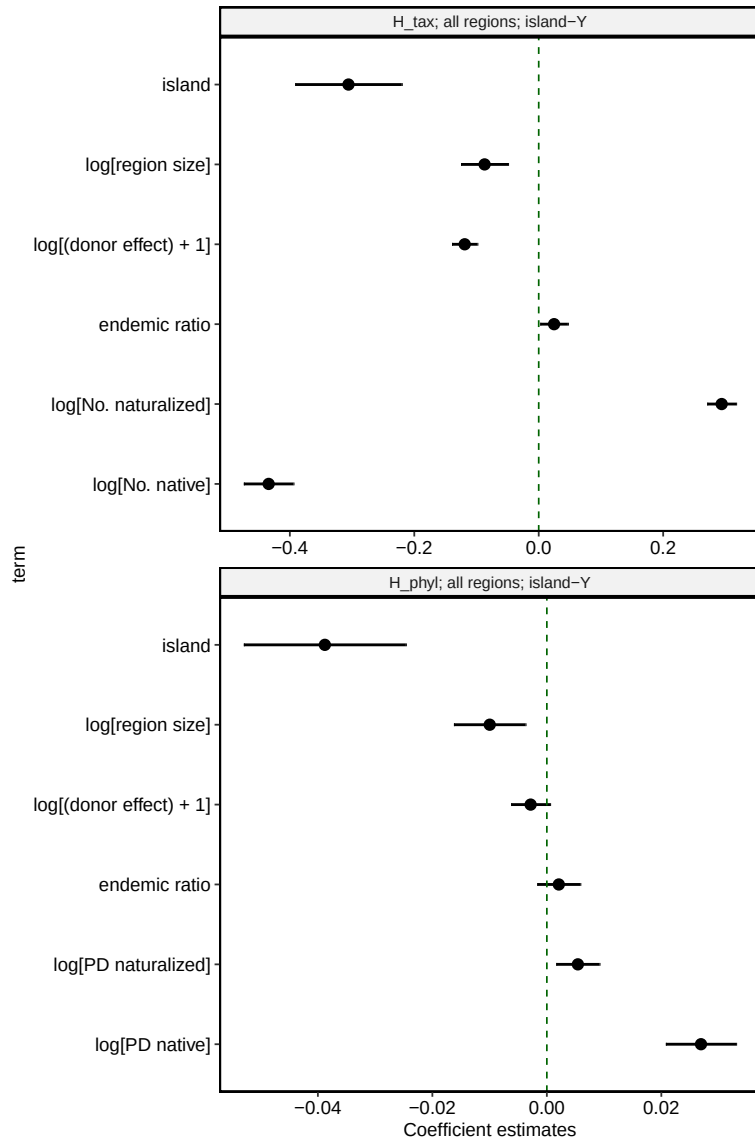
+   new.term = c(
+     "intercept",
+     "log[No. native]",
+     "log[No. naturalized]",
+     "endemic ratio",
+     "log[(donor effect) + 1]",
+     "log[region size]",
+     "island",
+     "log[PD native]",
+     "log[PD naturalized]",
+     "autocov"
+   )
+ )
> df.coefficient %<>% left_join(df.rename)
> df.coefficient %<>% select(-term) %>% rename(term = new.term)
>
> # relevel the facets and the predictors
> df.coefficient %<>%
+   mutate(term = fct_relevel(term, c(
+     "intercept",
+     "autocov",
+     "log[No. native]",
+     "log[No. naturalized]",
+     "log[PD native]",
+     "log[PD naturalized]",
+     "endemic ratio",
+     "log[(donor effect) + 1]",
+     "log[region size]",
+     "island"
+   ))) %>%
+   mutate(mod.name = fct_relevel(
+     mod.name,
+     unique(df.coefficient$mod.name)
+   ))
>
> df.coefficient$key[df.coefficient$key == "mod.coeff.orig"] <-
+   "without autocovariate"
> df.coefficient$key[df.coefficient$key == "mod.coeff.sac"] <-
+   "with autocovariate"
>
> df.coefficient %>%
+   filter(key == "with autocovariate") %>%
+   filter(term != "intercept") %>%
+   filter(term != "autocov") %>%
+   droplevels() %>%
+   ggplot(aes(x = term, y = estimate)) +
+   geom_point() +
+   geom_errorbar(aes(
+     ymin = estimate - 1.96 * std.error,
+     ymax = estimate + 1.96 * std.error
+   ),
+   width = 0
+ ) +

```

```

+ facet_wrap(~mod.name, scales = "free", nrow = 4) +
+ coord_flip() +
+ geom_hline(
+   yintercept = 0, col = "darkgreen",
+   linetype = "dashed", size = 0.3
+ ) +
+ scale_color_manual(values = c("blue", "gray", "red")) +
+ scale_shape_manual(values = c(5, 19)) +
+ theme_pubr(border = T, base_size = 7) +
+ ylab("Coefficient estimates")

```



Supplementary Table 1

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.regional.coredata.Rdata"))
>
> ## conduct fitting of distance decay using GLM
> ## family: binomial; link:log
>
> GLM.f <- function(response, predictor, formula_, n_sim, newdata_) {
+   df <- df.regional %>%
+     dplyr::select(one_of(c(response, predictor))) %>%
+     na.omit()
+   colnames(df) <- c("y", "x")
+   out <- glm(formula_ = formula_, family = binomial(link = "log"), data = df)
+   return(out)
+ }
> set.seed(1234) # about 1 hour to finish
>
> mod.native.tax.sim.to.geo.d <- GLM.f(
+   response = "beta.sim.native.identity",
+   predictor = "geographic_distance",
+   formula_ = y ~ x + 0, n_sim = 1000,
+   newdata_ = data.frame(x = seq(1, 20001,
+     length.out = 500
+   ))
+ )
> mod.natura.tax.sim.to.geo.d <- GLM.f(
+   response =
+     "beta.sim.native.and.naturalized.identity",
+   predictor = "geographic_distance",
+   formula_ = y ~ x + 0, n_sim = 1000,
+   newdata_ = data.frame(x = seq(1, 20001,
+     length.out = 500
+   ))
+ )
> mod.native.phy.sim.to.geo.d <- GLM.f(
+   response = "beta.sim.native.phylogeny",
+   predictor = "geographic_distance",
+   formula_ = y ~ x + 0, n_sim = 1000,
+   newdata_ = data.frame(x = seq(1, 20001,
+     length.out = 500
+   ))
+ )
> mod.natura.phy.sim.to.geo.d <- GLM.f(
+   response =
+     "beta.sim.native.and.naturalized.phylogeny",
+   predictor = "geographic_distance",
+   formula_ = y ~ x + 0, n_sim = 1000,
+   newdata_ = data.frame(x = seq(1, 20001,
+     length.out = 500
+   ))
+ )
> mod.native.tax.sim.to.cli.d <- GLM.f(

```

```

+   response = "beta.sim.native.identity",
+   predictor = "environmental_distance",
+   formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 15,
+     length.out = 500
+   ))
+ )
> mod.natura.tax.sim.to.cli.d <- GLM.f(
+   response =
+     "beta.sim.native.and.naturalized.identity",
+   predictor = "environmental_distance",
+   formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 15,
+     length.out = 500
+   ))
+ )
> mod.native.phy.sim.to.cli.d <- GLM.f(
+   response = "beta.sim.native.phylogeny",
+   predictor = "environmental_distance",
+   formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 15,
+     length.out = 500
+   ))
+ )
> mod.natura.phy.sim.to.cli.d <- GLM.f(
+   response =
+     "beta.sim.native.and.naturalized.phylogeny",
+   predictor = "environmental_distance",
+   formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 15,
+     length.out = 500
+   ))
+ )
>
> summary(mod.native.tax.sim.to.geo.d)$coefficients
> summary(mod.natura.tax.sim.to.geo.d)$coefficients
> summary(mod.native.phy.sim.to.geo.d)$coefficients
> summary(mod.natura.phy.sim.to.geo.d)$coefficients
> summary(mod.native.tax.sim.to.cli.d)$coefficients
> summary(mod.natura.tax.sim.to.cli.d)$coefficients
> summary(mod.native.phy.sim.to.cli.d)$coefficients
> summary(mod.natura.phy.sim.to.cli.d)$coefficients

```

Supplementary Table 2

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.regional.coredata.Rdata"))
>
> # MRM
> df.regional %<>%
+   select(homo.sim_by.naturalized_delta.0.001_id:administrative_relation)
> df.regional$administrative_relation[df.regional$administrative_relation ==
+   "others"] <- "dependency"
> df.regional$administrative_relation[df.regional$administrative_relation ==
+   "no administrative relation"] <- "no relation"
> df.regional %<>% na.omit()
>
> df.regional %<>%
+   mutate(administrative_relation = fct_relevel(
+     administrative_relation,
+     c("no relation", "dependency", "same country")
+   ))
>
> df.regional %<>%
+   mutate_at(c("geographic_distance", "environmental_distance"), scale)
>
> MRM.taxonomic <- lm(
+   homo.sim_by.naturalized_delta.0.001_id ~
+     geographic_distance * environmental_distance * administrative_relation,
+   data = df.regional
+ )
> MRM.phylogenetic <- lm(
+   homo.sim_by.naturalized_delta.0.001_phylo ~
+     geographic_distance * environmental_distance * administrative_relation,
+   data = df.regional
+ )
>
> # the coefficient estimate for taxonomic homogenization
> summary(MRM.taxonomic)$coefficients
>
> # the coefficient estimate for phylogenetic homogenization
> summary(MRM.phylogenetic)$coefficients

```

Supplementary Fig. 2

```

> data.path <- getwd()
> select <- dplyr::select
>
> # the big tree
> full.phylo <- read.tree(file = paste0(data.path, "/full.phylo.tre"))
>
> # order of species
> load(file = paste0(data.path,
+                     "/df.species.matching.and.addition.with.order.info.Rdata"))
>
> # the missing and present species
> load(file = paste0(data.path, "/present.species.Rdata"))
> df.present <- df.species.matching.and.addition.with.order.info %>%
+   dplyr::filter(tip.label.in.the.phylogeny %in% present.species)
> df.missing <- df.species.matching.and.addition.with.order.info %>%
+   anti_join(df.present, by = c("genus", "Species"))
> # select one tip for one binary species
> df.missing.subset <- df.missing %>%
+   group_by(genus, Species) %>%
+   slice_head(n = 1) %>%
+   ungroup()
> df.present %<>% mutate(present = "Y")
> df.missing.subset %<>% mutate(present = "N")
> df.present %<>% bind_rows(df.missing.subset)
>
> # exclude the gymnosperm species
> df.present %<>% dplyr::filter(group == "Angiosperm")
>
> # prune the tree to make every binary species only include one taxon
> full.phylo <- drop.tip(full.phylo,
+                         setdiff(full.phylo$tip.label,
+                                 df.present$tip.label.in.the.phylogeny))
> length(full.phylo$tip.label) # 304417
> table(df.present$present) # N:114655, Y:189762
>
>
> # the number of regions that each species is naturalized to
> load(file = "df.naturalized.Rdata")
>
> # find the the orders with more than 10000 species
> df.order <- df.present %>% count(Order)
> top.orders <- filter(df.order, n > 10000)$Order
> df.top.orders <- df.present %>%
+   select(Order, tip.label.in.the.phylogeny)
> df.top.orders <- left_join(
+   data.frame(tip.label.in.the.phylogeny = full.phylo$tip.label), df.top.orders)
> df.top.orders %<>% filter(Order %in% top.orders)
>
> df.present %<>% select(Order, tip.label.in.the.phylogeny, present) %>%
+   rename(species = tip.label.in.the.phylogeny)
>

```



```

> # rename the present column for displaying the number of missing species
> df.present %<>%
+   mutate(`all species (n = 304,417)` = map_chr(
+     present,
+     function(x) ifelse(x == "Y",
+       "included\n (n = 189,762; 62.3%)",
+       "not included\n (n = 114,655; 37.7%)")
+   ))
>
>
> all.species <- full.phylo$tip.label
>
> species.subset <- all.species
> tree <- drop.tip(full.phylo, setdiff(all.species, species.subset))
> df.present.subset <- df.present %>%
+   filter(species %in% species.subset)
> df.naturalized.subset <- df.naturalized %>%
+   filter(naturalized.tip.label %in% species.subset)
> df.top.oders.subset <- df.top.oders %>%
+   filter(tip.label.in.the.phylogeny %in% species.subset) %>%
+   as_tibble()
>
> # the base tree
> p <- ggtree(tree, layout = "circular", size = 0.01, color = "gray")
> # find the middle tip of each order
> p$data %<>%
+   left_join(df.top.oders, by = c("label" = "tip.label.in.the.phylogeny"))
>
> df.mid.tip <- p$data %>%
+   group_by(Order) %>%
+   summarise(mid.tip.node = round(median(node)))
> mid.tip.node <- df.mid.tip$mid.tip.node
>
>
> p +
+   geom_tiplab2(mapping = aes(subset = (node %in% mid.tip.node), label = Order),
+     offset = 160,
+     size = 4) +
+   geom_fruit(
+     data = df.present.subset %>% mutate(value = 1),
+     geom = geom_bar,
+     mapping = aes(x = value, y = species, fill = `all species (n = 304,417)`),
+     pwidth = 0.2,
+     orientation = "y",
+     stat = "identity"
+   ) +
+   scale_fill_manual(values = c("green", "orange")) +
+   new_scale_fill() +
+   geom_fruit(
+     data = df.naturalized.subset %>%
+       mutate(value = 1) %>%
+       rename(`No. naturalized regions` = n.n.g),
+     geom = geom_bar,

```

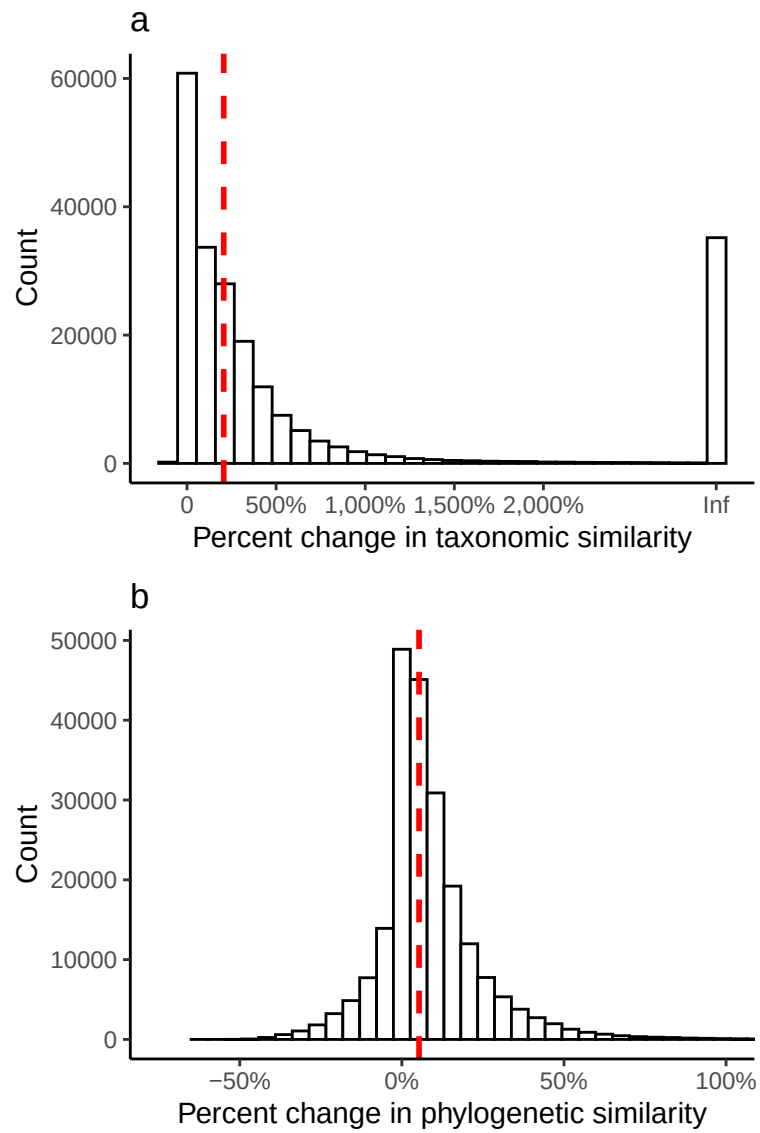
```
+ mapping = aes(x = value,  
+               y = naturalized.tip.label,  
+               fill = `No. naturalized regions`),  
+ pwidth = 0.2,  
+ orientation = "y",  
+ stat = "identity"  
+ ) +  
+ scale_fill_manual(  
+   values = RColorBrewer::brewer.pal(n = 9, name = "Blues")[5:9])
```

Supplementary Fig. 3

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.tidy.pairwise.dat.regional.level.Rdata"))
> df <- df.tidy.pairwise.dat.regional.level
>
> percentage.change.taxonomic <-
+   (df$beta.sim.native.and.naturalized.identity /
+    df$beta.sim.native.identity - 1) * 100
> percentage.change.phylogenetic <-
+   (df$beta.sim.native.and.naturalized.phylogeny /
+    df$beta.sim.native.phylogeny - 1) * 100
> percentage.change.taxonomic[is.na(percentage.change.taxonomic)] <- Inf
> percentage.change.taxonomic[percentage.change.taxonomic >= 3000] <- 3000
>
> p.taxonomic <-
+   data.frame(x = percentage.change.taxonomic) %>%
+   ggplot(aes(x = x)) +
+   geom_histogram(color = "black", fill = "white") +
+   theme_classic() +
+   geom_vline(xintercept = median(percentage.change.taxonomic),
+             color = "red", size = 1, linetype = "dashed") +
+   xlab("Percent change in taxonomic similarity") +
+   ylab("Count") +
+   scale_x_continuous(
+     breaks = c(0, 500, 1000, 1500, 2000, 2970),
+     labels = c("0", "500%", "1,000%", "1,500%", "2,000%", "Inf")
+   ) +
+   ggtitle("a")
>
> p.phylogenetic <-
+   data.frame(x = percentage.change.phylogenetic) %>%
+   ggplot(aes(x = x)) +
+   geom_histogram(color = "black", fill = "white", bins = 50) +
+   theme_classic() +
+   geom_vline(xintercept = median(percentage.change.phylogenetic),
+             color = "red", size = 1, linetype = "dashed") +
+   xlab("Percent change in phylogenetic similarity") +
+   ylab("Count") +
+   scale_x_continuous(
+     breaks = c(-50, 0, 50, 100),
+     labels = c("-50%", "0%", "50%", "100%")
+   ) +
+   coord_cartesian(xlim = c(-75, 100)) +
+   ggtitle("b")
>
> grid.arrange(p.taxonomic,
+   p.phylogenetic,
+   ncol = 1
+ )

```



Supplementary Fig. 4

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.MRM.permutation.Rdata"))
>
> # taxonomic homogenization
>
> df.taxonomic.coefficient.estimates <- df.MRM.permutation %>%
+   dplyr::select(perm.id, perm.note, MRM.L.perm.taxonomic.tidy) %>%
+   unnest(cols = MRM.L.perm.taxonomic.tidy) %>%
+   filter(term != "(Intercept)")
> df.term <- df.taxonomic.coefficient.estimates %>%
+   filter(perm.id == 1) %>%
+   dplyr::select(term) %>%
+   rename(old.term = term) %>%
+   mutate(term = c(
+     "Geo_dist",
+     "Clim_diss",
+     "Admin_dependency",
+     "Admin_same.country",
+     "Geo_dist : Clim_diss",
+     "Geo_dist : Admin_dependency",
+     "Geo_dist : Admin_same.country",
+     "Clim_diss : Admin_dependency",
+     "Clim_diss : Admin_same.country",
+     "Geo_dist : Clim_diss : Admin_dependency",
+     "Geo_dist : Clim_diss : Admin_same.country"
+   ))
> df.taxonomic.coefficient.estimates %<>%
+   rename(old.term = term) %>%
+   left_join(df.term) %>%
+   dplyr::select(-old.term)
> df.taxonomic.coefficient.estimates.obs <-
+   df.taxonomic.coefficient.estimates %>%
+   filter(perm.note == "obs") %>%
+   dplyr::select(perm.id, perm.note, term, estimate) %>%
+   rename(estimate.obs = estimate)
>
> df.taxonomic.coefficient.estimates %<>%
+   dplyr::select(estimate, term) %>%
+   mutate(term = fct_relevel(
+     term,
+     df.taxonomic.coefficient.estimates.obs$term
+   ))
> df.taxonomic.coefficient.estimates.obs %<>%
+   mutate(term = fct_relevel(
+     term,
+     df.taxonomic.coefficient.estimates.obs$term
+   ))
> p.taxonomic.coefficient.estimates <-
+   df.taxonomic.coefficient.estimates %>%
+   ggplot(aes(x = estimate)) +
+   geom_histogram(bins = 50, fill = "gray", color = "white") +

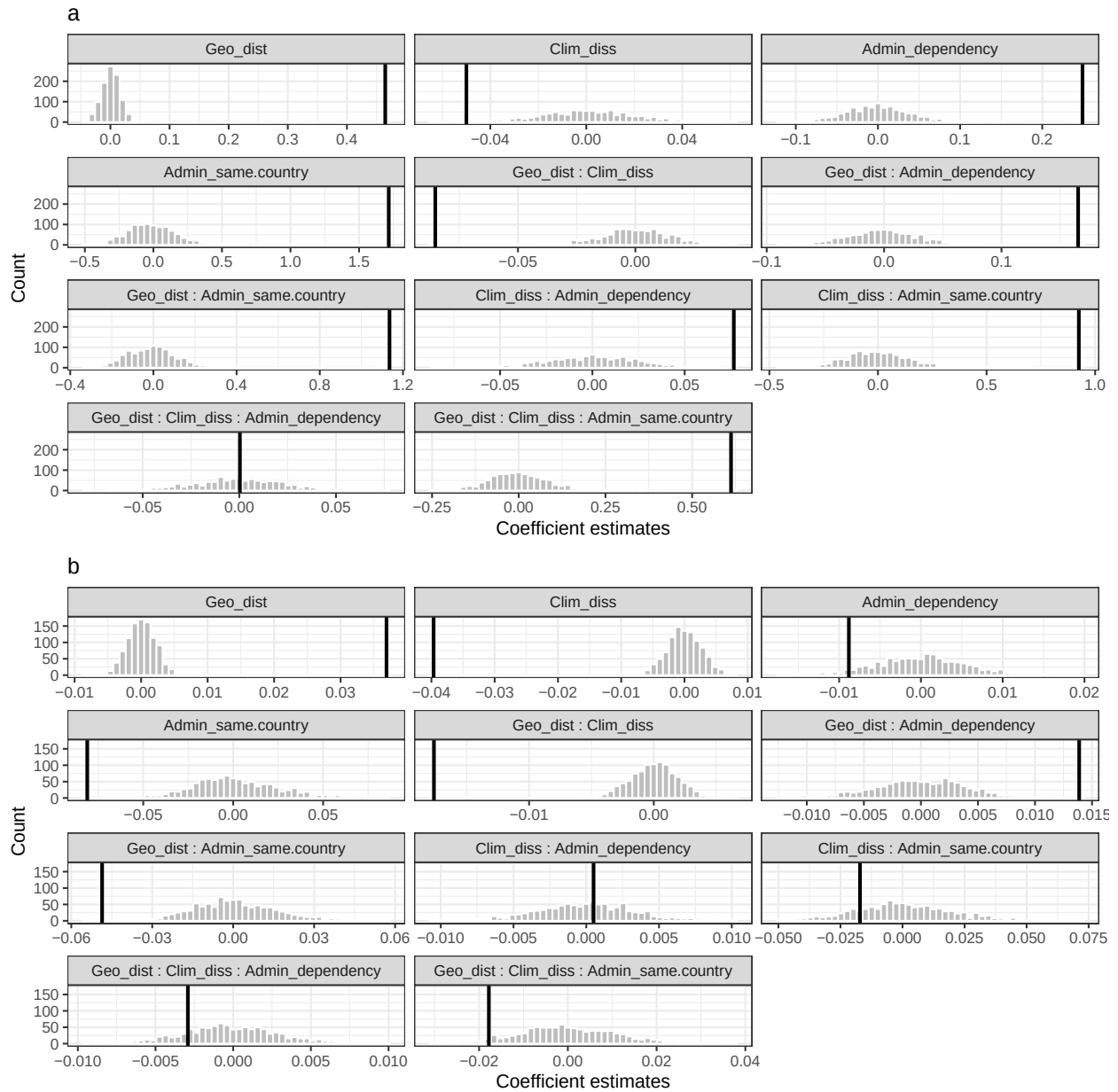
```

```

+   geom_vline(
+     data = df.taxonomic.coefficient.estimates.obs,
+     aes(xintercept = estimate.obs), size = 1
+   ) +
+   facet_wrap(~term, scales = "free_x", ncol = 3) +
+   theme_bw() +
+   scale_color_manual(values = c("red", "darkgreen", "blue")) +
+   xlab("Coefficient estimates") +
+   ylab("Count")
+
>
> # phylogenetic homogenization
>
> df.phylogenetic.coefficient.estimates <- df.MRM.permutation %>%
+   dplyr::select(perm.id, perm.note, MRM.L.perm.phylogenetic.tidy) %>%
+   unnest(cols = MRM.L.perm.phylogenetic.tidy) %>%
+   filter(term != "(Intercept)")
> df.term <- df.phylogenetic.coefficient.estimates %>%
+   filter(perm.id == 1) %>%
+   dplyr::select(term) %>%
+   rename(old.term = term) %>%
+   mutate(term = c(
+     "Geo_dist",
+     "Clim_diss",
+     "Admin_dependency",
+     "Admin_same.country",
+     "Geo_dist : Clim_diss",
+     "Geo_dist : Admin_dependency",
+     "Geo_dist : Admin_same.country",
+     "Clim_diss : Admin_dependency",
+     "Clim_diss : Admin_same.country",
+     "Geo_dist : Clim_diss : Admin_dependency",
+     "Geo_dist : Clim_diss : Admin_same.country"
+   ))
> df.phylogenetic.coefficient.estimates %<>%
+   rename(old.term = term) %>%
+   left_join(df.term) %>%
+   dplyr::select(-old.term)
> df.phylogenetic.coefficient.estimates.obs <-
+   df.phylogenetic.coefficient.estimates %>%
+   filter(perm.note == "obs") %>%
+   dplyr::select(perm.id, perm.note, term, estimate) %>%
+   rename(estimate.obs = estimate)
>
> df.phylogenetic.coefficient.estimates %<>%
+   dplyr::select(estimate, term) %>%
+   mutate(term = fct_relevel(
+     term, df.phylogenetic.coefficient.estimates.obs$term
+   ))
> df.phylogenetic.coefficient.estimates.obs %<>%
+   mutate(term = fct_relevel(
+     term, df.phylogenetic.coefficient.estimates.obs$term
+   ))
> p.phylogenetic.coefficient.estimates <-

```

```
+ df.phylogenetic.coefficient.estimate %>%
+ ggplot(aes(x = estimate)) +
+ geom_histogram(bins = 50, fill = "gray", color = "white") +
+ geom_vline(
+   data = df.phylogenetic.coefficient.estimate.obs,
+   aes(xintercept = estimate.obs), size = 1
+ ) +
+ facet_wrap(~term, scales = "free_x", ncol = 3) +
+ theme_bw() +
+ scale_color_manual(values = c("red", "darkgreen", "blue")) +
+ xlab("Coefficient estimate") +
+ ylab("Count")
>
> grid.arrange(p.taxonomic.coefficient.estimate + ggtitle("a"),
+   p.phylogenetic.coefficient.estimate + ggtitle("b"),
+   nrow = 2
+ )
```



Supplementary Fig. 5

perform the GAM

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.regional.coredata.Rdata"))
>
> # tidy data
> df.regional %<>%
+   dplyr::select(homo.sim_by.naturalized_delta.0.001_id:administrative_relation)
> df.regional$administrative_relation[df.regional$administrative_relation ==
+   "others"] <- "dependency"
> df.regional$administrative_relation[df.regional$administrative_relation ==
+   "no administrative relation"] <-
+   "no relation"
> df.regional %<>% na.omit()
>
> df.regional %<>%
+   mutate(administrative_relation = fct_relevel(
+     administrative_relation,
+     c("no relation", "dependency", "same country")
+   ))

> # define a gam function
> func.gam <- function(
+   Res = "homo.sim_by.naturalized_delta.0.001_id", k.value = 5) {
+   df <- df.regional %>%
+     select(one_of(Res, "geographic_distance",
+       "environmental_distance", "administrative_relation"))
+   colnames(df) <- c("Response", "geographic_distance",
+     "environmental_distance", "administrative_relation")
+   df %<>% as.data.frame()
+   gam.mod <- mgcv::gam(
+     Response ~ administrative_relation + s(geographic_distance, k = k.value) +
+     s(environmental_distance, k = k.value) +
+     s(geographic_distance, environmental_distance, k = k.value) +
+     s(geographic_distance, by = administrative_relation, k = k.value) +
+     s(environmental_distance, by = administrative_relation, k = k.value),
+     data = df, method = "ML"
+   )
+   return(gam.mod)
+ }
>
> # run gams
> set.seed(1234)
> gam.taxonomic.6 <- func.gam(
+   Res = "homo.sim_by.naturalized_delta.0.001_id", k.value = 6)
> gam.phylogenetic.6 <- func.gam(
+   Res = "homo.sim_by.naturalized_delta.0.001_phylo", k.value = 6)
> gams.LL <- list(gam.taxonomic.6, gam.phylogenetic.6)
> names(gams.LL) <- c("gam.taxonomic.6", "gam.phylogenetic.6")

> # load the new data of the predictor
> geo_range <- seq(0, max(df.regional$geographic_distance), length.out = 100) %>%

```

```

+ round()
> geo_range_predictor <- paste(geo_range, collapse = ", ")
> geo_range_predictor <- paste0("geographic_distance [", geo_range_predictor, "]")
> env_range <- seq(
+   0, max(df.regional$environmental_distance), length.out = 100) %>%
+   round(digits = 2)
> env_range_predictor <- paste(env_range, collapse = ", ")
> env_range_predictor <- paste0(
+   "environmental_distance [", env_range_predictor, "]")
> df.geo.range <- df.regional %>%
+   group_by(administrative_relation) %>%
+   summarise(
+     min.geo = quantile(geographic_distance, 0.001),
+     max.geo = quantile(geographic_distance, 0.999)
+   ) %>%
+   ungroup() %>%
+   mutate(geographic_distance = map2(min.geo, max.geo,
+                                     function(x1, x2)
+                                       seq(x1, x2, length.out = 100))) %>%
+   dplyr::select(-min.geo, -max.geo) %>%
+   unnest(cols = geographic_distance)
> environmental_distance <- c(1, 3, 5, 7, 9)
> new.data <- crossing(df.geo.range, environmental_distance)
> # define a function for figure
> gam.figure.func <- function(
+   gam.mod = gams.LL$gam.taxonomic.5,
+   yrange = c(-0.5, 4.5), ymin.legend1 = 2.5,
+   ymax.legend1 = 4.5, ymin.legend2 = -Inf,
+   ymax.legend2 = 1) {
+   df.partial.geographic_distance <- ggpredict(gam.mod, geo_range_predictor) %>%
+     as_tibble() %>%
+     dplyr::select(x, predicted, conf.low, conf.high)
+   p.partial.geographic_distance <- df.partial.geographic_distance %>%
+     ggplot(aes(x = x, y = predicted)) +
+     geom_line(size = 1) +
+     geom_ribbon(aes(ymax = conf.high, ymin = conf.low), alpha = 0.3) +
+     theme_bw() +
+     xlab("Geographic distance\n(km)") +
+     ylab("\nPredicted degree of homogenization") +
+     scale_x_continuous(breaks = seq(0, 20000, 10000)) +
+     coord_cartesian(ylim = yrange)
+   df.partial.environmental_distance <-
+     ggpredict(gam.mod, env_range_predictor) %>%
+     as_tibble() %>%
+     dplyr::select(x, predicted, conf.low, conf.high)
+   p.partial.environmental_distance <- df.partial.environmental_distance %>%
+     ggplot(aes(x = x, y = predicted)) +
+     geom_line(size = 1) +
+     geom_ribbon(aes(ymax = conf.high, ymin = conf.low), alpha = 0.3) +
+     theme_bw() +
+     xlab("Climatic dissimilarity\n") +
+     ylab("\nPredicted degree of homogenization") +

```

```

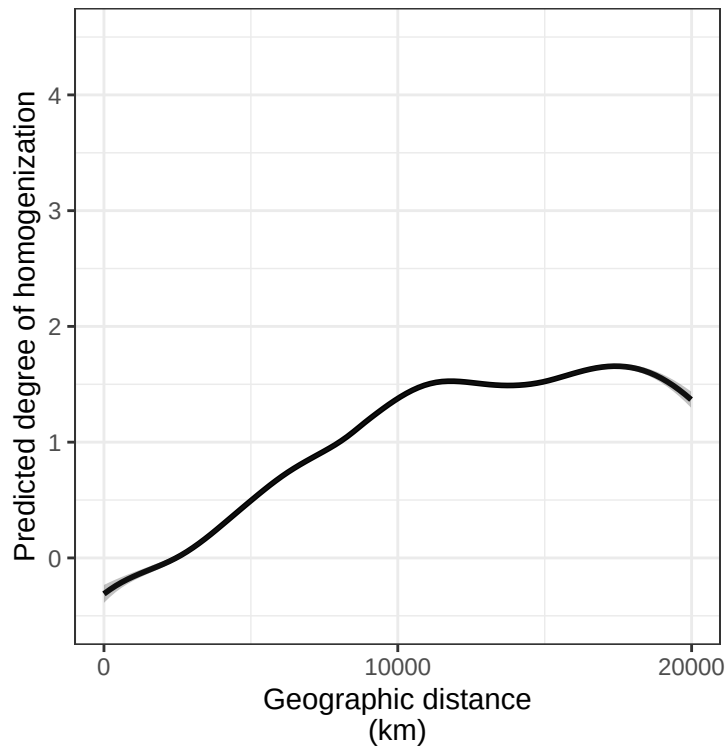
+   scale_x_continuous(breaks = c(0, 3, 6, 9, 12)) +
+   coord_cartesian(ylim = yrange)
+
+ df.partial.administrative_relation <-
+   ggpredict(gam.mod, "administrative_relation") %>% as_tibble()
+ df.partial.administrative_relation %<>%
+   mutate(xlabel = get_labels(x)) %>% as_tibble()
+ df.partial.administrative_relation %<>%
+   mutate(xlabel = fct_relevel(xlabel,
+                               c("no relation", "dependency", "same country")))
+ p.partial.administrative_relation <- df.partial.administrative_relation %>%
+   ggplot(aes(x = xlabel, y = predicted)) +
+   geom_point(size = 3) +
+   geom_errorbar(aes(ymin = conf.low, ymax = conf.high), width = 0.2) +
+   theme_bw() +
+   xlab("Administrative\n relation") +
+   ylab("\nPredicted degree of homogenization") +
+   scale_x_discrete(breaks = c("no relation", "dependency", "same country"),
+                   labels = c("no\n relation", "dependency",
+                              "same\n country")) +
+   coord_cartesian(ylim = yrange)
+
+ df.predicted <- new.data %>%
+   mutate(predicted = predict(object = gam.mod, newdata = new.data))
+ df.predicted %<>%
+   mutate(environmental_distance = as.factor(environmental_distance))
+ p.predicted <- df.predicted %>%
+   ggplot(aes(x = geographic_distance, y = predicted)) +
+   geom_line(size = 1,
+             aes(linetype = `administrative_relation`,
+                 color = `environmental_distance`)) +
+   theme_bw() +
+   scale_color_manual(values = scico(n = 5, palette = "batlow")) +
+   scale_linetype_manual(values = c("solid", "dashed", "dotted")) +
+   scale_x_continuous(breaks = seq(0, 20000, 10000)) +
+   xlab("Geographic distance\n(km)") +
+   ylab("\nPredicted degree of homogenization") +
+   coord_cartesian(ylim = yrange, xlim = c(0, 20000))
+
+ figure.LL <- list(p.partial.geographic_distance,
+                  p.partial.environmental_distance,
+                  p.partial.administrative_relation, p.predicted)
+ return(figure.LL)
+ }
+
+ >
+ > # make the figures
+ >
+ > figures.tax <- gam.figure.func(
+   gam.mod = gams.LL$gam.taxonomic.6,
+   yrange = c(-0.5, 4.5),
+   ymin.legend1 = 2.5, ymax.legend1 = 4.5,
+   ymin.legend2 = -Inf, ymax.legend2 = 1)
+ > figures.phyl <- gam.figure.func(

```

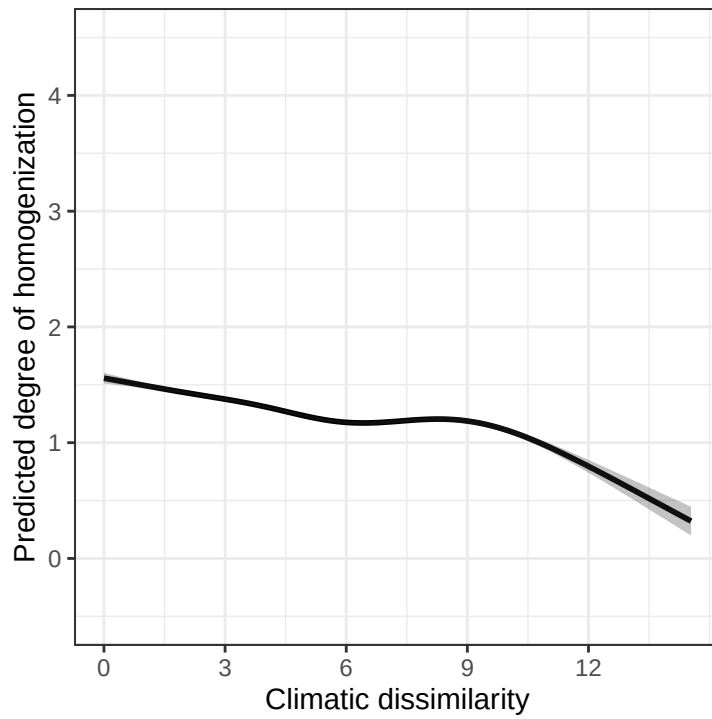
```
+ gam.mod = gams.LL$gam.phylogenetic.6,
+ yrange = c(-0.1, 0.28),
+ ymin.legend1 = 0.13, ymax.legend1 = 0.26,
+ ymin.legend2 = -Inf, ymax.legend2 = 0)
```

taxonomic homogenization

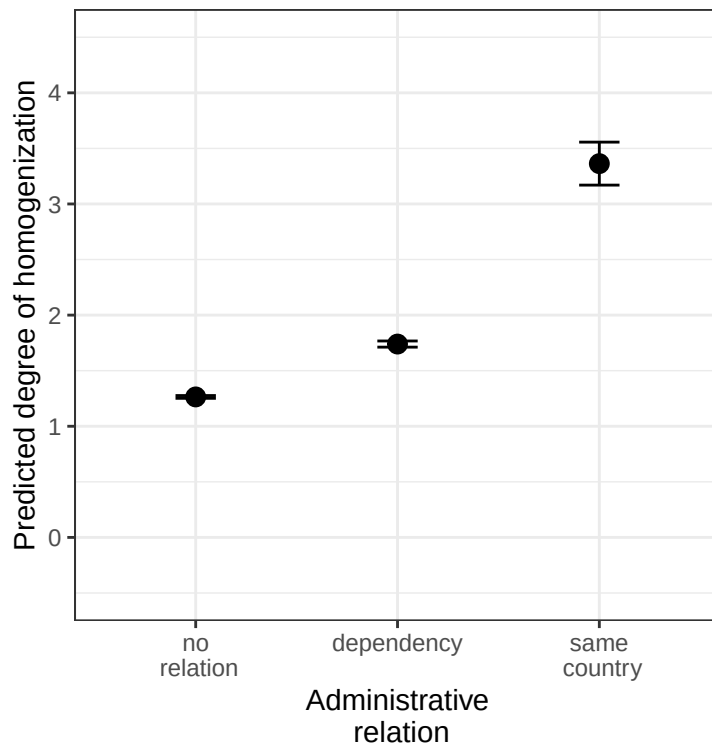
```
> figures.tax[[1]]
```



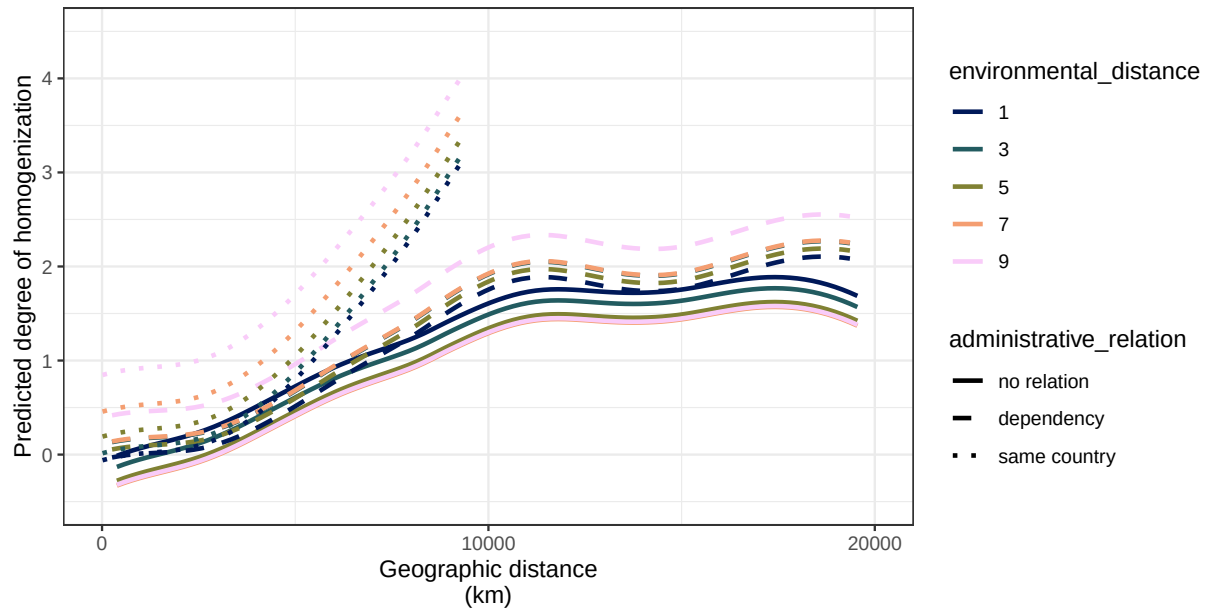
```
> figures.tax[[2]]
```



```
> figures.tax[[3]]
```

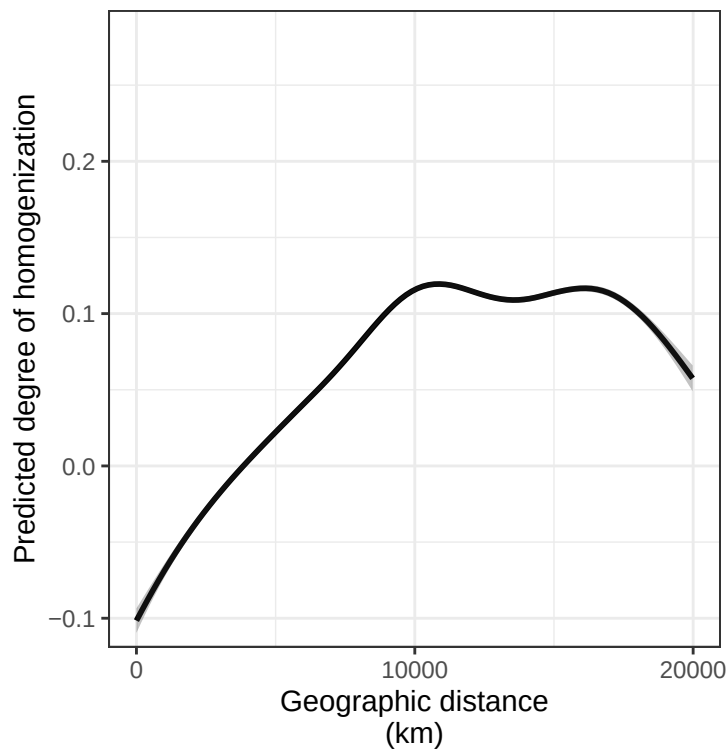


```
> figures.tax[[4]]
```

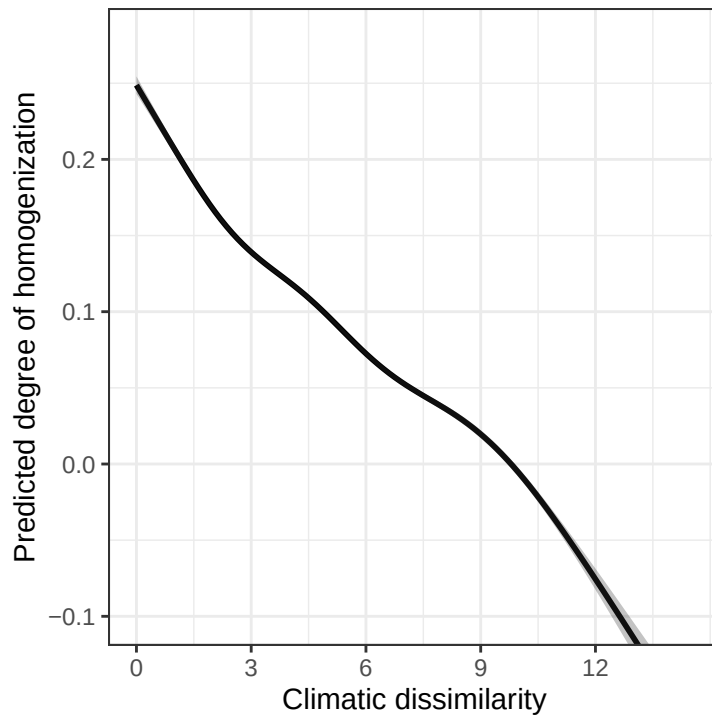


phylogenetic homogenization

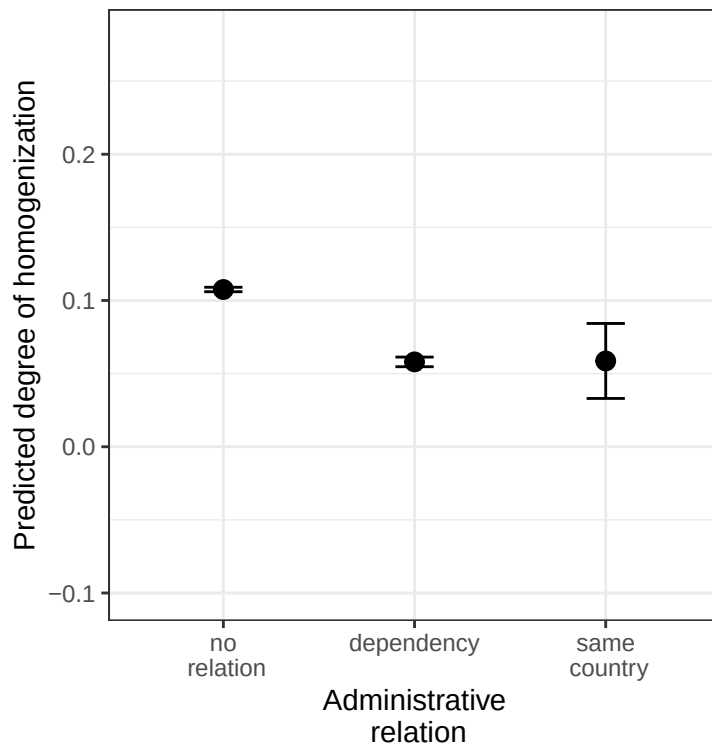
```
> figures.phyl[[1]]
```



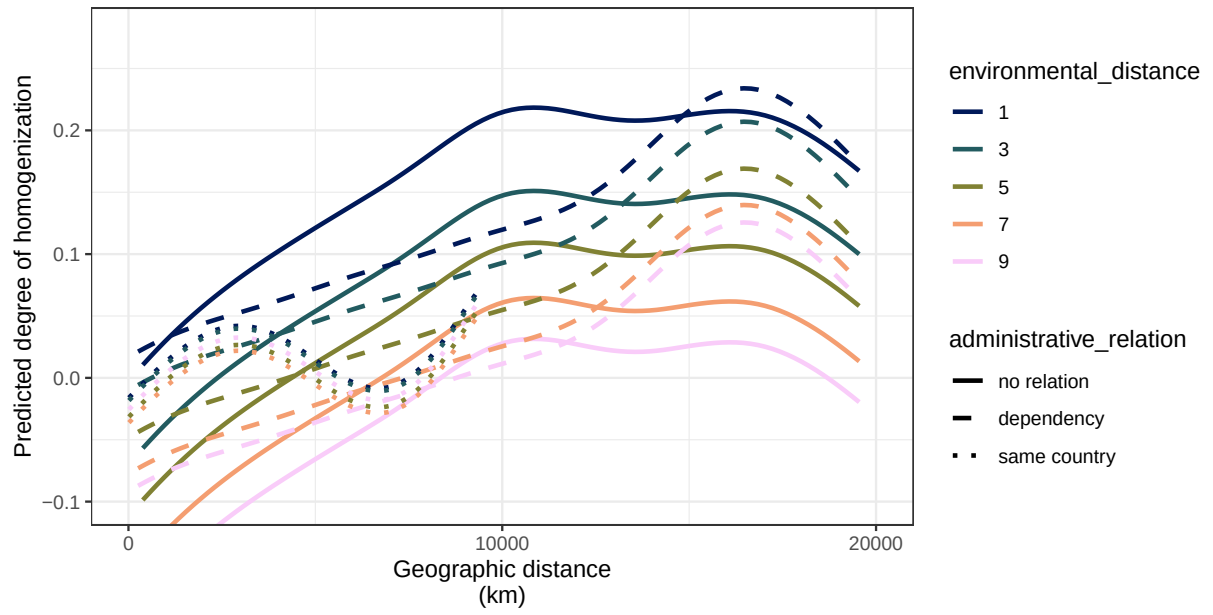
```
> figures.phyl[[2]]
```



```
> figures.phyl[[3]]
```



```
> figures.phyl[[4]]
```

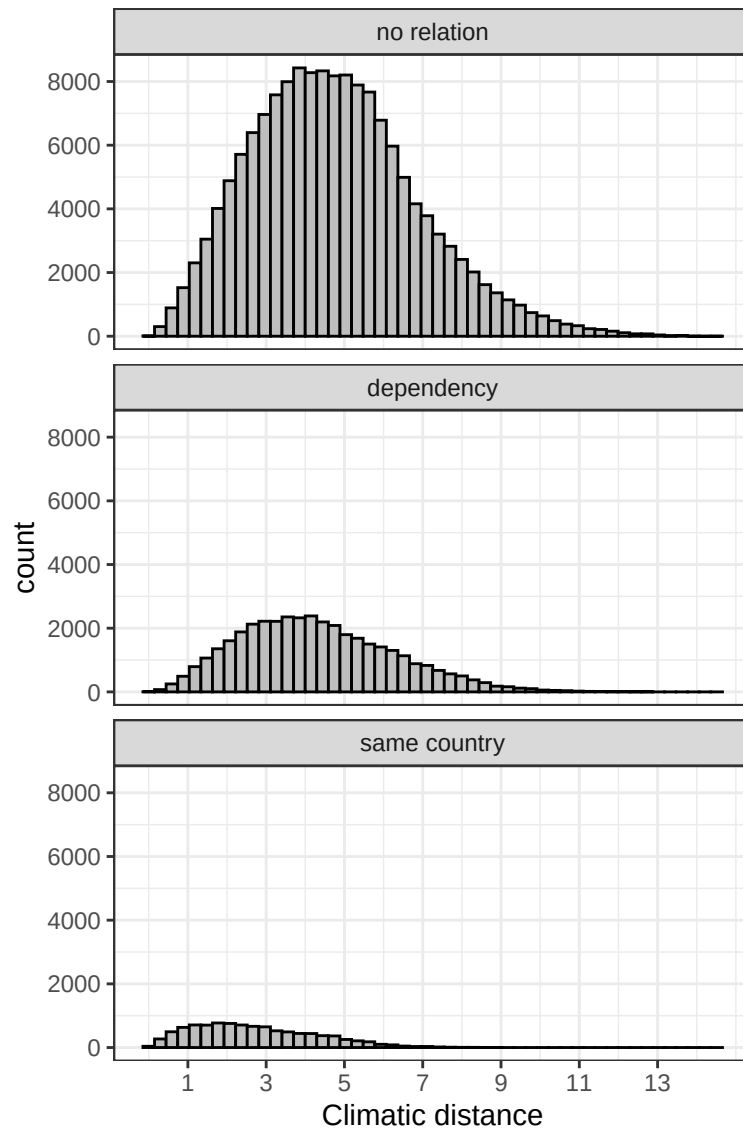


Supplementary Fig. 6

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.regional.coredata.Rdata"))
> df.regional %<>%
+   select(homo.sim_by.naturalized_delta.0.001_id:administrative_relation)
> df.regional$administrative_relation[df.regional$administrative_relation ==
+   "others"] <- "dependency"
> df.regional$administrative_relation[df.regional$administrative_relation ==
+   "no administrative relation"] <-
+   "no relation"
> df.regional %<>% na.omit()
> df.regional %<>%
+   mutate(administrative_relation = fct_relevel(
+     administrative_relation,
+     c("no relation", "dependency", "same country")
+   ))
> df.regional %>%
+   ggplot(aes(x = environmental_distance))+
+   geom_histogram(bins = 50, color="black", fill="gray")+
+   facet_wrap(~administrative_relation, nrow = 3)+
+   theme_bw()+
+   scale_x_continuous(breaks = seq(1,13,2))+
+   xlab("Climatic distance")+
+   scale_y_continuous(breaks = c(0,2000,4000,6000,8000))

```



Supplementary Fig. 7

Similarity along the gradient of PC1 and PC2

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.regional.coredata.Rdata"))
> # get the pc1 and pc2 data
> load(file = paste0(data.path, "/df.bioclimatic.pca.diff.all.cells.Rdata"))
> df.regional %<>% left_join(df.bioclimatic.pca.diff.all.cells)
>
> ## conduct fitting of distance decay using GLM
> ## family: binomial; link:log
>
> GLM.f <- function(response, predictor, formula_, n_sim, newdata_) {
+   df <- df.regional %>%
+     dplyr::select(one_of(c(response, predictor))) %>%
+     na.omit()
+   colnames(df) <- c("y", "x")
+   fit.obs <- glm(formula_ = formula_, family = binomial(link = "log"), data = df)
+   predicted.obs <- predict(fit.obs, newdata = newdata_, type = "response")
+   pred.M <- matrix(0, nrow = nrow(newdata_), ncol = n_sim)
+   for (i in 1:n_sim) {
+     fit.sim <- glm(
+       formula_ = formula_, family = binomial(link = "log"),
+       data = df %>% slice_sample(prop = .5)
+     )
+     predicted.sim <- predict(fit.sim, newdata = newdata_, type = "response")
+     pred.M[, i] <- predicted.sim
+     if (i %% 100 == 0) {
+       print(i)
+       print(Sys.time())
+     }
+   }
+   predicted.lwr <- apply(pred.M, 1, function(x) quantile(x, 0.025))
+   predicted.upr <- apply(pred.M, 1, function(x) quantile(x, 0.975))
+   out <- newdata_ %>%
+     mutate(
+       predicted.obs = predicted.obs,
+       predicted.lwr = predicted.lwr,
+       predicted.upr = predicted.upr
+     )
+   return(out)
+ }
> tictoc::tic()
> set.seed(1234) # about 1 hour to finish
> mod.native.tax.sim.to.pc1 <- GLM.f(
+   response = "beta.sim.native.identity",
+   predictor = "pc1_distance",
+   formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 13, length.out = 500))
+ )
> mod.natura.tax.sim.to.pc1 <- GLM.f(
+   response =

```

```

+   "beta.sim.native.and.naturalized.identity",
+   predictor = "pc1_distance",
+   formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 13, length.out = 500))
+ )
> mod.native.phy.sim.to.pc1 <- GLM.f(
+   response = "beta.sim.native.phylogeny",
+   predictor = "pc1_distance",
+   formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 13, length.out = 500))
+ )
> mod.natura.phy.sim.to.pc1 <- GLM.f(
+   response = "beta.sim.native.and.naturalized.phylogeny",
+   predictor = "pc1_distance", formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 13, length.out = 500))
+ )
> mod.native.tax.sim.to.pc2 <- GLM.f(
+   response = "beta.sim.native.identity",
+   predictor = "pc2_distance", formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 13, length.out = 500))
+ )
> mod.natura.tax.sim.to.pc2 <- GLM.f(
+   response = "beta.sim.native.and.naturalized.identity",
+   predictor = "pc2_distance", formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 13, length.out = 500))
+ )
> mod.native.phy.sim.to.pc2 <- GLM.f(
+   response = "beta.sim.native.phylogeny",
+   predictor = "pc2_distance", formula_ = y ~ x,
+   n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 13, length.out = 500))
+ )
> mod.natura.phy.sim.to.pc2 <- GLM.f(
+   response = "beta.sim.native.and.naturalized.phylogeny",
+   predictor = "pc2_distance", formula_ = y ~ x, n_sim = 1000,
+   newdata_ = data.frame(x = seq(0, 13, length.out = 500))
+ )
>
>
> ## figure: taxonomic similarity vs climatic distance (pc1) ####
>
> df.plot <- df.regional %>%
+   select(beta.sim.native.identity,
+          beta.sim.native.and.naturalized.identity, pc1_distance)
> colnames(df.plot) <- c("native", "native and naturalized", "x")
> df.plot %<>% gather(key = "key", value = "value", -x)
> df.plot %<>%
+   mutate(key = fct_relevel(key, c("native", "native and naturalized")))
> p.taxonomic.similarity.vs.pc1.distance <- df.plot %>%
+   ggplot(aes(x = x, y = value)) +
+   geom_point(aes(color = key), alpha = 0.02, pch = 1, size = 0.2) +
+   theme_bw() +
+   scale_color_manual(values = c("black", "#AA00FF")) +

```

```

+   ylab("Taxonomic similarity") +
+   xlab("Climatic distance (PC1)") +
+   scale_x_continuous(breaks = c(0, 4, 8, 12))
> # fitting curves
> fit.dat <- mod.native.tax.sim.to.pc1 %>%
+   mutate(group = "native") %>%
+   bind_rows(mod.natura.tax.sim.to.pc1 %>%
+     mutate(group = "native and naturalized")) %>%
+   set_colnames(c(
+     "pc1.distance", ".fitted",
+     "lower.ci", "upper.ci", "group"
+   ))
> p.taxonomic.similarity.vs.pc1.distance <-
+   p.taxonomic.similarity.vs.pc1.distance +
+   geom_line(
+     data = fit.dat,
+     aes(x = pc1.distance, y = .fitted, color = group),
+     size = 2
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF"))
>
> ##### -#####
>
> ## figure: phylogenetic similarity vs Climatic distance (PC1) #####
>
> df.plot <- df.regional %>%
+   select(beta.sim.native.phylogeny,
+     beta.sim.native.and.naturalized.phylogeny, pc1_distance)
> colnames(df.plot) <- c("native", "native and naturalized", "x")
> df.plot %<>% gather(key = "key", value = "value", -x)
> df.plot %<>%
+   mutate(key = fct_relevel(key, c("native", "native and naturalized")))
> p.phylogenetic.similarity.vs.pc1.distance <- df.plot %>%
+   ggplot(aes(x = x, y = value)) +
+   geom_point(aes(color = key), alpha = 0.02, pch = 1, size = 0.2) +
+   theme_bw() +
+   scale_color_manual(values = c("black", "#AA00FF")) +
+   ylab("Phylogenetic similarity") +
+   xlab("Climatic distance (PC1)") +
+   ggtitle("TBD") +
+   scale_x_continuous(breaks = c(0, 4, 8, 12))
> # fitting curves
> fit.dat <- mod.native.phy.sim.to.pc1 %>%
+   mutate(group = "native") %>%
+   bind_rows(mod.natura.phy.sim.to.pc1 %>%
+     mutate(group = "native and naturalized")) %>%
+   set_colnames(c(
+     "pc1.distance", ".fitted",
+     "lower.ci", "upper.ci", "group"
+   ))
> p.phylogenetic.similarity.vs.pc1.distance <-
+   p.phylogenetic.similarity.vs.pc1.distance +
+   geom_line(

```

```

+   data = fit.dat,
+   aes(x = pc1.distance, y = .fitted, color = group),
+   size = 2
+ ) +
+   scale_color_manual(values = c("black", "#AA00FF"))
>
> ## figure: taxonomic similarity vs climatic distance (pc2) ####
>
> df.plot <- df.regional %>%
+   select(beta.sim.native.identity,
+           beta.sim.native.and.naturalized.identity, pc2_distance)
> colnames(df.plot) <- c("native", "native and naturalized", "x")
> df.plot %<>% gather(key = "key", value = "value", -x)
> df.plot %<>%
+   mutate(key = fct_relevel(key, c("native", "native and naturalized")))
> p.taxonomic.similarity.vs.pc2.distance <- df.plot %>%
+   ggplot(aes(x = x, y = value)) +
+   geom_point(aes(color = key), alpha = 0.02, pch = 1, size = 0.2) +
+   theme_bw() +
+   scale_color_manual(values = c("black", "#AA00FF")) +
+   ylab("Taxonomic similarity") +
+   xlab("Climatic distance (PC2)") +
+   ggtitle("TBD") +
+   scale_x_continuous(breaks = c(0, 4, 8, 12))
> # fitting curves
> fit.dat <- mod.native.tax.sim.to.pc2 %>%
+   mutate(group = "native") %>%
+   bind_rows(mod.natura.tax.sim.to.pc2 %>%
+     mutate(group = "native and naturalized")) %>%
+   set_colnames(c(
+     "pc2.distance", ".fitted",
+     "lower.ci", "upper.ci", "group"
+   ))
> p.taxonomic.similarity.vs.pc2.distance <-
+   p.taxonomic.similarity.vs.pc2.distance +
+   geom_line(
+     data = fit.dat,
+     aes(x = pc2.distance, y = .fitted, color = group),
+     size = 2
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF"))
>
> #### -####
>
> ## figure: phylogenetic similarity vs Climatic distance (pc2) ####
>
> df.plot <- df.regional %>%
+   select(beta.sim.native.phylogeny,
+           beta.sim.native.and.naturalized.phylogeny, pc2_distance)
> colnames(df.plot) <- c("native", "native and naturalized", "x")
> df.plot %<>% gather(key = "key", value = "value", -x)
> df.plot %<>%
+   mutate(key = fct_relevel(key, c("native", "native and naturalized")))

```

```

> p.phylogenetic.similarity.vs.pc2.distance <- df.plot %>%
+   ggplot(aes(x = x, y = value)) +
+   geom_point(aes(color = key), alpha = 0.02, pch = 1, size = 0.2) +
+   theme_bw() +
+   scale_color_manual(values = c("black", "#AA00FF")) +
+   ylab("Phylogenetic similarity") +
+   xlab("Climatic distance (PC2)") +
+   ggtitle("TBD") +
+   scale_x_continuous(breaks = c(0, 4, 8, 12))
> # fitting curves
> fit.dat <- mod.native.phy.sim.to.pc2 %>%
+   mutate(group = "native") %>%
+   bind_rows(mod.natura.phy.sim.to.pc2 %>%
+     mutate(group = "native and naturalized")) %>%
+   set_colnames(c(
+     "pc2.distance", ".fitted",
+     "lower.ci", "upper.ci", "group"
+   ))
> p.phylogenetic.similarity.vs.pc2.distance <-
+   p.phylogenetic.similarity.vs.pc2.distance +
+   geom_line(
+     data = fit.dat,
+     aes(x = pc2.distance, y = .fitted, color = group),
+     size = 2
+   ) +
+   scale_color_manual(values = c("black", "#AA00FF"))

```

Homogenization as predicted by geographic distance, climatic distance (PC1 and PC2) and administrative relationship

```

> data.path <- getwd()
>
> # prepare the core data
> load(file = paste0(data.path, "/df.regional.coredata.Rdata"))
> load(file = paste0(data.path, "/df.bioclimatic.pca.diff.all.cells.Rdata"))
> load(file = paste0(data.path, "/df_geographic_distance_b2b.Rdata"))
>
> df <- df.regional %>%
+   left_join(df.bioclimatic.pca.diff.all.cells) %>%
+   left_join(df_geographic_distance_b2b) %>%
+   select(
+     homo.sim_by.naturalized_delta.0.001_id,
+     homo.sim_by.naturalized_delta.0.001_phylo,
+     geographic_distance,
+     environmental_distance,
+     administrative_relation,
+     pc1_distance,
+     pc2_distance,
+     pc3_distance,
+     geographic_distance_b2b
+   ) %>%
+   set_colnames(value = c(

```

```

+   "H.tax", "H.phyl", "geo.dist.centroid",
+   "env.dist.centroid", "admin",
+   "env.dist.pc1", "env.dist.pc2",
+   "env.dist.pc3", "geo.dist.b2b"
+ ))
>
> df2 <- df %>% na.omit()
>
> df2 %<>% select(-geo.dist.b2b, -env.dist.pc3, -env.dist.centroid)
> colnames(df2) <- c(
+   "H.tax", "H.phyl", "geo.dist",
+   "admin", "pc1", "pc2"
+ )
>
> # perform the regression model
> formula.tax <- formula(
+   H.tax ~ geo.dist + admin + pc1 + pc2 + geo.dist:admin +
+   geo.dist:pc1 + geo.dist:pc2 + admin:pc1 + admin:pc2
+ )
> formula.phyl <- formula(
+   H.phyl ~ geo.dist + admin + pc1 + pc2 + geo.dist:admin +
+   geo.dist:pc1 + geo.dist:pc2 + admin:pc1 + admin:pc2
+ )
>
> mod.tax <- lm(formula = formula.tax, data = df2)
> mod.phyl <- lm(formula = formula.phyl, data = df2)
>
> # make the predictions
> geo_dist_predictor <- seq(100, 20000, 100)
> geo_dist_predictor <- paste(geo_dist_predictor, collapse = ", ")
> geo_dist_predictor <- paste0("geo.dist [", geo_dist_predictor, "]")
> df.tax.predicted.pc1 <- ggpredict(
+   model = mod.tax,
+   terms = c(geo_dist_predictor, "admin", "pc1 [1,3,5,7,9]")
+ )
> df.tax.predicted.pc2 <- ggpredict(
+   model = mod.tax,
+   terms = c(geo_dist_predictor, "admin", "pc2 [1,3,5,7,9]")
+ )
> df.phyl.predicted.pc1 <- ggpredict(
+   model = mod.phyl,
+   terms = c(geo_dist_predictor, "admin", "pc1 [1,3,5,7,9]")
+ )
> df.phyl.predicted.pc2 <- ggpredict(
+   model = mod.phyl,
+   terms = c(geo_dist_predictor, "admin", "pc2 [1,3,5,7,9]")
+ )
> df.tax.predicted.pc1 %<>%
+   rename(
+     `geographic distance` = x,
+     `administrative relation` = group,
+     `climatic distance (pc1)` = facet
+   ) %>%

```



```

+   as.data.frame()
> df.tax.predicted.pc2 %<>%
+   rename(
+     `geographic distance` = x,
+     `administrative relation` = group,
+     `climatic distance (pc2)` = facet
+   ) %>%
+   as.data.frame()
> df.phyl.predicted.pc1 %<>%
+   rename(
+     `geographic distance` = x,
+     `administrative relation` = group,
+     `climatic distance (pc1)` = facet
+   ) %>%
+   as.data.frame()
> df.phyl.predicted.pc2 %<>%
+   rename(
+     `geographic distance` = x,
+     `administrative relation` = group,
+     `climatic distance (pc2)` = facet
+   ) %>%
+   as.data.frame()
>
>
> df.predictor.range <- df2 %>%
+   group_by(admin) %>%
+   summarise(
+     min.geo = quantile(geo.dist, 0.001),
+     max.geo = quantile(geo.dist, 0.999),
+     min.pc1 = min(pc1),
+     max.pc1 = max(pc1),
+     min.pc2 = min(pc2),
+     max.pc2 = max(pc2)
+   ) %>%
+   ungroup()
> df.predictor.range

## # A tibble: 3 x 7
##   admin                min.geo max.geo  min.pc1 max.pc1 min.pc2 max.pc2
##   <chr>                <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 no administrative relation  384.   19549. 0.000761  12.6  0.00847  12.6
## 2 others                  258.   19357. 0.0131    12.1  0.00687  12.1
## 3 same country             12.1    9340. 0.00340    8.12 0.0205    8.87

> range.correct.f <- function(df_ = df.tax.predicted.pc1) {
+   df_no <- df_ %>%
+     filter(`administrative relation` == "no administrative relation")
+   df_others <- df_ %>%
+     filter(`administrative relation` == "others")
+   df_same <- df_ %>%
+     filter(`administrative relation` == "same country")
+   df_no %<>%
+     filter(`geographic distance` > 384, `geographic distance` < 19549)
+   df_others %<>%

```

```

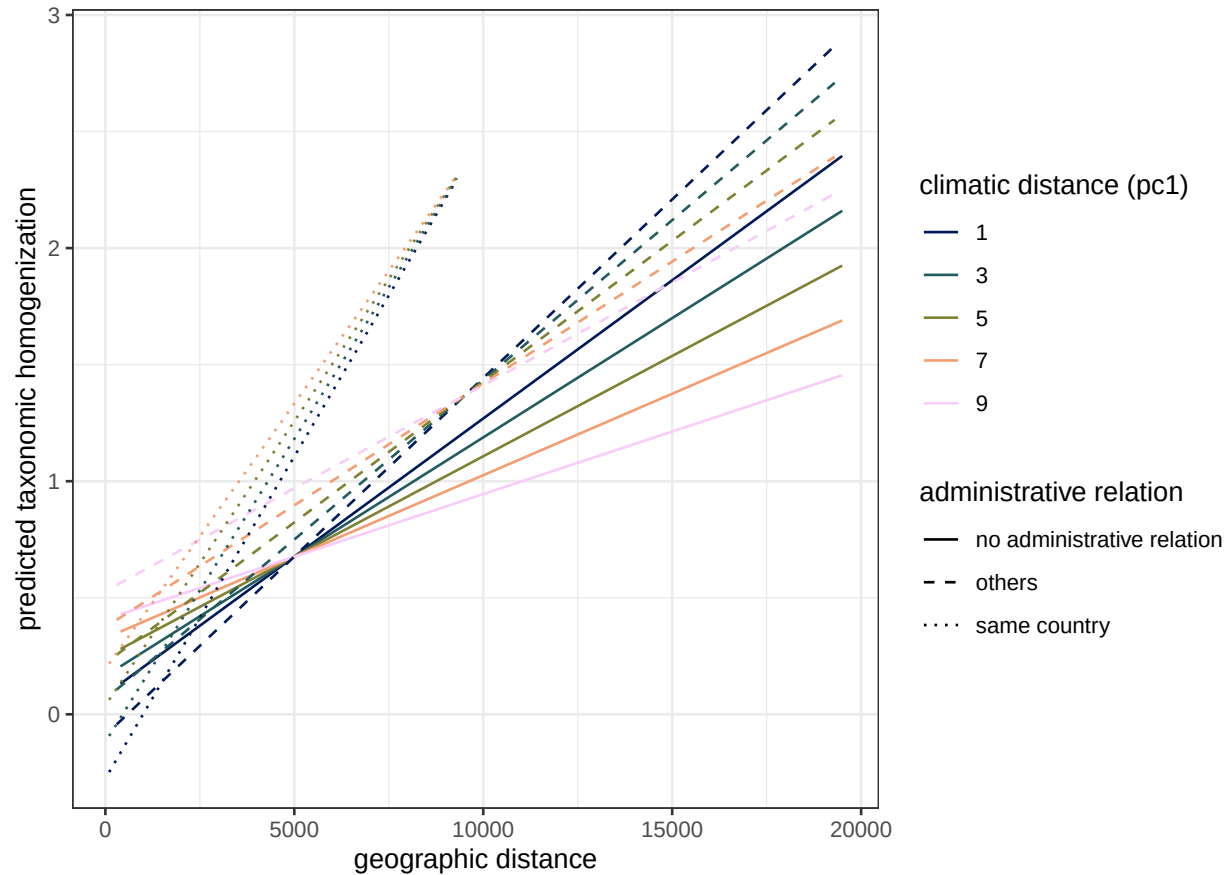
+   filter(`geographic distance` > 258, `geographic distance` < 19357)
+   if (any(str_detect(colnames(df_), "pc1"))){
+     df_same %<>%
+       filter(
+         `geographic distance` > 12.1,
+         `geographic distance` < 9340,
+         `climatic distance (pc1)` != 9
+       )
+   } else {
+     df_same %<>%
+       filter(
+         `geographic distance` > 12.1,
+         `geographic distance` < 9340,
+         `climatic distance (pc2)` != 9
+       )
+   }
+   out <- df_no %>%
+     bind_rows(df_others) %>%
+     bind_rows(df_same)
+   return(out)
+ }
>
> df.tax.predicted.pc1 %<>% range.correct.f()
> df.tax.predicted.pc2 %<>% range.correct.f()
> df.phyl.predicted.pc1 %<>% range.correct.f()
> df.phyl.predicted.pc2 %<>% range.correct.f()
> # releve administrative relationship
> df.tax.predicted.pc1 %<>%
+   mutate(`administrative relation` = fct_relevel(
+     `administrative relation`,
+     c("no administrative relation", "others", "same country")
+   ))
> df.tax.predicted.pc2 %<>%
+   mutate(`administrative relation` = fct_relevel(
+     `administrative relation`,
+     c("no administrative relation", "others", "same country")
+   ))
> df.phyl.predicted.pc1 %<>%
+   mutate(`administrative relation` = fct_relevel(
+     `administrative relation`,
+     c("no administrative relation", "others", "same country")
+   ))
> df.phyl.predicted.pc2 %<>%
+   mutate(`administrative relation` = fct_relevel(
+     `administrative relation`,
+     c("no administrative relation", "others", "same country")
+   ))
>
> # make the figures
> figure.f <- function(dat = df.tax.predicted.pc1,
+   pc.name = "climatic distance (pc1)") {
+   dat <- dat %>%

```

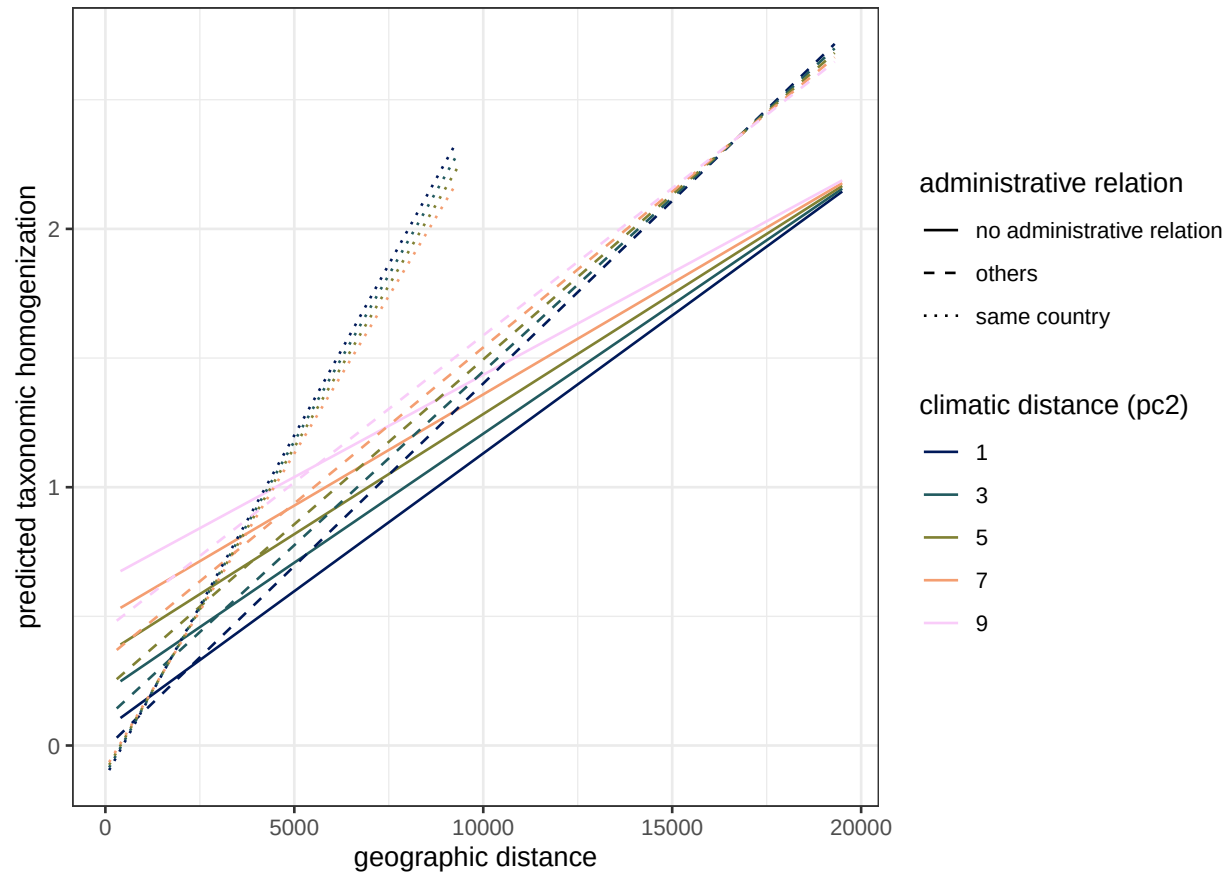
```

+   select(one_of(c(
+     "geographic distance", "predicted",
+     "administrative relation", pc.name
+   )))
+   if (pc.name == "climatic distance (pc1)") {
+     p <- dat %>%
+       ggplot(aes(x = `geographic distance`, y = predicted)) +
+       geom_line(aes(
+         linetype = `administrative relation`,
+         color = `climatic distance (pc1)`
+       ))
+   } else {
+     p <- dat %>%
+       ggplot(aes(x = `geographic distance`, y = predicted)) +
+       geom_line(aes(
+         linetype = `administrative relation`,
+         color = `climatic distance (pc2)`
+       ))
+   }
+   p +
+     scale_color_manual(values = scico(n = 5, palette = "batlow")) +
+     scale_linetype_manual(values = c("solid", "dashed", "dotted")) +
+     theme_bw() +
+     scale_x_continuous(
+       breaks = seq(0, 20000, 5000)
+     )
+ }
+
>
> (p.tax.predicted.pc1 <- figure.f(
+   df.tax.predicted.pc1,
+   pc.name = "climatic distance (pc1)"
+ ) +
+   ylab("predicted taxonomic homogenization"))

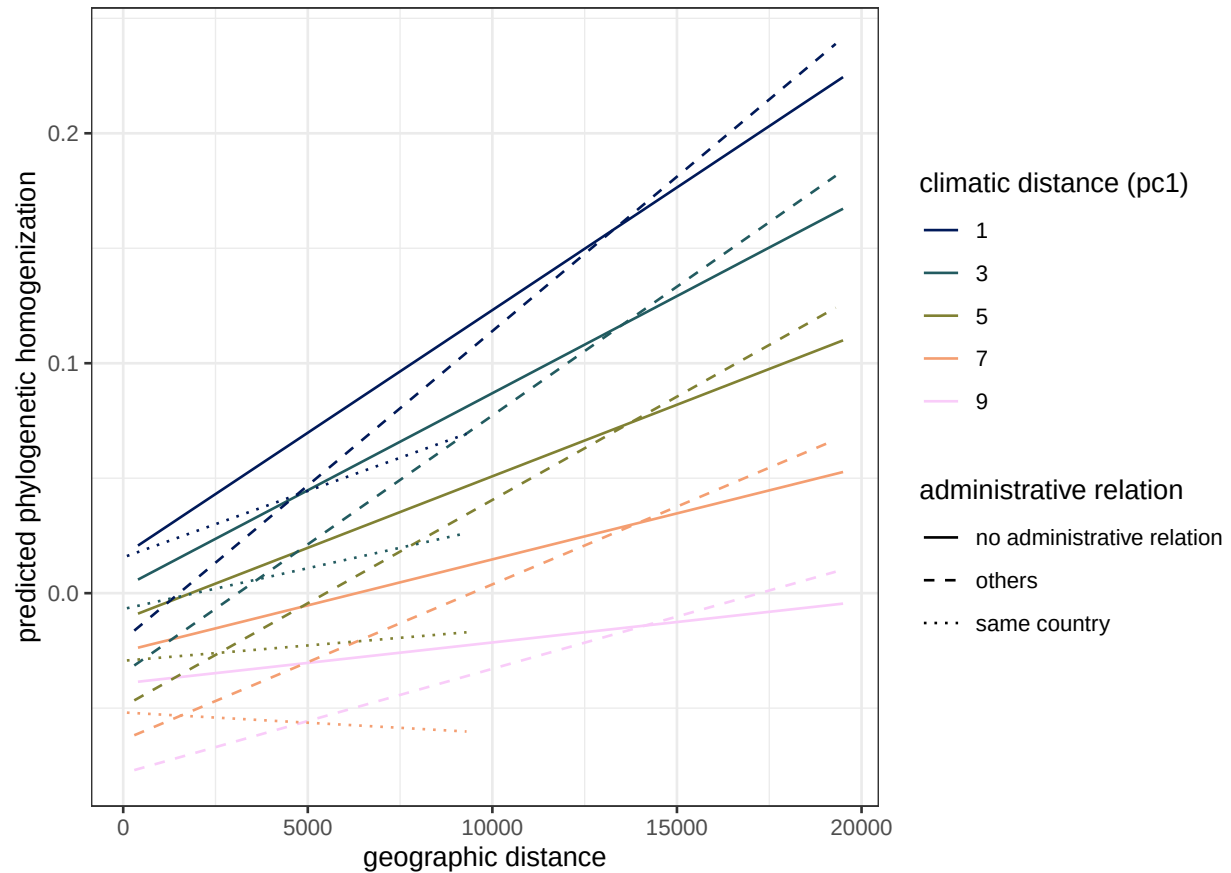
```



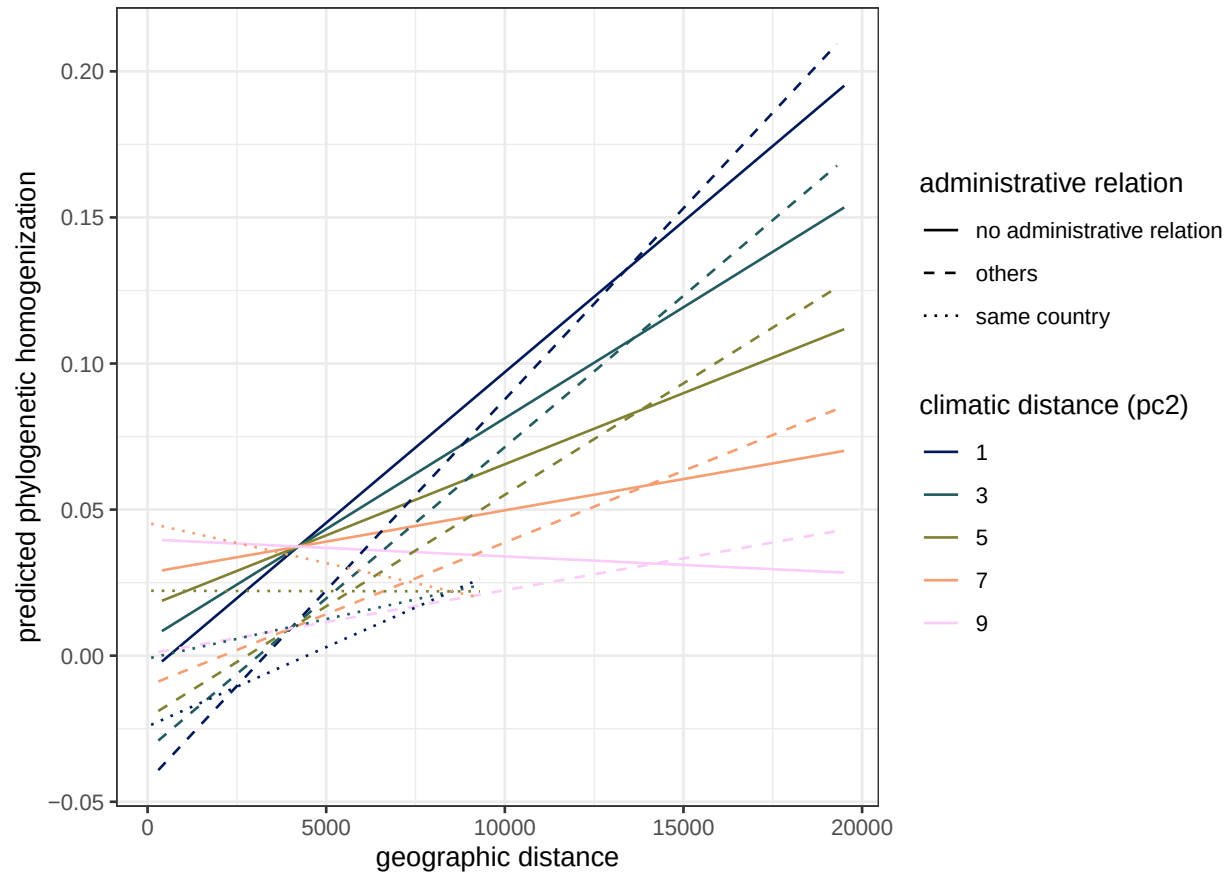
```
> (p.tax.predicted.pc2 <- figure.f(
+   df.tax.predicted.pc2,
+   pc.name = "climatic distance (pc2)"
+ ) +
+   ylab("predicted taxonomic homogenization"))
```



```
> (p.phyl.predicted.pc1 <- figure.f(
+   df.phyl.predicted.pc1,
+   pc.name = "climatic distance (pc1)"
+ ) +
+   ylab("predicted phylogenetic homogenization"))
```



```
> (p.phyl.predicted.pc2 <- figure.f(
+   df.phyl.predicted.pc2,
+   pc.name = "climatic distance (pc2)"
+ ) +
+   ylab("predicted phylogenetic homogenization"))
```



Supplementary Fig. 8

taxonomic homogenization

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.MRM.permutation (pc1&pc2).all.cells.Rdata"))
>
>
> df.taxonomic.coefficient.estimates <- df.MRM.permutation %>%
+   dplyr::select(perm.id, perm.note, MRM.L.perm.taxonomic.tidy) %>%
+   unnest(cols = MRM.L.perm.taxonomic.tidy) %>%
+   filter(term != "(Intercept)")
> df.term <- df.taxonomic.coefficient.estimates %>%
+   filter(perm.id == 1) %>%
+   dplyr::select(term) %>%
+   rename(old.term = term) %>%
+   mutate(term = c(
+     "Geo_dist",
+     "PC1_dist",
+     "PC2_dist",
+     "Admin_dependency",
+     "Admin_same.country",
+     "Geo_dist : PC1_dist",
+     "Geo_dist : PC2_dist",
+     "Geo_dist : Admin_dependency",
+     "Geo_dist : Admin_same.country",
+     "PC1_dist : Admin_dependency",
+     "PC1_dist : Admin_same.country",
+     "PC2_dist : Admin_dependency",
+     "PC2_dist : Admin_same.country"
+   ))
> df.taxonomic.coefficient.estimates %<>%
+   rename(old.term = term) %>%
+   left_join(df.term) %>%
+   dplyr::select(-old.term)
> df.taxonomic.coefficient.estimates.obs <- df.taxonomic.coefficient.estimates %>%
+   filter(perm.note == "obs") %>%
+   dplyr::select(perm.id, perm.note, term, estimate) %>%
+   rename(estimate.obs = estimate)
>
> df.taxonomic.coefficient.estimates %<>%
+   dplyr::select(estimate, term) %>%
+   mutate(term = fct_relevel(term, df.term$term))
> df.taxonomic.coefficient.estimates.obs %<>%
+   mutate(term = fct_relevel(term, df.term$term))
> (p.taxonomic.coefficient.estimates <-
+   df.taxonomic.coefficient.estimates %>%
+   ggplot(aes(x = estimate)) +
+   geom_histogram(bins = 50, fill = "gray", color = "white") +
+   geom_vline(
+     data = df.taxonomic.coefficient.estimates.obs,
+     aes(xintercept = estimate.obs), size = 1
+   ) +

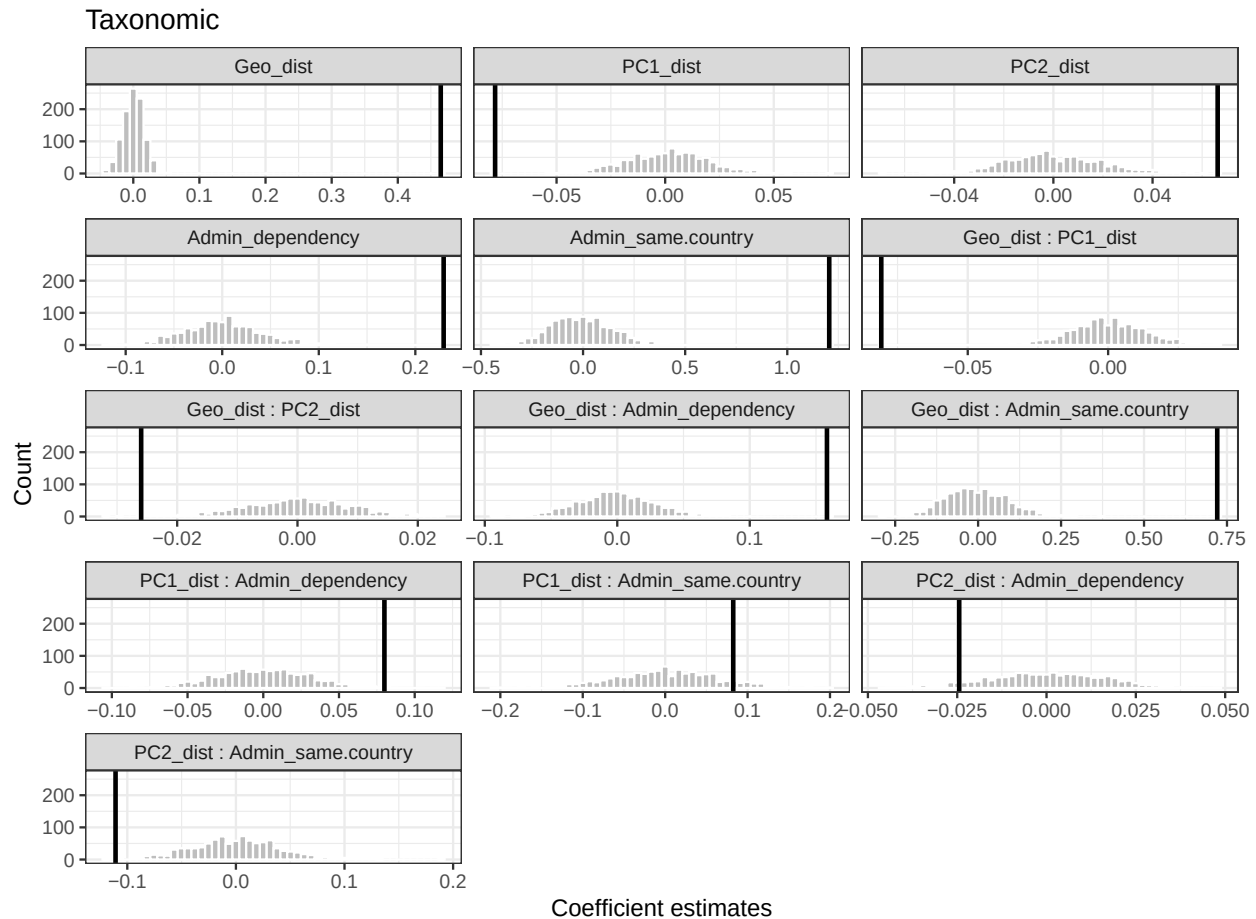
```



```

+ facet_wrap(~term, scales = "free_x", ncol = 3) +
+ theme_bw() +
+ scale_color_manual(values = c("red", "darkgreen", "blue")) +
+ xlab("Coefficient estimates") +
+ ylab("Count") +
+ ggtitle("Taxonomic")

```



phylogenetic homogenization

```

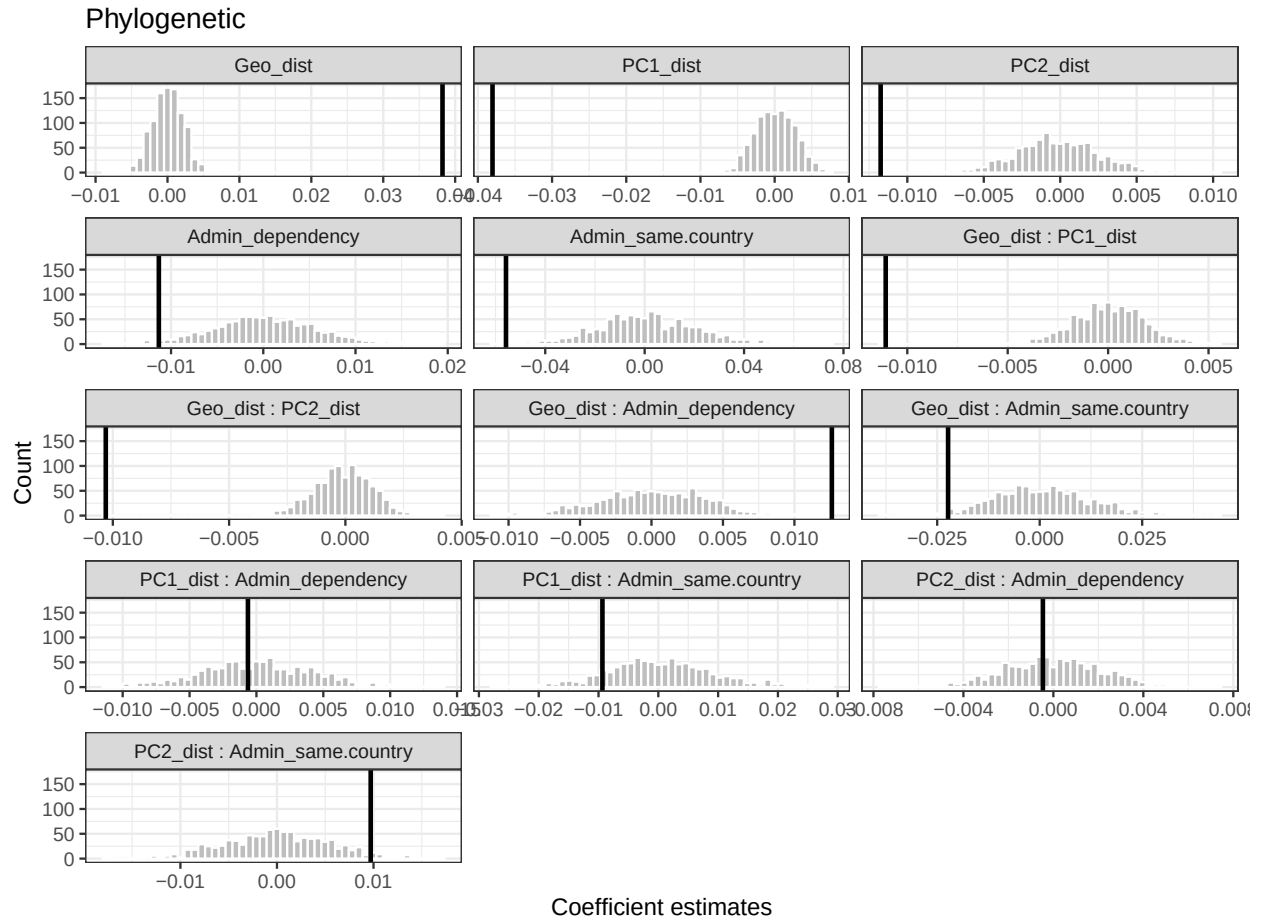
> df.phylogenetic.coefficient.estimates <- df.MRM.permutation %>%
+ dplyr::select(perm.id, perm.note, MRM.L.perm.phylogenetic.tidy) %>%
+ unnest(cols = MRM.L.perm.phylogenetic.tidy) %>%
+ filter(term != "(Intercept)")
> df.term <- df.phylogenetic.coefficient.estimates %>%
+ filter(perm.id == 1) %>%
+ dplyr::select(term) %>%
+ rename(old.term = term) %>%
+ mutate(term = c(
+   "Geo_dist",
+   "PC1_dist",
+   "PC2_dist",
+   "Admin_dependency",

```

```

+   "Admin_same.country",
+   "Geo_dist : PC1_dist",
+   "Geo_dist : PC2_dist",
+   "Geo_dist : Admin_dependency",
+   "Geo_dist : Admin_same.country",
+   "PC1_dist : Admin_dependency",
+   "PC1_dist : Admin_same.country",
+   "PC2_dist : Admin_dependency",
+   "PC2_dist : Admin_same.country"
+ ))
> df.phylogenetic.coefficient.estimates %<>%
+   rename(old.term = term) %>%
+   left_join(df.term) %>%
+   dplyr::select(-old.term)
> df.phylogenetic.coefficient.estimates.obs <-
+   df.phylogenetic.coefficient.estimates %>%
+   filter(perm.note == "obs") %>%
+   dplyr::select(perm.id, perm.note, term, estimate) %>%
+   rename(estimate.obs = estimate)
>
> df.phylogenetic.coefficient.estimates %<>%
+   dplyr::select(estimate, term) %>%
+   mutate(term = fct_relevel(term, df.term$term))
> df.phylogenetic.coefficient.estimates.obs %<>%
+   mutate(term = fct_relevel(term, df.term$term))
> (p.phylogenetic.coefficient.estimates <-
+   df.phylogenetic.coefficient.estimates %>%
+   ggplot(aes(x = estimate)) +
+   geom_histogram(bins = 50, fill = "gray", color = "white") +
+   geom_vline(
+     data = df.phylogenetic.coefficient.estimates.obs,
+     aes(xintercept = estimate.obs), size = 1
+   ) +
+   facet_wrap(~term, scales = "free_x", ncol = 3) +
+   theme_bw() +
+   scale_color_manual(values = c("red", "darkgreen", "blue")) +
+   xlab("Coefficient estimates") +
+   ylab("Count") +
+   ggtitle("Phylogenetic"))

```



Supplementary Fig. 9

```

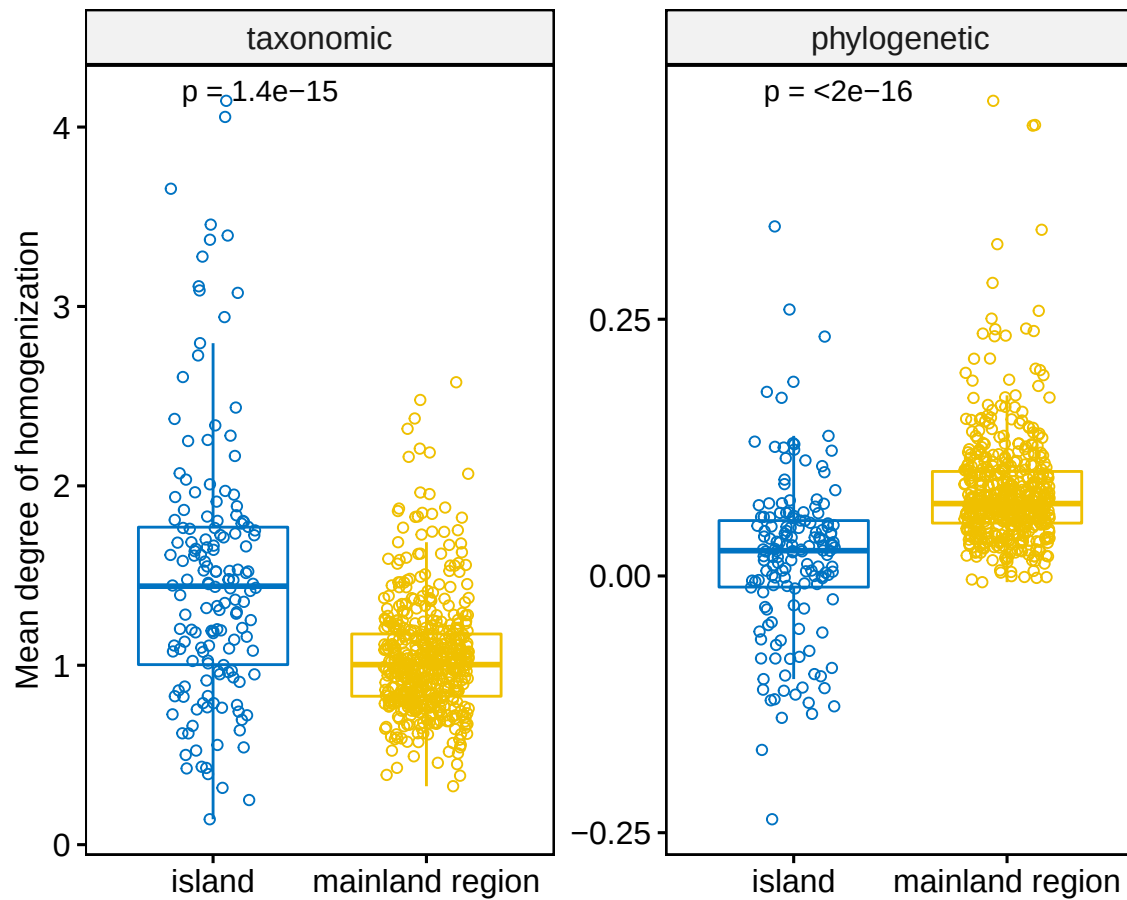
> data.path <- getwd()
> load(file = paste0(data.path, "/df.tidy.pairwise.dat.regional.level.Rdata"))
> load(file = paste0(data.path, "/df_island.Rdata"))
> load(file = paste0(data.path, "/df.region.code.Rdata"))
>
> # calculate average homogenization of one region to all the other regions
> df.regional <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(
+     new_region_id_a, new_region_id_b,
+     homo.sim_by.naturalized_delta.0.001_id,
+     homo.sim_by.naturalized_delta.0.001_phylo
+   )
> df.pairwise <- df.regional
> df.pairwise2 <- df.pairwise %>%
+   dplyr::select(new_region_id_b, new_region_id_a, everything())
> colnames(df.pairwise2) <- colnames(df.pairwise)
> df.pairwise %<>% bind_rows(df.pairwise2)
> df.average <- df.pairwise %>%
+   group_by(new_region_id_a) %>%
+   summarise(
+     mean.homo.id = mean(homo.sim_by.naturalized_delta.0.001_id),
+     mean.homo.phylo = mean(homo.sim_by.naturalized_delta.0.001_phylo)
+   ) %>%
+   rename(new_region_id = new_region_id_a) %>%
+   ungroup()
>
> # get the island information for each region
> df.region.island <- df.region.code %>% left_join(df_island)
>
> # add island information to df.average (excluding the region Chile)
> df.average %<>%
+   filter(new_region_id != 647) %>%
+   left_join(df.region.island) %>%
+   dplyr::select(new_region_id, island) %>% distinct()
>
> # make the graph
> df <- df.average %>%
+   dplyr::select(new_region_id, mean.homo.id, mean.homo.phylo, island) %>%
+   rename(taxonomic = mean.homo.id, phylogenetic = mean.homo.phylo) %>%
+   gather(key = "Group", value = "value", taxonomic, phylogenetic)
> df %<>% mutate(Group = fct_relevel(Group, c("taxonomic", "phylogenetic")))
> df %<>% mutate(region = map_chr(
+   island, function(x) ifelse(x == 1, "island", "mainland region")
+ ))
> p <- ggboxplot(df,
+   x = "region", y = "value",
+   color = "region", palette = "jco",
+   add = "jitter", outlier.shape = NA,
+   add.params = list(shape = 1),
+   facet.by = "Group", short.panel.labs = FALSE
+ )

```

```

> # Use only p.format as label. Remove method name.
> p <- p + stat_compare_means(label = "p.format")
> p <- facet(p, facet.by = "Group", scales = "free_y")
> (p <- p + ylab("Mean degree of homogenization") + xlab("") +
+   theme(
+     strip.text = element_text(size = 12),
+     legend.position = "none"
+   ))

```



Supplementary Fig. 10

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.tidy.pairwise.dat.regional.level.Rdata"))
> load(file = paste0(data.path, "/df_island.Rdata"))
> load(file = paste0(data.path, "/df.region.code.Rdata"))
>
> # calculate average homogenization of one region to all the other regions
> df.regional <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(
+     new_region_id_a, new_region_id_b,
+     homo.sim_by.naturalized_delta.0.001_id,
+     homo.sim_by.naturalized_delta.0.001_phylo
+   )
> df.pairwise <- df.regional
> df.pairwise2 <- df.pairwise %>%
+   dplyr::select(new_region_id_b, new_region_id_a, everything())
> colnames(df.pairwise2) <- colnames(df.pairwise)
> df.pairwise %<>% bind_rows(df.pairwise2)
> df.average <- df.pairwise %>%
+   group_by(new_region_id_a) %>%
+   summarise(
+     mean.homo.id = mean(homo.sim_by.naturalized_delta.0.001_id),
+     mean.homo.phylo = mean(homo.sim_by.naturalized_delta.0.001_phylo)
+   ) %>%
+   rename(new_region_id = new_region_id_a) %>%
+   ungroup()
>
> # get the island information for each region
> df.region.island <- df.region.code %>% left_join(df_island)
>
>
> # add island information to df.average (excluding the region Chile)
> df.average %<>%
+   filter(new_region_id != 647) %>%
+   left_join(df.region.island %>%
+     dplyr::select(new_region_id, island) %>% distinct())
>
> df.average %<>% left_join(df_island)
> df_homo <- df.average
>
> # correlation coefficient for mainland
> df.mainland <- df_homo %>%
+   filter(island == 0) %>%
+   dplyr::select(mean.homo.id, mean.homo.phylo)
> lm.mainland <- lm(mean.homo.phylo ~ mean.homo.id, data = df.mainland)
> summary(lm.mainland)

##
## Call:
## lm(formula = mean.homo.phylo ~ mean.homo.id, data = df.mainland)
##
## Residuals:
##      Min       1Q   Median       3Q      Max

```

```
## -0.10229 -0.01464 -0.00118 0.01599 0.24327
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.064312  0.004764 -13.50  <2e-16 ***
## mean.homo.id 0.140762  0.004364  32.26  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03155 on 501 degrees of freedom
## Multiple R-squared:  0.675, Adjusted R-squared:  0.6743
## F-statistic: 1040 on 1 and 501 DF, p-value: < 2.2e-16

> p.compare.taxonomic.and.phylogenetic.homogenization.mainland <- df_homo %>%
+   filter(island == 0) %>%
+   ggplot(., aes(x = mean.homo.id, y = mean.homo.phylo)) +
+   geom_point(cex = 2, shape = 1) +
+   geom_smooth(method = "lm", color = "blue", fill = "blue") +
+   theme_bw() +
+   xlab("Mean taxonomic homogenization") +
+   ylab("\n Mean phylogenetic homogenization") +
+   ggtitle("a")
>
> # correlation coefficient for island
> df.island <- df_homo %>%
+   filter(island == 1) %>%
+   dplyr::select(mean.homo.id, mean.homo.phylo)
> lm.island <- lm(mean.homo.phylo ~ mean.homo.id, data = df.island)
> summary(lm.island)

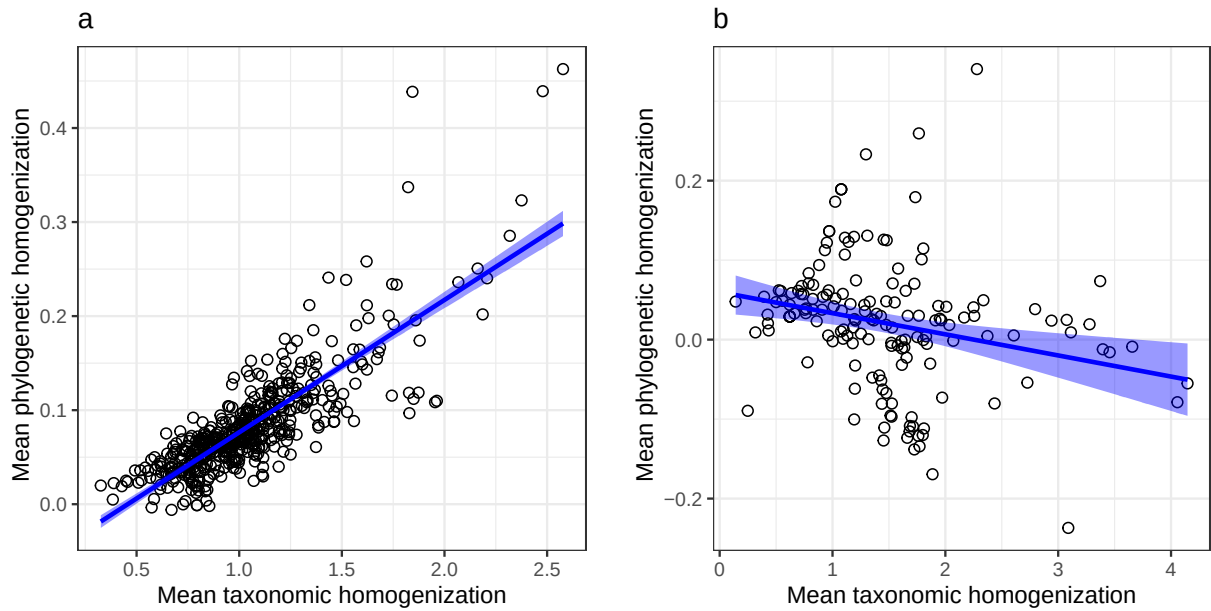
##
## Call:
## lm(formula = mean.homo.phylo ~ mean.homo.id, data = df.island)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.21461 -0.02857  0.00063  0.02921  0.34112
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.059847  0.013483  4.439 1.67e-05 ***
## mean.homo.id -0.026584  0.008255 -3.221  0.00155 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07826 on 161 degrees of freedom
## Multiple R-squared:  0.06052, Adjusted R-squared:  0.05469
## F-statistic: 10.37 on 1 and 161 DF, p-value: 0.001548

> p.compare.taxonomic.and.phylogenetic.homogenization.island <- df_homo %>%
+   filter(island == 1) %>%
+   ggplot(., aes(x = mean.homo.id, y = mean.homo.phylo)) +
+   geom_point(cex = 2, shape = 1) +
+   geom_smooth(method = "lm", color = "blue", fill = "blue") +
+   theme_bw() +
```

```

+   xlab("Mean taxonomic homogenization") +
+   ylab("\n Mean phylogenetic homogenization") +
+   ggtitle("b")
+
>
> grid.arrange(p.compare.taxonomic.and.phylogenetic.homogenization.mainland,
+   p.compare.taxonomic.and.phylogenetic.homogenization.island,
+   nrow = 1
+ )

```



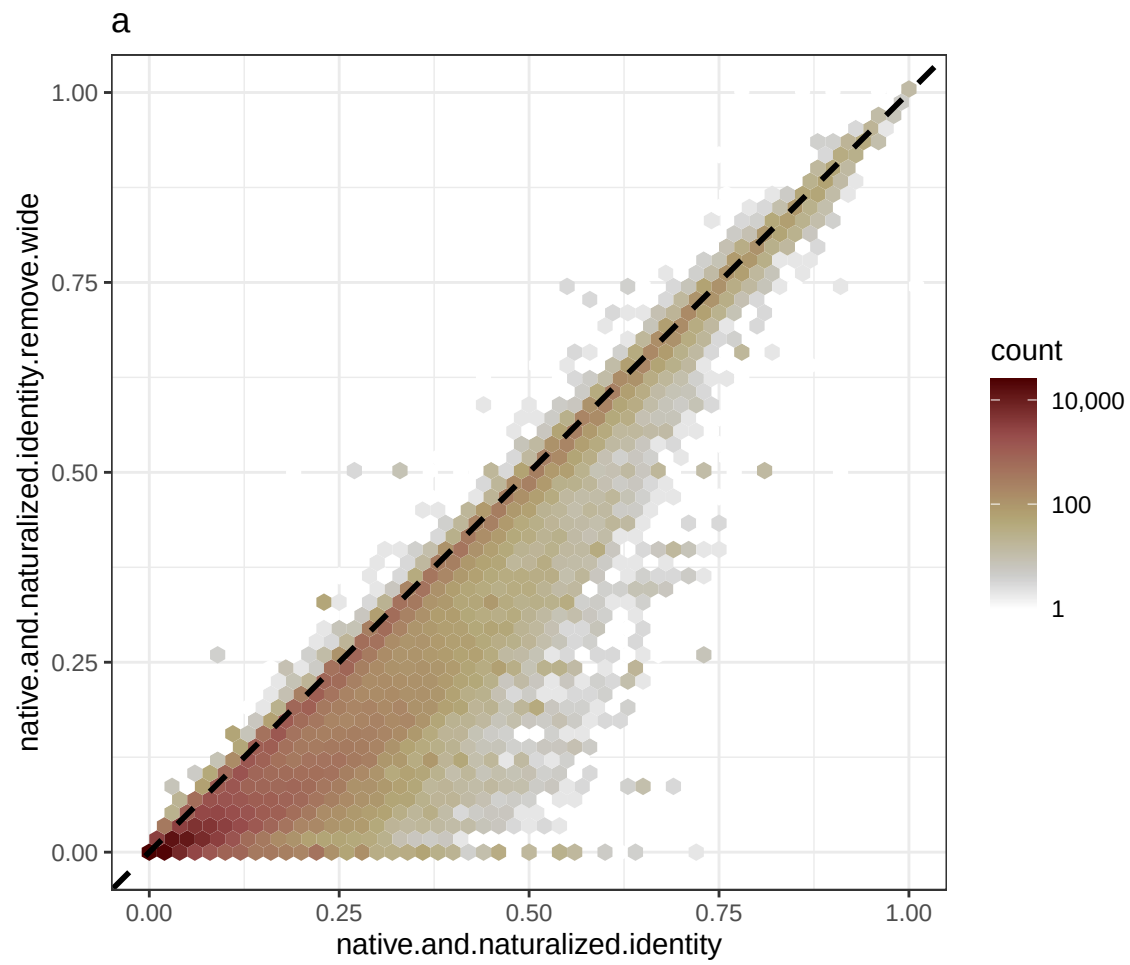
Supplementary Fig. 11

```

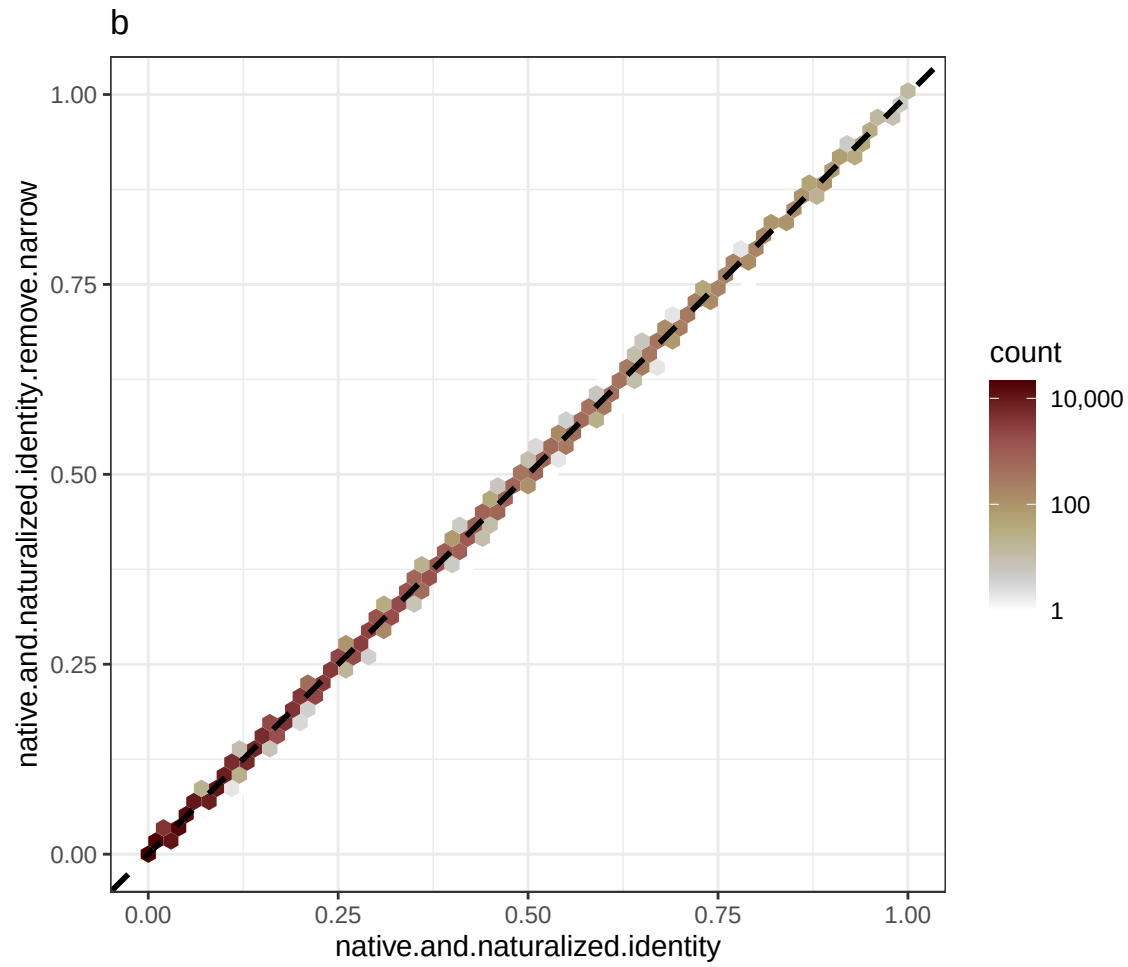
> data.path <- getwd()
>
> load(file = paste0(data.path,
+                     "/df.pairwise.dat.regional.level.removal.experiment.Rdata"))
> load(file = paste0(data.path,
+                     "/df.tidy.pairwise.dat.regional.level.Rdata"))
>
> df.maindata.subset <- df.tidy.pairwise.dat.regional.level %>%
+   select(
+     new_region_id_a, new_region_id_b,
+     beta.sim.native.and.naturalized.identity,
+     beta.sim.native.and.naturalized.phylogeny
+   )
> df.removal.subset <- df.pairwise.dat.regional.level.removal.experiment %>%
+   select(
+     new_region_id_a, new_region_id_b,
+     contains("beta.sim")
+   )
> df <- df.maindata.subset %>% left_join(df.removal.subset)
>
>
> f.hex <- function(dat, xvar, yvar) {
+   dat %>%
+     select(one_of(c(xvar, yvar))) %>%
+     set_colnames(value = c("x", "y")) %>%
+     ggplot(aes(x = x, y = y)) +
+     geom_hex(bins = 50, aes(fill = stat(count))) +
+     scale_fill_scico(palette = "bilbao", trans = "log10",
+       labels = scales::label_comma(accuracy = 1)
+     ) +
+     geom_abline(
+       slope = 1, intercept = 0, color = "black",
+       size = 1, linetype = "dashed"
+     ) +
+     coord_fixed(ratio = 1, xlim = c(0, 1), ylim = c(0, 1)) +
+     theme_bw()
+ }
>
> p1 <- f.hex(
+   dat = df,
+   xvar = "beta.sim.native.and.naturalized.identity",
+   yvar = "beta.sim.native.and.naturalized.identity.remove.wide"
+ ) +
+   xlab("native.and.naturalized.identity") +
+   ylab("native.and.naturalized.identity.remove.wide")
>
> p2 <- f.hex(
+   dat = df,
+   xvar = "beta.sim.native.and.naturalized.identity",
+   yvar = "beta.sim.native.and.naturalized.identity.remove.narrow"
+ ) +

```

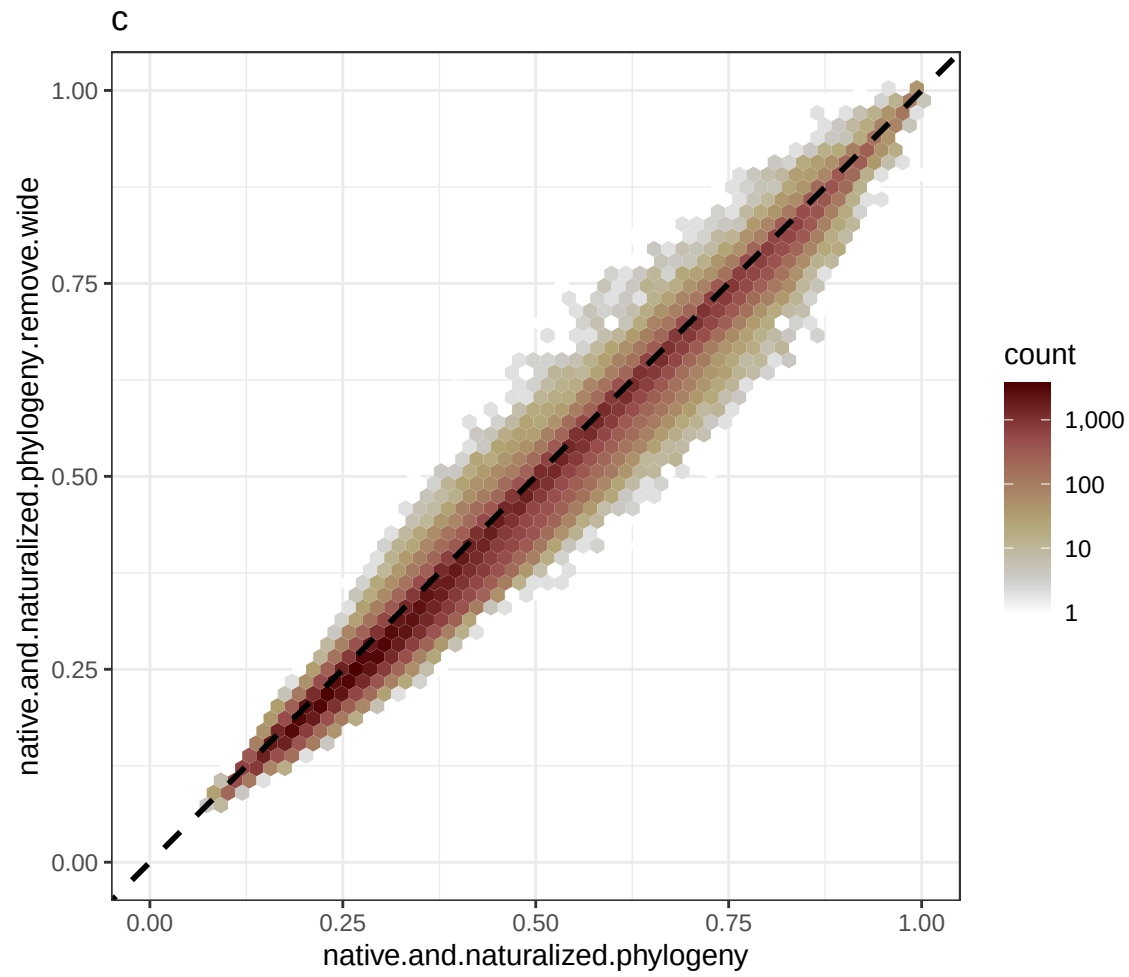
```
+ xlab("native.and.naturalized.identity") +
+ ylab("native.and.naturalized.identity.remove.narrow")
>
> p3 <- f.hex(
+   dat = df,
+   xvar = "beta.sim.native.and.naturalized.phylogeny",
+   yvar = "beta.sim.native.and.naturalized.phylogeny.remove.wide"
+ ) +
+ xlab("native.and.naturalized.phylogeny") +
+ ylab("native.and.naturalized.phylogeny.remove.wide")
>
> p4 <- f.hex(
+   dat = df,
+   xvar = "beta.sim.native.and.naturalized.phylogeny",
+   yvar = "beta.sim.native.and.naturalized.phylogeny.remove.narrow"
+ ) +
+ xlab("native.and.naturalized.phylogeny") +
+ ylab("native.and.naturalized.phylogeny.remove.narrow")
>
> p1 + ggtitle("a")
```



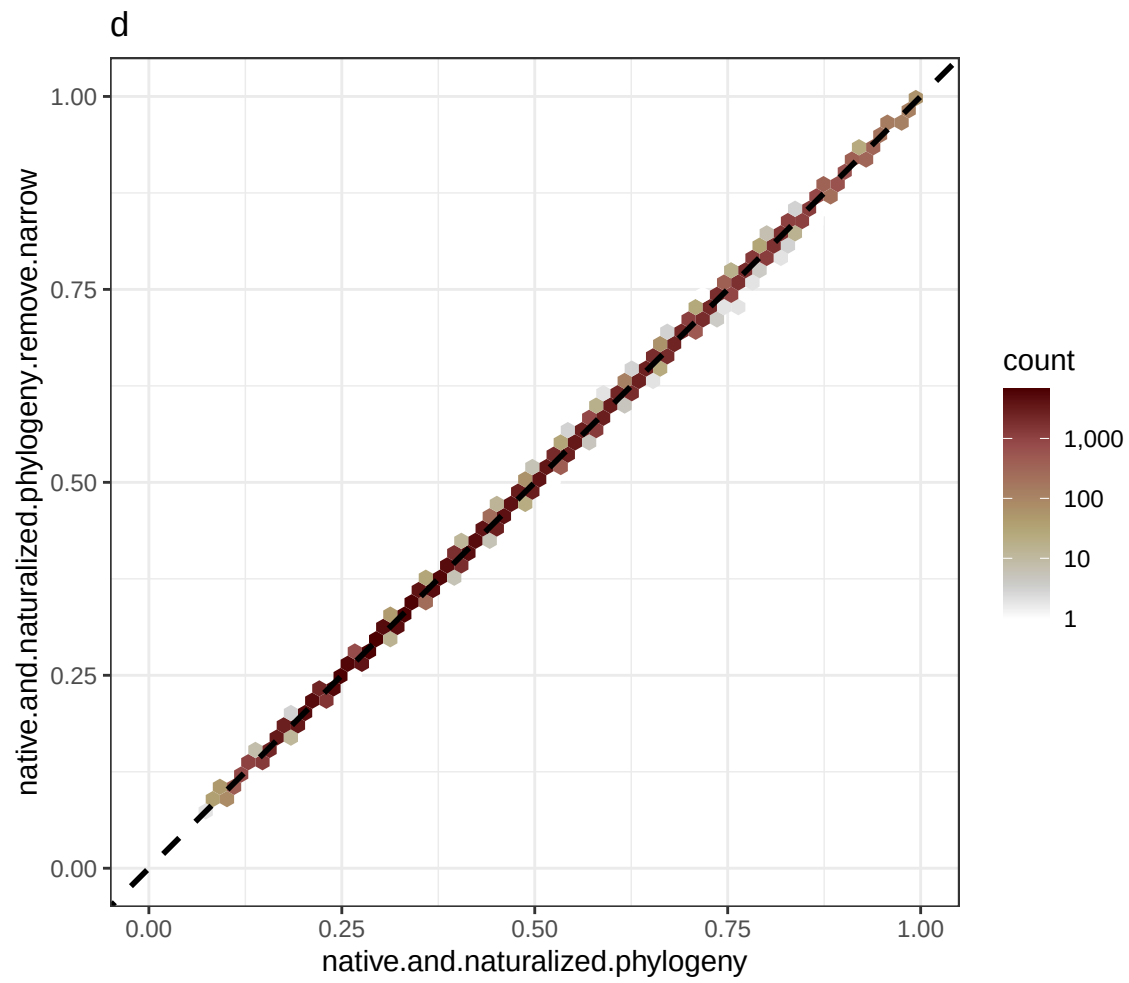
```
> p2 + ggtitle("b")
```



```
> p3 + ggtitle("c")
```



```
> p4 + ggtitle("d")
```



Supplementary Fig. 12

```

> data.path <- getwd()
> load(file = paste0(data.path, "/df.tidy.pairwise.dat.regional.level.Rdata"))
> load(file = paste0(data.path, "/df.region.code.Rdata")) # new region id, code
> load(paste0(data.path, "/df_island.Rdata")) # code island
>
> wrld <- readOGR(
+   dsn = paste0(data.path, "/regions_2020-08-11"),
+   layer = "regions_2020-08-11"
+ )
>
> # calculate the mean value of similarity index
> df.regional <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(
+     new_region_id_a, new_region_id_b,
+     beta.sim.native.identity, beta.sim.native.phylogeny
+   ) %>%
+   rename(place_a = new_region_id_a, place_b = new_region_id_b)
> df.pairwise <- df.regional
> df.pairwise2 <- df.pairwise %>% dplyr::select(place_b, place_a, everything())
> colnames(df.pairwise2) <- colnames(df.pairwise)
> df.pairwise %<>% bind_rows(df.pairwise2)
> df.average <- df.pairwise %>%
+   group_by(place_a) %>%
+   summarise(
+     mean.sim.id = mean(beta.sim.native.identity),
+     mean.sim.phylo = mean(beta.sim.native.phylogeny)
+   ) %>%
+   rename(place_id = place_a) %>%
+   ungroup()
>
> # join average data with shape files
> wrld.sf <- wrld %>% st_as_sf()
> # there are two KOR in wrld.sf, remove one
> wrld.sf %<>% mutate(sf.id = 1:nrow(wrld.sf)) %>% filter(sf.id != 911)
> df.average.sim.regional <- df.average %>%
+   left_join(df.region.code, by = c("place_id" = "new_region_id"))
> wrld.regional.level <- wrld.sf %>%
+   left_join(df.average.sim.regional, by = c("IDregion" = "code")) %>%
+   filter(!is.na(place_id))
>
>
> # now cut the sim value into pieces based on their range
> breaks_mean_id <- c(
+   min(wrld.regional.level$mean.sim.id),
+   seq(0.03, 0.18, 0.03),
+   max(wrld.regional.level$mean.sim.id)
+ )
> breaks_mean_phylo <- c(
+   min(wrld.regional.level$mean.sim.phylo),
+   seq(0.3, 0.6, 0.06),
+   max(wrld.regional.level$mean.sim.phylo)

```

```

+ )
> labels_mean_id <- round(breaks_mean_id[-1], 2)
> labels_mean_id_full <- round(breaks_mean_id, 2)
> labels_mean_phylo <- round(breaks_mean_phylo[-1], 2)
> labels_mean_phylo_full <- round(breaks_mean_phylo, 2)
> wrld.regional.level %<>%
+   mutate(mean.sim.id.group = cut(mean.sim.id,
+     breaks = breaks_mean_id,
+     include.lowest = TRUE,
+     labels = labels_mean_id
+   )) %>%
+   mutate(mean.sim.phylo.group = cut(mean.sim.phylo,
+     breaks = breaks_mean_phylo,
+     include.lowest = TRUE,
+     labels = labels_mean_phylo
+   ))
>
> df_island %<>% dplyr::select(code, island) %>%
+   rename(IDregion = code) %>%
+   filter(IDregion %in% wrld.regional.level$IDregion) %>%
+   distinct()
> wrld.regional.level %<>% left_join(df_island)
> small.island.centroids <- wrld.regional.level %>%
+   filter(island == 1) %>%
+   filter(GeodAREA < 10000) %>%
+   st_centroid()
>
>
> # define color gradients
> colors1 <- rev(magma(7)) # color gradients for taxonomic similarity
> colors2 <- colors1 # color gradients for phylogenetic similarity
> plot(1:7, col = colors2, pch = 19, cex = 5)
>
>
> # make the map for mean taxonomic similarity
> scale_manual_id <- list(
+   scale_fill_manual(values = colors1, drop = FALSE),
+   scale_color_manual(values = colors1, drop = FALSE)
+ )
>
> p.mean.id <-
+   ggplot() +
+   geom_sf(data = wrld.sf, fill = "gray", color = "gray") +
+   geom_sf(data = wrld.regional.level %>%
+     filter(island == 0), aes(fill = mean.sim.id.group), color = NA) +
+   geom_sf(data = wrld.regional.level %>%
+     filter(island == 1), aes(fill = mean.sim.id.group), color = NA) +
+   geom_sf(data = small.island.centroids,
+     aes(color = mean.sim.id.group), shape = 1) +
+   coord_sf(crs = "+proj=wag4 +lon_0=0 +datum=WGS84 +units=m +no_defs") +
+   theme_minimal() +
+   scale_manual_id +
+   labs(title = "a")

```



```

>
>
> # make the figure for mean phylogenetic similarity
> scale_manual_phylo <- list(
+   scale_fill_manual(values = colors2, drop = FALSE),
+   scale_color_manual(values = colors2, drop = FALSE)
+ )
>
>
> p.mean.phylo <-
+   ggplot() +
+   geom_sf(data = wrld.sf, fill = "gray", color = "gray") +
+   geom_sf(data = wrld.regional.level %>% filter(island == 0),
+           aes(fill = mean.sim.phylo.group), color = NA) +
+   geom_sf(data = wrld.regional.level %>% filter(island == 1),
+           aes(fill = mean.sim.phylo.group), color = NA) +
+   geom_sf(data = small.island.centroids,
+           aes(color = mean.sim.phylo.group), shape = 1) +
+   coord_sf(crs = "+proj=wag4 +lon_0=0 +datum=WGS84 +units=m +no_defs") +
+   theme_minimal() +
+   scale_manual_phylo +
+   labs(title = "b")

```

Supplementary Fig. 13

```

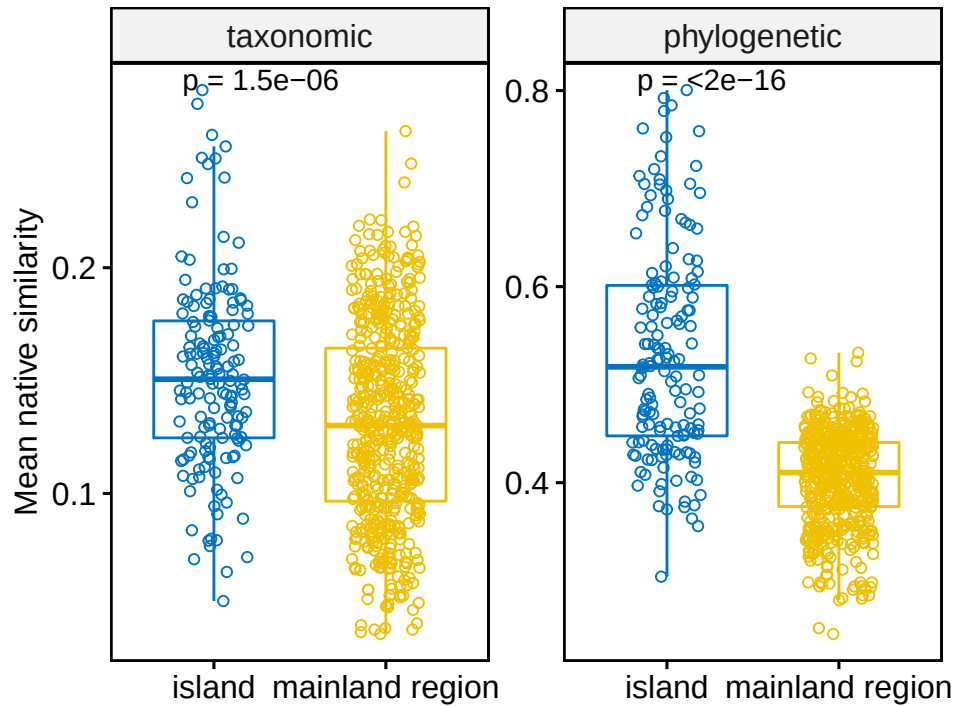
> data.path <- getwd()
> load(file = paste0(data.path, "/df.tidy.pairwise.dat.regional.level.Rdata"))
> load(file = paste0(data.path, "/df_island.Rdata"))
> load(file = paste0(data.path, "/df.region.code.Rdata"))
>
> # calculate average similarity of one region to all the other regions
> df.regional <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(
+     new_region_id_a, new_region_id_b,
+     beta.sim.native.and.naturalized.identity,
+     beta.sim.native.and.naturalized.phylogeny
+   )
> df.pairwise <- df.regional
> df.pairwise2 <- df.pairwise %>%
+   dplyr::select(new_region_id_b, new_region_id_a, everything())
> colnames(df.pairwise2) <- colnames(df.pairwise)
> df.pairwise %<>% bind_rows(df.pairwise2)
> df.average <- df.pairwise %>%
+   group_by(new_region_id_a) %>%
+   summarise(
+     mean.sim.id = mean(beta.sim.native.and.naturalized.identity),
+     mean.sim.phylo = mean(beta.sim.native.and.naturalized.phylogeny)
+   ) %>%
+   rename(new_region_id = new_region_id_a) %>%
+   ungroup()
>
> # get the island information for each region
> df.region.island <- df.region.code %>% left_join(df_island)
>
>
> # add island information to df.average (excluding the region Chile)
> df.average %<>%
+   filter(new_region_id != 647) %>%
+   left_join(df.region.island) %>%
+   dplyr::select(new_region_id, island) %>% distinct()
>
> # make the graph
> df <- df.average %>%
+   dplyr::select(new_region_id, mean.sim.id, mean.sim.phylo, island) %>%
+   rename(taxonomic = mean.sim.id, phylogenetic = mean.sim.phylo) %>%
+   gather(key = "Group", value = "value", taxonomic, phylogenetic)
> df %<>% mutate(Group = fct_relevel(Group, c("taxonomic", "phylogenetic")))
> df %<>% mutate(region = map_chr(
+   island, function(x) ifelse(x == 1, "island", "mainland region")
+ ))
> p <- ggboxplot(df,
+   x = "region", y = "value",
+   color = "region", palette = "jco",
+   add = "jitter", outlier.shape = NA,
+   add.params = list(shape = 1),
+   facet.by = "Group", short.panel.labs = FALSE

```

```

+ )
> # Use only p.format as label. Remove method name.
> p <- p + stat_compare_means(label = "p.format")
> p <- facet(p, facet.by = "Group", scales = "free_y")
> p <- p + ylab("Mean native similarity") + xlab("") +
+   theme(
+     strip.text = element_text(size = 12),
+     legend.position = "none"
+   )
> p

```



Supplementary Fig. 14

define a function for making the hex plot

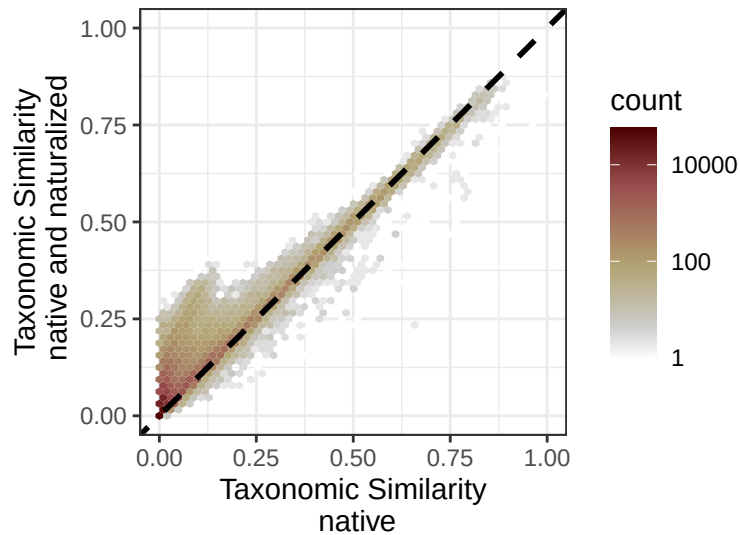
```
> f.hex <- function(dat) {
+   dat %>%
+     ggplot(aes(x = native, y = native.and.naturalized)) +
+     geom_hex(bins = 50, aes(fill = stat(count))) +
+     scale_fill_scico(palette = "bilbao", trans = "log10") +
+     geom_abline(
+       slope = 1, intercept = 0, color = "black", size = 1,
+       linetype = "dashed"
+     ) +
+     coord_fixed(ratio = 1, xlim = c(0, 1), ylim = c(0, 1)) +
+     theme_bw()
+ }
```

define a function for making the histogram

```
> f.hist <- function(dat) {
+   dat %>%
+     ggplot(aes(x = homo.value)) +
+     geom_histogram(bins = 20, colour = "darkgray", fill = NA, size = 0.6) +
+     theme_classic() +
+     geom_vline(xintercept = 0, color = "red", size = .6)
+ }
```

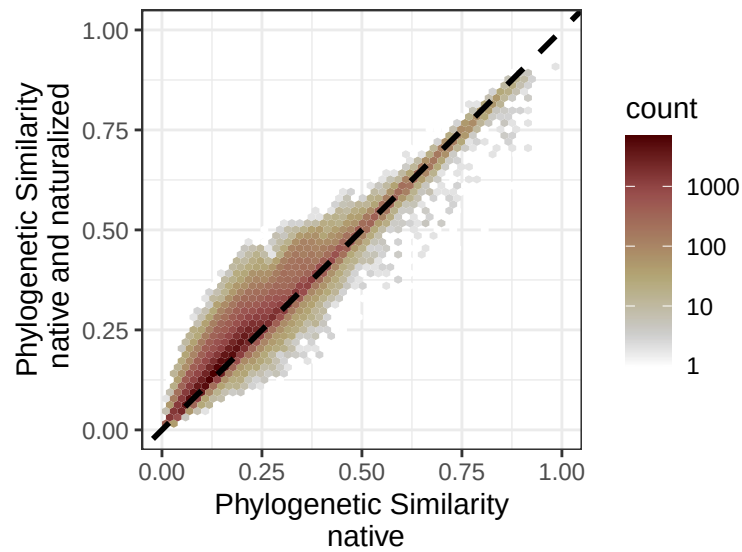
the hex plot in Supplementary Fig. 13a

```
> load(file = "df.tidy.pairwise.dat.regional.level.Rdata")
> select <- dplyr::select
>
> df.tidy.pairwise.dat.regional.level %>%
+   select(beta.sor.native.identity, beta.sor.native.and.naturalized.identity) %>%
+   rename(
+     native = beta.sor.native.identity,
+     native.and.naturalized = beta.sor.native.and.naturalized.identity
+   ) %>%
+   f.hex() +
+   scale_fill_scico(palette = "bilbao", trans = "log10",
+     breaks = c(1, 100, 10000)
+   ) +
+   xlab("Taxonomic Similarity\n native") +
+   ylab("Taxonomic Similarity\n native and naturalized")
```



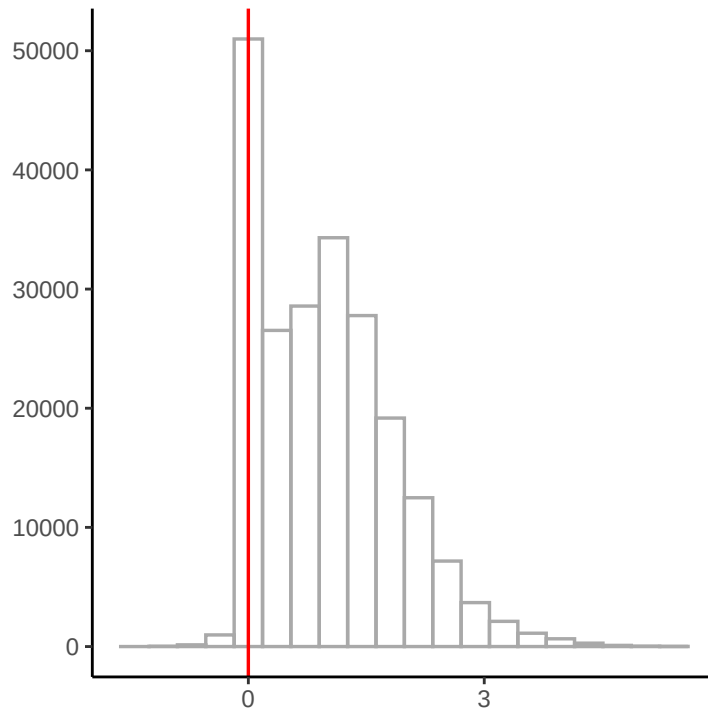
the hex plot in Supplementary Fig. 13b

```
> df.tidy.pairwise.dat.regional.level %>%
+   select(
+     beta.sor.native.phylogeny,
+     beta.sor.native.and.naturalized.phylogeny
+   ) %>%
+   rename(
+     native = beta.sor.native.phylogeny,
+     native.and.naturalized = beta.sor.native.and.naturalized.phylogeny
+   ) %>%
+   f.hex() +
+   scale_fill_scico(palette = "bilbao", trans = "log10",
+     breaks = c(1, 10, 100, 1000)
+   ) +
+   xlab("Phylogenetic Similarity\n native") +
+   ylab("Phylogenetic Similarity\n native and naturalized")
```



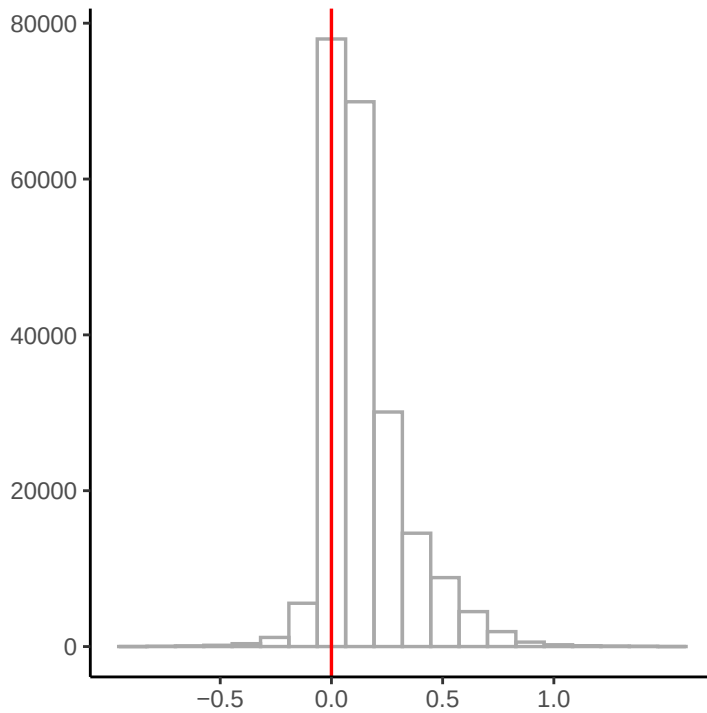
the histogram in Supplementary Fig. 14a

```
> dat <- df.tidy.pairwise.dat.regional.level %>%
+   select(homo.sor_by.naturalized_delta.0.001_id) %>%
+   rename(homo.value = homo.sor_by.naturalized_delta.0.001_id)
> f.hist(dat) +
+   scale_x_continuous(breaks = c(0, 3, 6)) +
+   xlab("") +
+   ylab("")
```



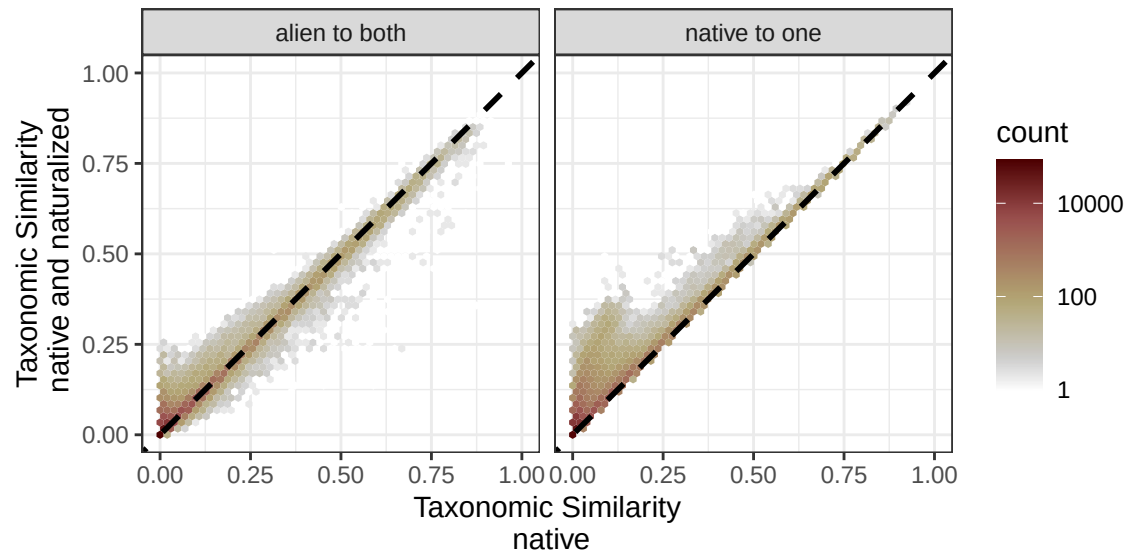
the histogram in Supplementary Fig. 14b

```
> dat <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(homo.sor_by.naturalized_delta.0.001_phylo) %>%
+   rename(homo.value = homo.sor_by.naturalized_delta.0.001_phylo)
> f.hist(dat) +
+   scale_x_continuous(breaks = c(-0.5, 0, 0.5, 1)) +
+   xlab("") +
+   ylab("")
```



the hex plot in Supplementary Fig. 14c

```
> df.plot.dat.identity <- df.tidy.pairwise.dat.regional.level %>%
+   select(
+     beta.sor.native.identity:beta.sor.native.and.outside.species.identity) %>%
+   mutate(row.id = 1:nrow(df.tidy.pairwise.dat.regional.level)) %>%
+   rename(
+     `native to one` = beta.sor.native.and.exchange.identity,
+     `alien to both` = beta.sor.native.and.outside.species.identity
+   ) %>%
+   gather(key = "group", value = "Similarity",
+     -row.id, -beta.sor.native.identity) %>%
+   filter(group %in% c("native to one", "alien to both"))
> df.plot.dat.identity %>%
+   ggplot(aes(x = beta.sor.native.identity, y = Similarity)) +
+   facet_wrap(~group, nrow = 1) +
+   geom_hex(bins = 50, aes(fill = stat(count))) +
+   scale_fill_scico(palette = "bilbao", trans = "log10",
+     breaks = c(1, 100, 10000)) +
+   geom_abline(slope = 1, intercept = 0, size = 1, linetype = "dashed") +
+   coord_fixed(ratio = 1, xlim = c(0, 1), ylim = c(0, 1)) +
+   theme_bw() +
+   xlab("Taxonomic Similarity\n native") +
+   ylab("Taxonomic Similarity\n native and naturalized")
```

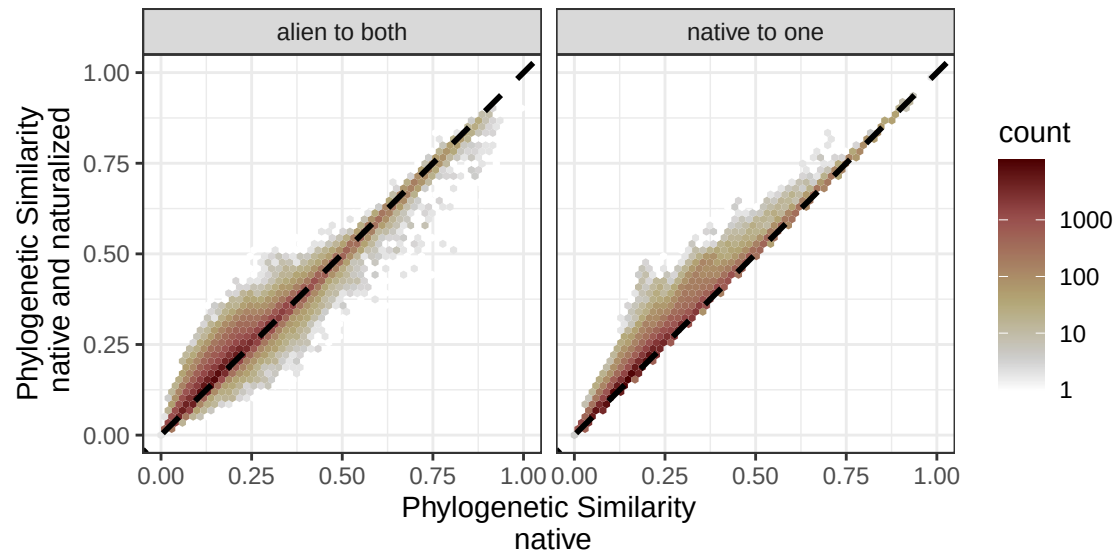



the hex plot in Supplementary Fig. 14d

```

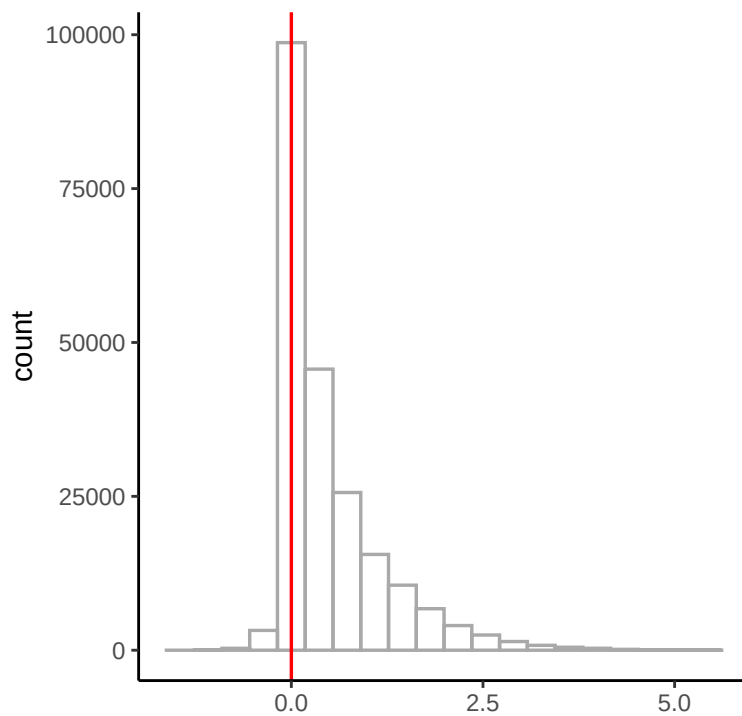
> df.plot.dat.phylogeny <- df.tidy.pairwise.dat.regional.level %>%
+   select(
+     beta.sor.native.phylogeny:beta.sor.native.and.outside.species.phylogeny) %>%
+   mutate(row.id = 1:nrow(df.tidy.pairwise.dat.regional.level)) %>%
+   rename(
+     `native to one` = beta.sor.native.and.exchange.phylogeny,
+     `alien to both` = beta.sor.native.and.outside.species.phylogeny
+   ) %>%
+   gather(key = "group", value = "Similarity",
+     -row.id, -beta.sor.native.phylogeny) %>%
+   filter(group %in% c("native to one", "alien to both"))
> df.plot.dat.phylogeny %>%
+   ggplot(aes(x = beta.sor.native.phylogeny, y = Similarity)) +
+   facet_wrap(~group, nrow = 1) +
+   geom_hex(bins = 50, aes(fill = stat(count))) +
+   scale_fill_scico(palette = "bilbao", trans = "log10",
+     breaks = c(1, 10, 100, 1000)) +
+   geom_abline(slope = 1, intercept = 0, size = 1, linetype = "dashed") +
+   coord_fixed(ratio = 1, xlim = c(0, 1), ylim = c(0, 1)) +
+   theme_bw() +
+   xlab("Phylogenetic Similarity\n native") +
+   ylab("Phylogenetic Similarity\n native and naturalized")

```



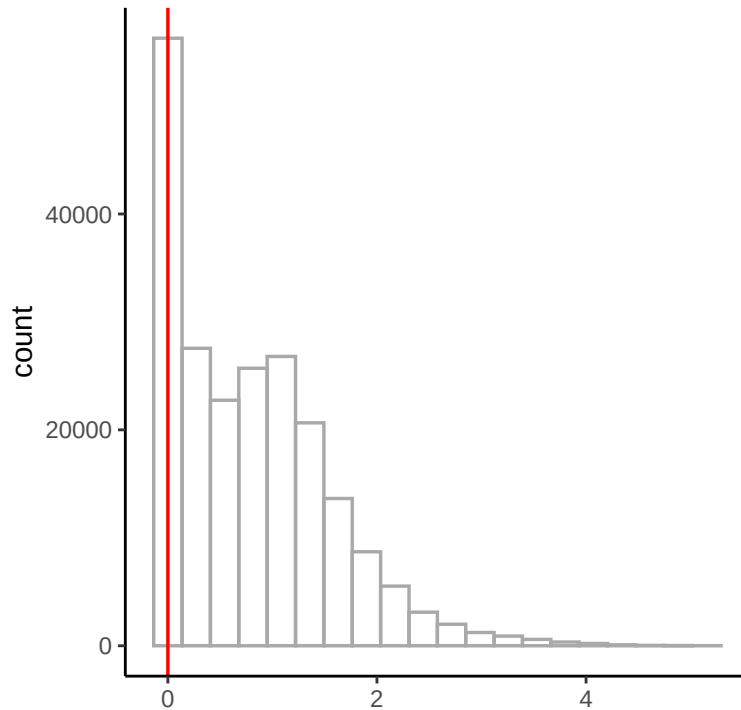
the histogram in Supplementary Fig. 14c (left panel)

```
> dat.taxonomic.outside <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(homo.sor_by.outside.species_delta.0.001_id) %>%
+   rename(homo.value = homo.sor_by.outside.species_delta.0.001_id)
> f.hist(dat = dat.taxonomic.outside) +
+   scale_x_continuous(breaks = c(0, 2.5, 5)) +
+   xlab("")
```



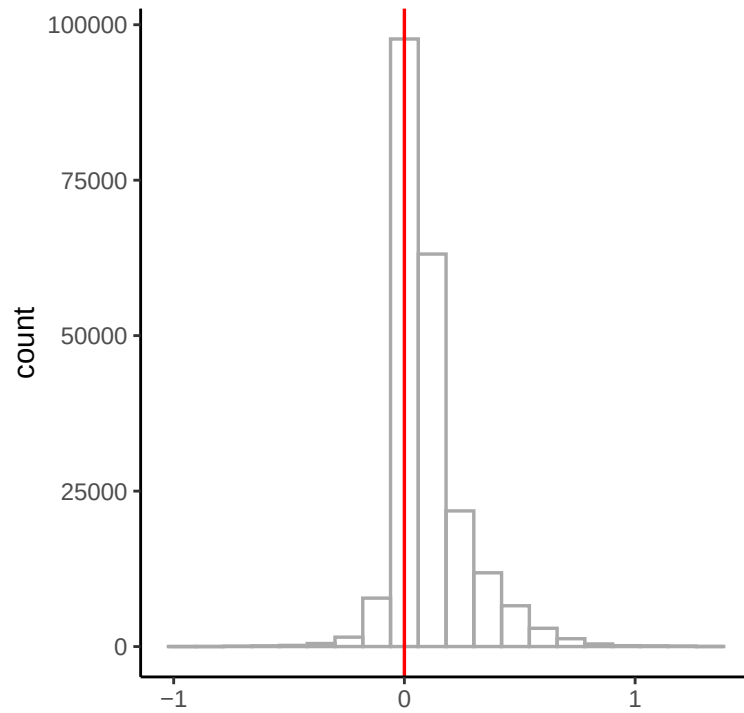
the histogram in Supplementary Fig. 14c (right panel)

```
> dat.taxonomic.within <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(homo.sor_by.exchange_delta.0.001_id) %>%
+   rename(homo.value = homo.sor_by.exchange_delta.0.001_id)
> f.hist(dat = dat.taxonomic.within) +
+   scale_x_continuous(breaks = c(0, 2, 4, 6)) +
+   xlab("")
```



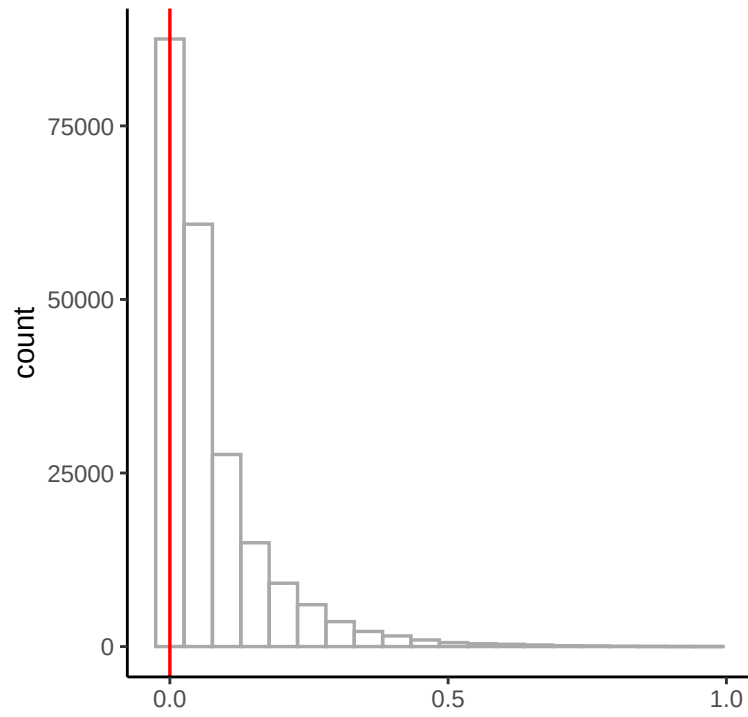
the histogram in Supplementary Fig. 14d (left panel)

```
> dat.phylogenetic.outside <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(homo.sor_by.outside.species_delta.0.001_phylo) %>%
+   rename(homo.value = homo.sor_by.outside.species_delta.0.001_phylo)
> f.hist(dat = dat.phylogenetic.outside) +
+   scale_x_continuous(breaks = c(-1, 0, 1)) +
+   xlab("")
```



the histogram in Supplementary Fig. 14d (right panel)

```
> dat.phylogenetic.within <- df.tidy.pairwise.dat.regional.level %>%
+   dplyr::select(homo.sor_by.exchange_delta.0.001_phylo) %>%
+   rename(homo.value = homo.sor_by.exchange_delta.0.001_phylo)
> f.hist(dat = dat.phylogenetic.within) +
+   scale_x_continuous(breaks = c(0, 0.5, 1)) +
+   xlab("")
```

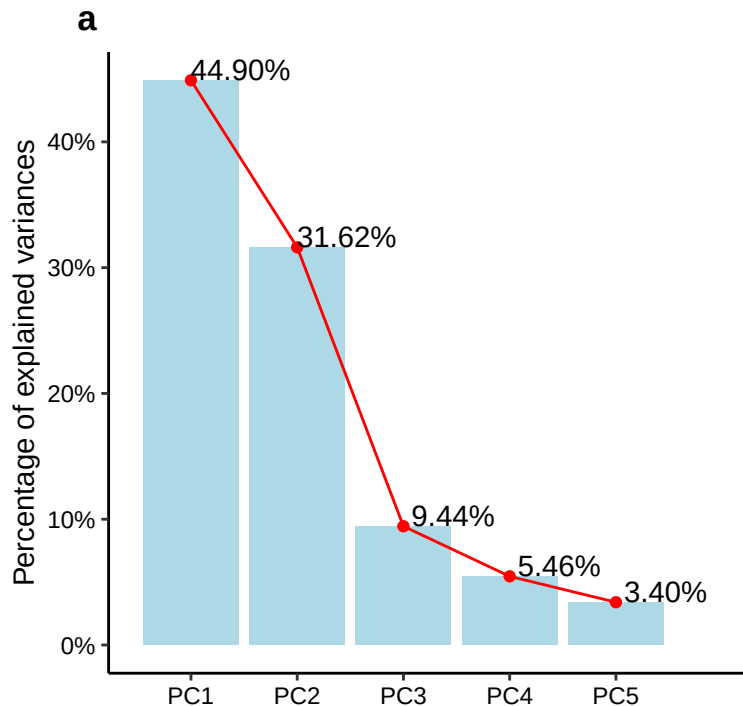


Supplementary Fig. 15

```

> load(file = "df.eig.val.Rdata")
> (
+   p.eig.val <- df.eig.val %>%
+     filter(PC.axis <= 5) %>%
+     ggplot(aes(x = PC.axis, y = variance.percent)) +
+     geom_bar(stat = "identity", fill = "lightblue") +
+     geom_point(color = "red") +
+     geom_line(color = "red") +
+     geom_text(aes(label = var.text), vjust = 0, hjust = 0) +
+     scale_x_continuous(limits = c(0.5, 6), breaks = 1:5,
+       label = paste0("PC", 1:5)) +
+     scale_y_continuous(breaks = seq(0, 40, 10),
+       label = paste0(seq(0, 40, 10), "%")) +
+     theme_classic() +
+     theme(
+       axis.text = element_text(color = "black"),
+       plot.title = element_text(hjust = -.05, face = "bold")
+     ) +
+     xlab("") +
+     ylab("Percentage of explained variances") +
+     ggtitle("a")
+ )

```



```

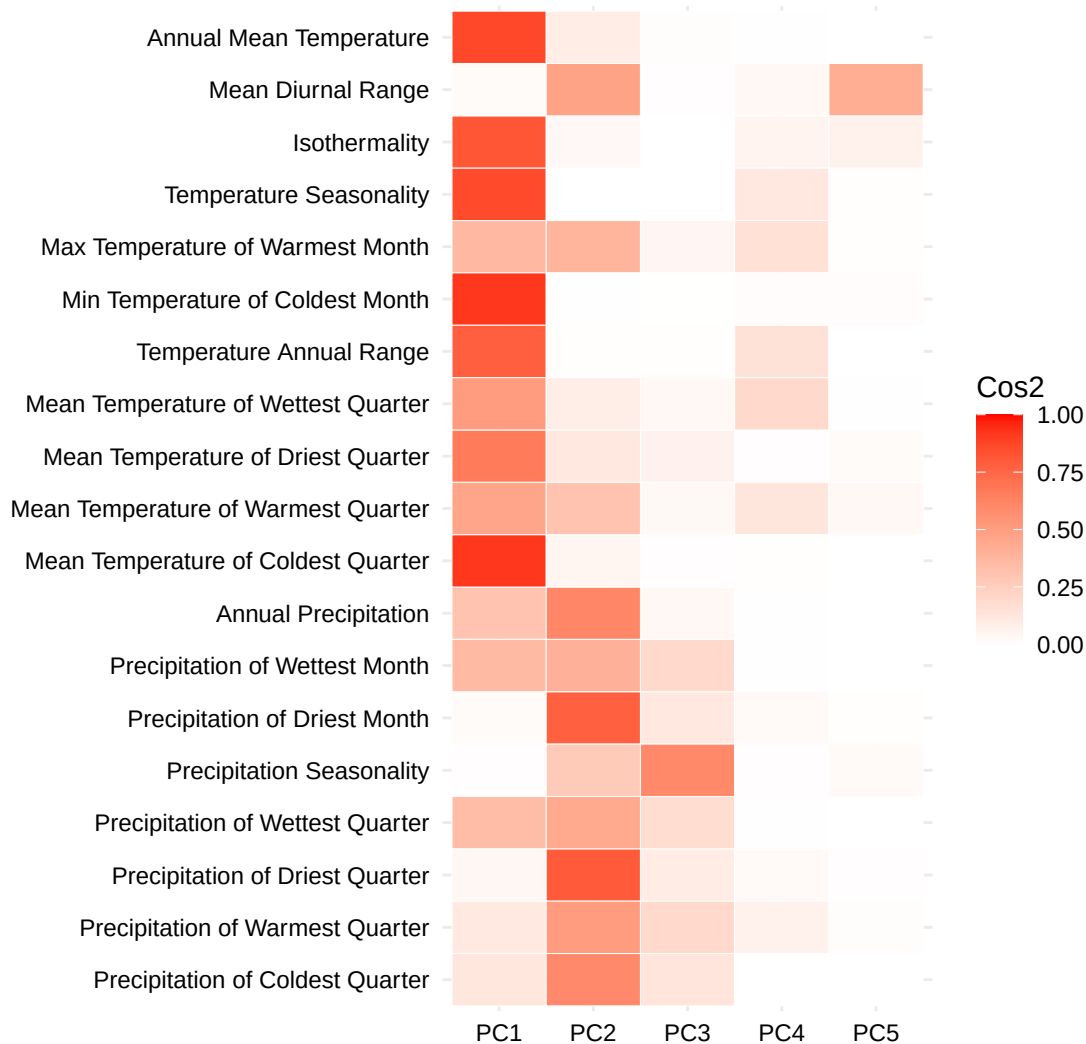
> load(file = "df.cos2.Rdata")
> (
+   p.pca.cos2 <- df.cos2 %>%
+     mutate(Var2 = str_replace_all(Var2, "Dim.", "PC")) %>%

```

```

+   ggplot(., aes(x = Var2, y = Var1, fill = cos2)) +
+   geom_tile(color = "white") +
+   scale_fill_gradient2(
+     low = "white", high = "red", limit = c(0, 1), space = "Lab",
+     name = "Cos2"
+   ) +
+   theme_minimal() +
+   theme(
+     axis.text = element_text(color = "black", size = 9),
+     plot.title = element_text(hjust = -1, face = "bold")
+   ) +
+   xlab("") +
+   ylab("") +
+   ggtitle("b")
+ )

```

b

Supplementary Fig. 16

```

> load(file = "df.semivariance.set.Rdata")
> (p.semivariance <- df.semivariance.set %>%
+   ggplot(aes(x = mid.dist, y = semi_variance)) +
+   facet_wrap(~mod.name, nrow = 1, scales = "free") +
+   geom_line(aes(color = key)) +
+   geom_point(aes(color = key)) +
+   scale_x_continuous(
+     limits = c(0, 20000),
+     breaks = c(0, 5000, 10000, 15000, 20000)
+   ) +
+   scale_color_brewer(palette = "Dark2") +
+   scale_shape_manual(values = c(5, 19)) +
+   theme_bw() +
+   theme(legend.position = "top") +
+   xlab("Lag distance (km)") +
+   ylab("Semivariance")
+ )

```

